A map is a phase without a clock: what path-integrated attention learns, and what decides whether training finds it

Main paper, draft 2 (2026-10-21; revised 2026-10-26 per `docs/audits/2026-10-26/CRITICAL_REVIEW.md` *Part B and cut to about 11k words). Appendix:* `docs/paper/MAIN_APPENDIX2.md`*. Word counts, the source of every number and what changed:* `docs/paper/MAIN_DRAFT2_notes.md`*. Delete this line before submission.*

Abstract

Path-integrated rotary attention replaces RoPE's token index with a learned phase that sums a content-dependent step per token. We measure what such models learn and what decides whether training finds a map. On 1024-move torus walks, where a map-free rule that retraces the last run of moves scores 0.843, one path-integrated layer scores 0.998 (8 of 8 seeds solved) against 0.731 for index RoPE; at matched depth one path layer is worth about three index layers on Dyck-2 and a biased 1D walk, and more than four on the torus, where four index layers score what retracing scores. A map code is a phase without a clock, the phase an out-and-back adds, and we bound how much a clock erodes one attention pass's position match. Two choices of the step map decide whether training removes the clock. Per-head rank: rank 2 solves the torus on at most 2 of 8 seeds against 8 of 8 at rank 4, also at twice the budget and at three recipes, although a rank-2 solution exists and is kept by training; forbidding the clock by construction makes rank 2 solve on every seed. Parameterisation: steps written |s| or softplus(s), which put a barrier at zero, solve the torus in one layer on 0-1 of 8 seeds against 8 of 8 for signed steps from small starts; the |.| cost holds with each step at its best learning rate on a ring and the torus and in a one-layer diagonal complex state-space model, although the |.| solution exists and is kept or re-found. Where the angles start relative to pi matters too: a |.| step started past pi learns the ring map by wrapping, and the LRU-style exp step shows no cost in the state-space model on the ring at one rate, consistent with its past-pi start, but pays in attention on the torus. Told in English, the walk is learned by one path layer (0.979 vs 0.505), and the result survives a pixel front end.

1. Introduction

Rotary position encodings are integrators. Index RoPE (Su et al. 2024) adds a constant angle per token; content-dependent variants let each token set its own step (CARoPE, Veisi et al. 2025; CoPE, Golovneva et al. 2024; Selective RoPE, Movahedi et al. 2026; Mamba-3, Lahoti et al. 2026; MapFormer, Rambaud et al. 2026). When actions drive the step, the summed phase is a position on a map. MapFormer built this into a transformer and showed that it helps navigation and learns opposite latents for opposite actions. We add a measurement of how much of the task needs a map, the exchange rate between path integration and index depth, and an account of what decides whether training finds the map.

The account turns on one decomposition. Split the phase a channel gains over an interval into a part that depends on net displacement, the *map*, and a part that grows with the

number of moves whatever their direction, the *clock*. A map code is one whose used channels carry no clock: an out-and-back returns the phase to where it started, and one attention layer finds the same place at any gap and across the wrap of the torus. Every model we see fail keeps short-gap revisits and loses long-gap and wrap-only ones (Fig. 2), the signature of a clock (Proposition 2).

Depth does not buy the map (Sec. 6), so it has to come from the step map, and whether training finds it depends on whether training removes the clock. A signed step can remove it collectively, driving the latents of opposite moves to sum to zero; above rank D, the map's dimension, each channel can also stop reading the clock on its own. A |.| or softplus step from a small start has neither route, and at rank D no channel can be both clock-free and a full map while a clock remains (Proposition 4). On both axes the constrained classes represent the map exactly, keep or re-find an installed solution, and from scratch rarely find it (Fig. 1).

Contributions.

1. *Measuring map use.* Map-free floors and revisit strata. Retrace scores 0.934 on MapFormer's 128-move walks and 0.843 at 1024 moves, where one path layer scores 0.998 against index RoPE's 0.731 (8/8 vs 0/8 SOLVED, p 0.0002).
2. *What depth buys index attention.* Registered exchange rates at matched length and depth: about three index layers on Dyck-2 and a biased ring, more than four on the torus (replicated on fresh seeds); trained index models learn local return rules whose reach grows with depth in direction runs (analysis).
3. *What decides whether the map is found.* The clock/map decomposition and a proved route inventory per step class. Registered support on both axes: the solution exists, is kept or re-found, and is not found from scratch at rank 2 (at most 2/8, also at twice the budget and three recipes) or with a |.| step (0/8 against 8/8 on the torus, at most 2/8 on the ring, also at each step's best learning rate and in a one-layer diagonal complex SSM, where softplus pays too). Causal tests: forbidding the clock lets rank 2 solve (8/8 against 1/8); freezing the move latents blocks the signed step's collective route (1/8 against 7/8); a signed increment under a non-negative readout restores it (16/16); a |.| step started past pi learns the ring map (8/8).
4. *Readers and codes.* Loops, extra layers and a score rule change how a code is read, never the code family (proved). PoPE's polar score pins every kernel's peak at zero phase difference (proved); freeing its offset raises the index model by +0.122 and leaves the path model at ceiling (registered).
5. *Maps from words.* One path layer learns navigation told in English (registered), 0.956 against 0.411 for two index layers on the revisits retrace cannot reach, and files a word by its role, not its identity.

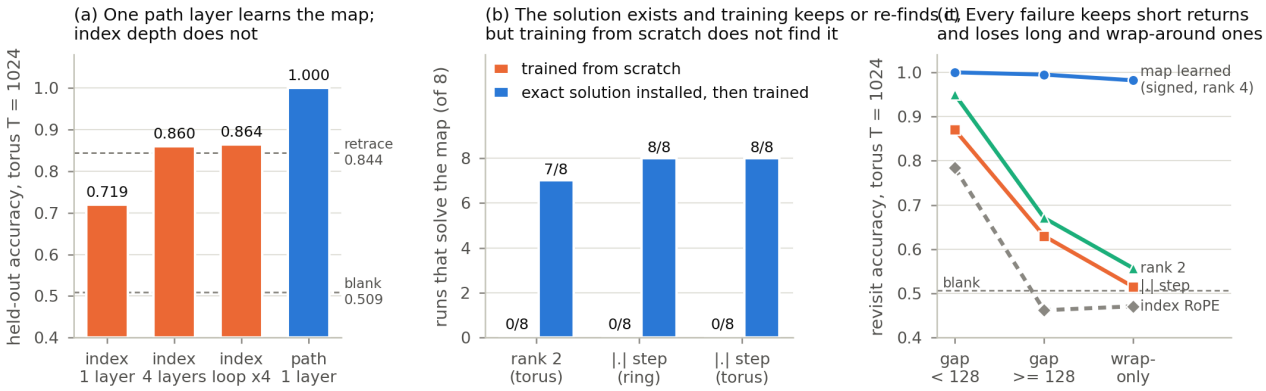


Figure 1 (`fig_summary`). Torus, T = 1024 unless marked. (a) One path-integrated layer learns the map; index attention at one or four layers or looped four times does not (n = 4 shown; fresh-seed replication in App. F7; dashed: retrace and always-blank floors). (b) For both constrained classes a solution exists and training keeps it (rank 2: 7/8; |.| on the ring: 8/8) or re-finds it (|.| on the torus: 8/8), while training from scratch solves 0/8. (c) Every failing one-layer model keeps short-gap revisits and loses long-gap and wrap-only ones; the model that learns the map holds all three (n = 8 per arm; dashed: always-blank within each stratum).

2. Background

Rotary position as an integrator. Write the phase of rotary channel c (a head and a two-dimensional block) at token t as theta_{c,t} = omega_c Phi_{c,t}, with Phi_{c,t} = sum_{u <= t} Delta_{c,u}. The channel's contribution to a score depends on position only through exp(-i omega_c (Phi_{c,t} - Phi_{c,t'})) (Mamba-3's Proposition 3; App. E1). Howard et al. (2014) drive one bank of integrators by a rate alpha(t): a constant rate codes elapsed time (Case I), the speed codes distance travelled (Case II), and the signed velocity codes position (Case III). With an imaginary rate the rotary families, as organised by GRAPE (Zhang et al. 2026), Selective RoPE and Mamba-3, fall into these cases: index RoPE adds a constant angle; CoPE, CARoPE and non-negative content steps a non-negative one; MapFormer, Selective RoPE and Mamba-3 a signed one. Velocity-modulated integrators have been trained with a real Laplace variable (Alipour et al. 2025; Mochizuki-Freeman et al. 2023; Kabir et al. 2025), and RoPE derived as a velocity-controlled oscillator (GridPE, Li et al. 2024). TEM and TEM-t (Whittington et al. 2020, 2022) frame the Case III code as a "where" learned apart from the "what".

MapFormer. MapFormer's working-memory model (MapWM) rotates content queries and keys by theta_t = omega * cumsum(W_out W_in x_t), a low-rank content-to-angle map; observations learn near-zero steps and opposite actions opposite steps (their Fig. 3d). In their Table 2, one-layer MapWM with rank 2 solves 2D navigation at 128 steps, rank 1 does not, and RoPE, TAPE and CoPE with up to four layers fall short. Our implementation reproduces that result (0.989 for MapWM and 0.987 for MapEM, against 1.00 in v4 and 0.99 / 1.0 in v1; App. A4).

Representability. Real non-negative eigenvalues cannot track parity (Sarrof et al. 2024, Thm. 2; Grazzi et al. 2025, Thm. 1). Rotations escape this: rotations by non-negative angles generate cyclic groups (Grazzi et al., Thm. 3; Mamba-3, Sec. 3.2). Proposition 3

extends this to our rank-limited step maps, so every difference between signed and non-negative steps in this paper is a difference in what training finds.

3. Setup and how map use is measured

Models. MapWM with one layer (up to four where stated), 2 heads, d_model 128 (about 204k parameters), and step Delta = W_out W_in x through a rank-r latent; unless stated r = 4, shared across heads (MapFormer's latent is per head; per-head variants are named wherever rank is compared). *Per-head rank* is the rank of the map from token to one head's angles. Step parameterisations: *signed* (Delta = s = W_out W_in x), |.| (Delta = |s|), *softplus* (every step starting at ln 2) and *exp* (the LRU's form, every step starting at 1). |.| and softplus are *barrier parameterisations*: no gradient step carries Delta through zero. *Observation-step-free* variants give steps to the move tokens only. Each head has 32 two-dimensional rotary blocks (*channels*) with frequencies omega_i = 2 pi (1/Delta_max)^(i/31), Delta_max the grid size, as in MapFormer; omega is free. The index baseline is RoPE at 1-4 layers or one block looped four times. We also pair PoPE (Gopalakrishnan et al. 2025), a rotary variant with a polar score, with its path-integrated version MapPoPE, and compare faithful CARoPE, CoPE and Selective RoPE (App. I).

Training and evaluation. AdamW (learning rate 1e-3, weight decay 0.05), warm-up then cosine decay, cross-entropy on revisited observations only (Dyck-2: next-token loss; App. A2). Torus models are scored on a held-out map, so every torus number is transfer to a new map; ring models on fresh maps.

Tasks. (i) *2D torus*: 64 x 64, walked by runs of 1-10 repeated moves interleaved with observations drawn from 16 objects; T = 1024 moves (2,048 tokens), 900 epochs; MapFormer's T = 128 for the floor analysis and the PoPE pair. (ii) *1D ring* of 32 cells, T = 128, forward-step probability p_plus (0.5 balanced; 1.0 never reverses), 300 epochs. (iii) *Dyck-2* at length 32 and nesting depth 12, scored by Hewitt et al.'s (2020) closing accuracy (chance 0.5). (iv) *Text world*: the torus walk rendered as English (58 words), T = 1024 words. (v) 2D and 3D tori of other sizes for rank against dimension. Loss and accuracy are on *revisits*, observations at cells visited before.

Floors and strata. For each task we measure the best map-free predictor: always-blank, the best n-gram over orders 0-5, and *retrace*: while the current run of moves undoes the previous run, copy the observation seen when the walk was last at the cell it is returning to. *Short-gap* revisits come fewer than 128 moves after the last visit; *long-gap* revisits later; *wrap-only* revisits are to a cell seen before only from a different unwrapped position, a lap away. Runs are at most 10 moves, so retrace never reaches a long-gap or wrap-only revisit: those strata separate a map from a local rule (Fig. 2).

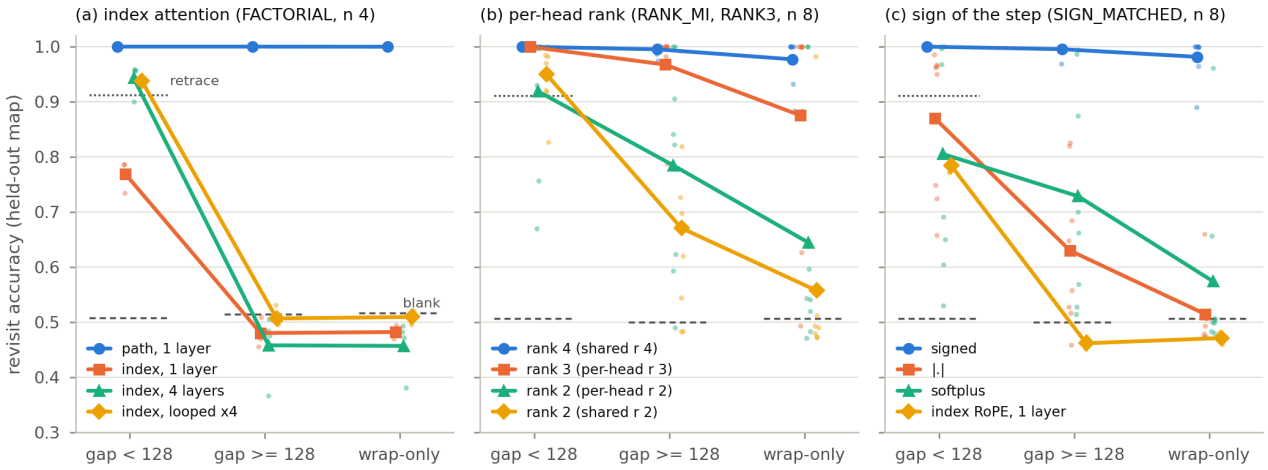


Figure 2 (`fig_strata_modes`). Revisit accuracy by stratum, torus, T = 1024; lines: arm means, dots: runs. (a) Index attention (n = 4): depth and a loop help short-gap revisits only. (b) Per-head rank (n = 8, same initial weights): rank 4 near 1.000 in every stratum; rank 2 loses accuracy with the gap, most at wrap-only revisits. (c) Step parameterisation (n = 8): signed near 1.000 everywhere; |.| and softplus fall from short gaps to wrap-only revisits; one-layer index RoPE for reference. Dashed: always-blank within each stratum; dotted: retrace on short gaps (0.91).

Run classes and protocol. A run is *SOLVED* if its mean training loss over the final 5% of epochs is below 0.05; otherwise *STALLED* if its final-10% mean loss is within 5% of the preceding 10%, and *DESCENDING* otherwise. Every contrast not labelled "(analysis)" is a registered decision or declared secondary of a pre-registered batch unless marked "not registered", with n = 8 seeds per arm unless stated, every model trained and tested at the same length and depth, and exact permutation (accuracy) and Fisher (SOLVED) tests. Numbers marked "(analysis)" were computed post hoc on stored streams and checkpoints. App. A lists every batch with its seeds, its independent code check and its reproduction status; App. O lists every analysis script.

4. Theory: maps and clocks in the phase

Proofs are in App. E; Table 1 gives the status of each statement. A channel c has angle a_c(x) = omega_c step_c(x) for token x; for a step map the latent of token x is z_x = W_in e_x and the channel's readout row is w_c, so step_c(x) = w_c . z_x before any |.| or softplus. The *usage* q_c of a channel is its mean query norm at move tokens times its mean key norm at observation tokens, normalised to sum to 1. A channel is *blind* to an inverse pair if both its angles are 0 mod 2 pi.

Proposition 1 (the prefix sum is in the phase; proved). The rotary score of channel c is Re(conj(q~*c) k~_c exp(-i omega_c (Phi*{c,t} - Phi_{c,s}))). If every move token has a fixed step and every observation step is zero, theta_t = omega P_t, with P_t the unwrapped displacement after move t, and the score depends on position only through omega (P_t - P_s). One layer then compares two positions at any gap, and wrapping costs nothing in a channel whose angle per move is a multiple of 2 pi / N.

Remark 1 (cited). Index attention sees position only through t - s, so "the moves in (s, t] sum to zero" must be computed in the residual stream. A global count gives it in two

layers in principle (Liu et al. 2023, Lemma 6; Yao et al. 2021), with a normaliser, precision 1/n_t (1/2048 per axis here) and MLP width of order 4nT (64 x 2048 per axis, against our 512); count-free rules (retrace, a one-lap offset) are available to one layer. Measured exchange rates are therefore what training finds, not depth bounds.

Clock and map. For an inverse pair of moves (x, y) in channel c write the *clock* k_c = a_c(x) + a_c(y) and the *map* m_c = a_c(y). Over an interval with n_x moves x and n_y moves y, displacement d = n_y - n_x, the phase is m_c d + k_c n_x exactly. A *map code* has k_c = 0 mod 2 pi in every channel attention uses. The *clock / map ratio* is the usage-weighted rms of wrap(k_c) over the usage-weighted rms of the map, with the map taken symmetrically, (wrap(a_c(x))^2 + wrap(a_c(y))^2) / 2, over the inverse pairs and the channels of both heads.

Proposition 2 (position-factor bound and tick rule; proved). (a) A zero-displacement revisit after n backward moves reads phase k_c n in channel c. If a usage share s of the channels has kappa <= |k_c| <= K, the same-cell position factor sum_c q_c cos(k_c n) is at most 1 - s (1 - cos(kappa n)) for n <= pi / K: the bound says how fast a clock erodes the position match while n |k| stays below pi, not that every clocked code fails at long gaps. (b) If the target is the key at the last visit to the same cell and returns come at lags with greatest common divisor g (g = 2 on the even torus), a channel can serve place-keyed retrieval only if its per-move *tick* is 0 mod 2 pi / g.

Proposition 3 (representability by wrapping; proved). If at most r tokens step, a |.| model of rank r reproduces any signed angle table exactly mod 2 pi; annihilating the observation embeddings in W_in leaves only the moves stepping, so every |.| model in this paper can represent the map. Softplus and exp can too: with the observations annihilated each steps ln 2 or 1, and omega' = 2 pi / ln 2 or 2 pi makes that step a full turn.

Lemma 1 (relative-sign gauge; proved). Negating a channel's steps (through omega_c) together with the second coordinate of that channel's query and key projections leaves every score unchanged, since J R(phi) J = R(-phi) for J = diag(1, -1). A non-negative step therefore fixes only the relative signs of the tokens that step in a channel. For an inverse pair it removes cancellation through zero, angles (a, -a), while the wrapped code (a, 2 pi - a), the same function, remains. Lemma 1 is a statement about representation; its learnability reading, that a |.| step should learn freely when no step need cross zero, was tested and not confirmed (Sec. 7.3).

Proposition 4 (route inventory; proved). (a) *Signed, collective route.* The clock vector is k = diag(omega) W u with u = z_x + z_y. The set {u = 0} cancels every channel at once and leaves the map free; gradient flow on a quadratic clock surrogate converges to it from every start, and for almost every readout it is the only latent set that cancels every channel. (b) |.|. The same latent move ends at a(y) = a(x), a clock. At a generic readout no latent choice cancels more than 2r channels non-blindly; per channel, an unwrapped |.| pair has |k_c| >= |m_c|, and its non-blind zero-clock points lie at |k_c| = 2 pi n, across |k_c| = pi from a small start. (c) *Rank.* Let c be the latent step common to all moves (with the mean observation step added when observations step) and U the map frame. At per-head rank D, c lies in the plane U spans, so a channel is clock-free only by losing map resolution; above rank D the part of c outside the map plane can be cancelled channel by channel with the map untouched (*parking*). (d) *One dimension.* At D = 1 only the collective route acts.

Further statements (App. E9-E11). *Proposition 5* (proved to first order): at rank D >= 2 a channel can lower the drift loss at no first-order cost to the map by turning its resolution off the drift direction (*hiding*), which slows the collective route; *Conjecture 1*: this keeps the drift in place at rank D >= 2 (in 1D there is nothing to hide along). That parking is why rank above D helps (the *clock-exit account*) is a conjecture with a PARTIAL registered test. *Remark 2*: "a solution exists and is kept, but is not found from scratch" means a non-empty, stable solution set whose basin the initialisation rarely hits. *Proposition 6* (proved): theta is computed once from the embeddings, so a loop or extra layers widen the reader and never change the code family; a score rule acts only on the readout term of target coherence, not on the code term C0.

Proposition 7 (peak pinning; proved given delta = 0). PoPE's score sets beta_c = -delta_c with delta content-free and, in trained models, in its clamp's dead zone (delta = 0). Then every head's kernel, for every content pair, peaks at dphi = 0: with a path phase the same place, the revisit target; with an index phase the current token, and with PoPE's frequencies (at most 1 rad per token) no fixed far offset can be preferred in one layer.

Table 1. Status of the theory (proofs in App. E).

statement	status	checked against
Prop. 1; Remark 1	proved; cited	long-gap same-cell attention; no count in torus index models (Sec. 6)
Prop. 2, bound and tick rule	proved	consistent with every failing model's strata (Fig. 2); text-world channels (Sec. 8)
Prop. 3, Lemma 1	proved	constructions exact to 1.4e-6 rad (App. D)
Lemma 1 read as learnability	registered test UNMEASURED	+1/+2: |.| 2/8, no better than on +1/-1 (Sec. 7.3)
Prop. 4, route inventory	proved	end-state clocks; frozen latents; hybrid readout; route class (Sec. 7.3)
At rank D the clock is the obstacle (torus)	registered causal support	clock forbidden 8/8 against 1/8 (Sec. 7.2)
Prop. 5; Conjecture 1; clock-exit account	first order; conjecture; registered PARTIAL	drift axis; clock exits 3/8 against 1/8 (Sec. 7.2)
Props. 6-8	proved (7 given delta = 0)	reader fits; delta-freedom batch; CARoPE models (Secs. 7.3-7.4)

5. One path layer solves the long-walk map

Short walks mostly do not need a map (analysis). On MapFormer's T = 128 walks retrace scores 0.934 and no revisit is wrap-only: a run that reverses the previous one returns, step by step, to cells seen moments earlier. At T = 1024 retrace falls to 0.843 and 7.9% of revisits are wrap-only (Table 2). At T = 128 the path model still beats index attention (position effect +0.243 at rank 2), but index RoPE falls below retrace and the rank-2 path model clears it by only 0.04.

Table 2. Torus floors (analysis, on the models' evaluation walks) and accuracies (held-out map; n = 8 per model).

T (moves)	wrap-only revisits	blank	n-gram	retrace	index RoPE, 1 layer	path, rank 2	path, rank 4
128	0.000	0.507	0.598	0.934	0.805	0.971	1.000
1024	0.079	0.506	0.575	0.843	0.731	0.894	0.998

At T = 1024 one path layer solves the map and index attention does not. One path layer scores 0.998 +/- 0.004 (8/8 SOLVED) against 0.731 +/- 0.008 for index RoPE (0/8), +0.267 (permutation and Fisher p 0.0002); index RoPE sits below retrace on every run (analysis). Two further registered batches give 1.000 against 0.722 (n = 8) and 1.000 against 0.719 (n = 4). By stratum (short gap / long gap / wrap-only) the path model scores 1.000 / 0.995 / 0.982 and index RoPE 0.784 / 0.462 / 0.471, against stratum floors of 0.506 / 0.500 / 0.507 and retrace 0.911 on short gaps (analysis; Fig. 2c). Long-gap and wrap-only revisits require the map, and only the path model has it.

6. What index depth buys

We ask how many index layers one path layer is worth, both trained and tested at the same length and nesting depth.

Dyck-2: about three. A bracket string is a walk on a stack. One path layer scores 0.996, index attention 0.648 / 0.889 / 0.964 / 0.982 at 1-4 layers, against a stack-free floor of 0.594 (Fig. 3a). The position main effect over two encodings (MapWM vs RoPE, MapPoPE vs PoPE; App. F) is +0.353 / +0.130 / +0.045 / +0.024 at 1-4 layers, positive on 8/8 seeds at each depth (sign-flip p 0.0078). With three times the budget four index layers reach the ceiling (0.997-0.998); training on depths 4-12 keeps +0.110 at depth 12.

A biased 1D walk: the same rate. On fresh seeds (n = 6) one path layer scores 1.000 at every bias, and index attention with 1 / 2 / 3 layers 0.710 / 0.949 / 0.998 at p_plus 0.5, 0.677 / 0.962 / 0.998 at 0.75 and 0.824 / 0.987 / 0.999 at 0.9 (Fig. 3b; declared secondary). Wherever a step can be undone three index layers come within 0.002 of the path layer; at p_plus 1.0 one index layer reaches 1.000.

The 2D torus: more than four. At T = 1024 (n = 4) one path layer scores 1.000 (4/4) against index attention at one layer 0.719, four layers 0.860 and one block looped four times 0.864 (each 0/4; complete separations, p 0.029; Fig. 3c). A registered replication on four fresh seeds gives every reading again (App. F7): 0.999 (4/4) against 0.729 / 0.830 / 0.859 (0/4 each, p 0.029); over both batches (n = 8, descriptive) 0.999 against 0.724 / 0.845 / 0.862. Four index layers score what retrace scores on the same walks (-0.000 over both batches; 0.913 on short gaps against retrace's 0.911) and fall to 0.41 on long gaps, below the constant 0.51: depth buys index attention about what a map-free rule gives.

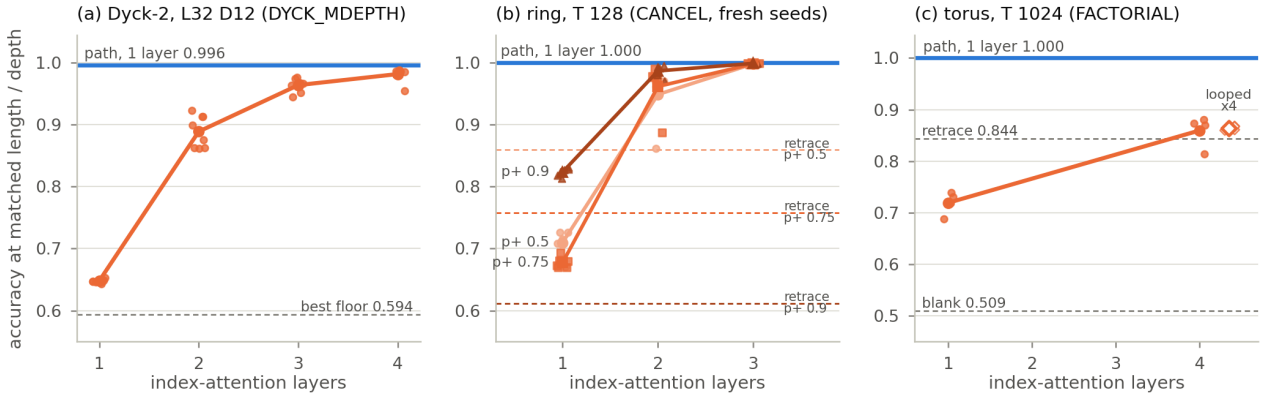


Figure 3 (`fig_exchange_rate`). Accuracy of index RoPE against its number of layers (orange; dots: runs) beside one path layer (blue), trained and tested at the same length and depth. (a) Dyck-2, length 32, depth 12 (n = 8). (b) Ring of 32 cells, three move biases (n = 6 fresh seeds). (c) Torus, T = 1024 (n = 4; fresh-seed replication in App. F7): four index layers (0.860) or a loop of four (0.864) stay just above this batch's retrace floor (0.844). Dashed: map-free floors (analysis).

Why the rate differs by task (analysis; App. F). Of Remark 1's options training finds only the count-free rules, with a reach that differs by task. One index layer learns retrace and the one-lap offset, so at p_plus 1.0, where every revisit is a one-lap return, it matches the path layer. Three index layers on the ring stay flat in the number of direction runs a revisit spans (Fig. 4b) and carry a displacement code over the last ~64 moves (probe R^2 0.72; older displacement R^2 0.01), which covers most revisits of a 128-move walk. On the torus no index model carries a count (R^2 at most 0.21); four layers or the loop solve retraces (0.996 / 1.000) and decay with the runs spanned, also at a fixed gap (Fig. 4a). Nearly all long-gap revisits span 17 or more runs (98.5%), beyond that reach; there the last layer puts 0.00-0.01 of its attention on the remembered cell, where the path layer puts 0.81. So the torus rate is "more than four" because its long-range revisits lie beyond a local reach and the global count is not found. Heo et al. (2026) show that the error of constant-depth parallelizable models falls exponentially with depth, an expressivity account; this is what training finds at each depth. On a rank-2 path code the same four passes lift long-gap revisits from 0.671 to 0.927-0.990 (Sec. 7.4): extra passes add accuracy where the code carries position information one pass cannot read.

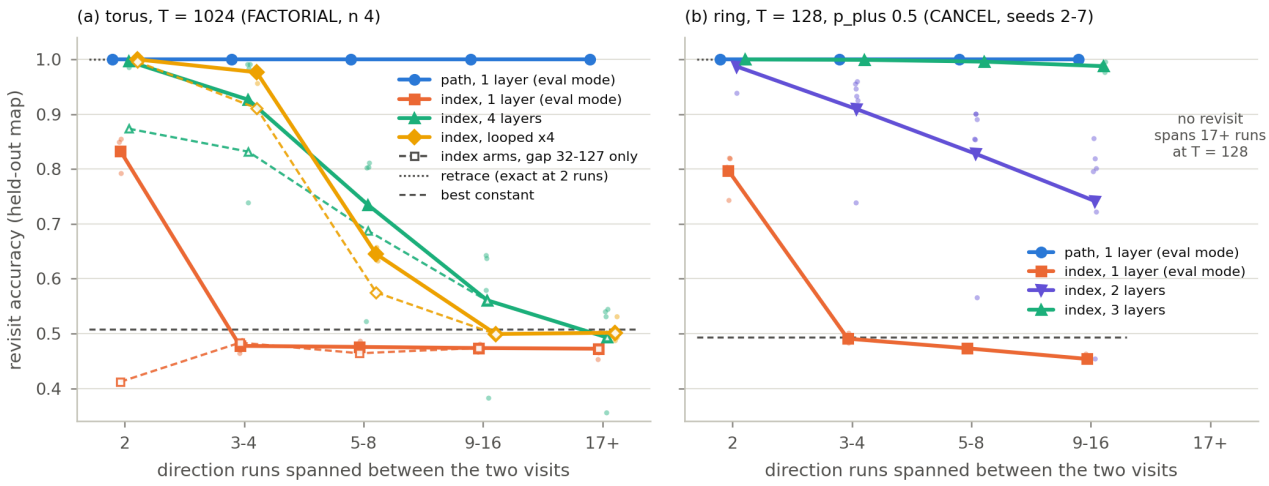


Figure 4 (`fig_segments`; analysis of registered batches, eval mode). Revisit accuracy against the direction runs spanned between the two visits (2 runs = retrace); lines: means of per-seed accuracies, dots: seeds. (a) Torus, T = 1024 (n = 4), plain revisits under 128 moves (solid) and at gaps of 32-127 moves only (dashed): four index layers fall from 0.996 at 2 runs to 0.493 at 17+, the loop from 1.000 to 0.502; one path layer is at 1.000 in every bin (table in App. F1). (b) Ring, p_plus 0.5 (n = 6), plain revisits: three index layers stay flat, two decay, one learns retrace only. Floors on each batch's own walks: retrace exact on 2-run revisits (dotted), the best constant elsewhere (dashed).

7. What decides whether training finds the map

Two choices of the content-to-angle map decide whether the torus map is learned: its per-head rank and how its step is parameterised. For each, the constrained class fails from scratch, while a constructed solution, installed and trained, is kept or re-found.

7.1 The clock and the signature

From a small start training ends in one of two places on either axis (analysis; table and figure in App. G3). Models that succeed end with clock-free heads: signed and rank-3/4 arms at clock / map 0.001-0.051 (arm means), every solved rank-2 run with at least one clean head. Models that fail end with a clock of the order of their map: the barrier-parameterised arms at 1.25-1.57 (for an unwrapped |.| channel at least 1 by construction, so the informative part is that the channels did not wrap), signed steps with frozen move latents at 1.18, and the unsolved rank-2 runs at 0.54-0.69; 4 of the 20 unsolved rank-2 runs instead end clock-free with their two map axes collapsed onto one direction. The strata show the signature Proposition 2 leads one to expect: rank 2 scores 0.950 / 0.671 / 0.557 on short-gap / long-gap / wrap-only revisits and |.| 0.870 / 0.630 / 0.515, against 1.000 / 0.995 / 0.982 for the signed rank-4 model (Fig. 2b-c). The bound alone does not predict failure: two signed torus runs started past pi solve with ratios 1.39 and 1.78, carrying their usage on channels whose per-pair clock is near pi, where the bound says nothing (App. H7).

7.2 Per-head rank of the content-to-angle map

Per-head rank decides it. On the torus at T = 1024 six bottlenecks built from the same initial weights at each seed (Table 3, Fig. 5a): every arm whose heads each read a rank-2 map solves 0-2 of 8; rank 3 solves 6/8 and rank 4 8/8, whether the latent is shared or per head. At matched sharing (per-head rank 4 against a block-diagonal rank-4 latent in which each head reads rank 2) the split is 8/8 against 2/8 (Fisher and permutation p 0.0070). Neither total latent size nor the initial angle scale tracks success. With MapFormer's Table 2: rank 1 cannot represent a 2D map; rank 2 can, and learns 128-step walks; at 1024 moves reliable learning needs per-head rank above the map's dimension.

Table 3. SOLVED (of 8) and accuracy at T = 1024 by per-head rank; torus, 900 epochs, same initial weights per seed.

per-head rank	arm	SOLVED	accuracy
2	one rank-2 latent shared by the heads	0/8	0.894
2	per-head rank-2 latents	2/8	0.885

per-head rank	arm	SOLVED	accuracy
2	block-diagonal rank 4 (each head reads rank 2)	2/8	0.948
3	per-head rank 3	6/8	0.987
4	shared rank 4	8/8	0.998
4	per-head rank 4	8/8	0.999

Rank 2 still fails at twice the budget and at three recipes. At 1800 epochs from scratch rank 2 solves 0/8 (0.908) against rank 4's 7/8 (Fisher p 0.0014; accuracy +0.086, p 0.0003). PoPE's polar score does not rescue it (3/12 under either score; Sec. 7.4). On fresh seeds rank 2 was trained at the default, at three times the learning rate and at four times the weight decay (App. G9): it solves 0/8 at each (0/24), while rank 4 solves 8/8 at the default; rank 2 at its best recipe (3e-3, selected on training loss) scores 0.910 against 1.000 (decision p 0.0002 on SOLVED and accuracy). The selection cannot hurt rank 2, but the search is limited: neither alternative is a recipe at which rank 4 is known to learn (at 3e-3 the related signed rank-4 observation-step-free model solves 0/8 on this torus within 900 epochs, App. H12; the higher decay shrinks the angles most and leaves the most rank-2 runs stalled). The result reads: rank 2 fails at each of three recipes. Within 900 epochs rank 4, in turn, learned only with weight decay (8/8 against 1/8 without; App. G8).

The rank-2 solution exists and is held. The rank-2 projection of each solved rank-4 model scores 0.9955 frozen. Trained from there it solves 7/8, as many as the rank-4 control from its own solution (7/8), against 1/8 for rank 2 trained further from its own stalled weights (accuracy p 0.0009; Fig. 6a). A rank-2 map that solves the torus exists, is kept by training, and is not reached from scratch.

Rank against the map's dimension. Rank D is hard in 2D and 3D and not in 1D (Fig. 5b): on a 2D torus of 32 cells per side rank 2 solves 1/8 against rank 3's 8/8 (Fisher p 0.0014); on a 3D torus of 10 per side rank 3 solves 1/8 (rank 4, 4/8), and rank 4 solves an 18-per-side 3D torus 8/8; on the ring rank 1 solves 7/8 against rank 2's 8/8. Larson (2026) finds that gradient descent recruits the minimal faithful dimension of a group-composition task, with a fixed decoder and dense state supervision. Here too the rank-D solution exists and is held, so the two agree on capacity; what differs is search: with a sparse revisit loss and a shared readout, training reached the map reliably only with one dimension more.

On the torus the rank-2 obstacle is the clock (registered). A batch on fresh seeds (shared rank, seven arms from the same initial weights; App. G7) replicates the split (rank 2 1/8, rank 4 8/8). In the *drift-forbidden* rank-2 arm the steps of each inverse pair are tied to exact negatives and every observation step is 0, so the clock is zero by construction: it solves 8/8 at accuracy 1.000, like rank 4. In the *actions-only* rank-2 arm only the observation steps are 0 and the moves' own clock remains: it solves 1/8 (Fisher p 0.0014 against 8/8). Removing the clock is sufficient, removing its observation part is not: what stops rank 2 is the clock, not capacity.

Why the extra rank helps is open. The same batch's registered primary gives rank 2 an explicit clock exit without extra map rank (a learned per-channel slot per move axis): it solves 3/8 against rank 2's 1/8 (Fisher p 0.57), as does a per-token slot, registered PARTIAL with 4 of the 5 unsolved per-axis runs still descending. The checkpoints are consistent with parking but do not discriminate it from removal through the latent

(analysis; App. G4): in solved heads above rank D the residual clock lies almost entirely off the map plane, and drifting rank-D heads drift along their least-resolved axis (50 of 55 runs, against 25.1 at random), the geometry Proposition 5 leads to.

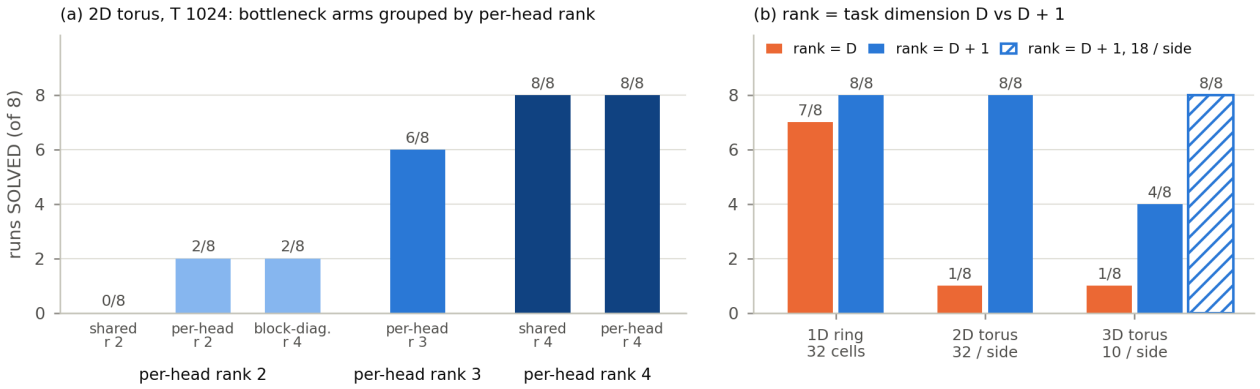


Figure 5 (`fig_rank_solved`). Runs (of 8) that solve the map, by per-head rank. (a) Torus, T = 1024, six bottlenecks from the same initial weights: rank-2 arms 0-2/8, rank 3 6/8, rank 4 8/8. (b) Rank equal to the task's dimension D against D + 1 (separate batches and recipes): ring, rank 1 7/8 (rank 2, 8/8); 2D torus, rank 2 1/8 (rank 3, 8/8); small 3D torus, rank 3 1/8 and rank 4 4/8; rank 4 solves an 18-per-side 3D torus 8/8 (hatched). Torus budget 900 epochs; at 1800 epochs rank 2 still solves 0/8.

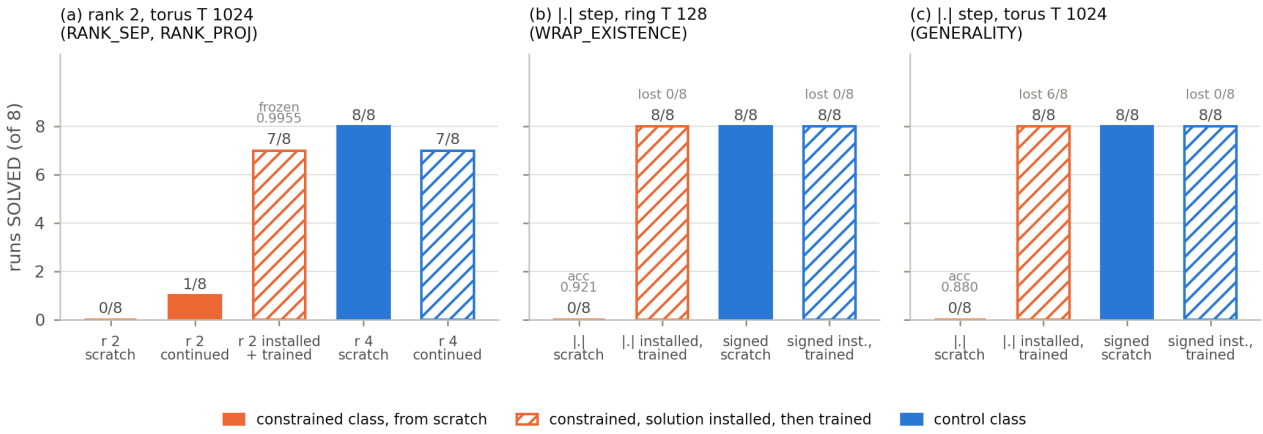


Figure 6 (`fig_existence`). A solution exists in each constrained class and training keeps or re-finds it, but training from scratch does not find it (bars: runs SOLVED of 8; hatched: installed and then trained). (a) Rank 2 on the torus: the rank-2 projection of a solved rank-4 model is held (7/8); from scratch 0/8, stalled runs trained further 1/8. (b) |.| on the ring: the installed exact solution is held (8/8, lost on 0/8); from scratch 0/8 against signed 8/8. (c) |.| on the torus: the installed solution is perturbed (loss above 0.05 on 6/8) and re-found (8/8); from scratch 0/8 against 8/8.

7.3 How the step is parameterised, and where its angles start

Barrier parameterisations rarely learn the map from a small start. On the torus at T = 1024 the |.| step scores 0.821 against 0.998 for the signed step (-0.177, permutation p 0.0002) and solves 0/8 against 8/8; softplus scores 0.781 and solves 1/8 (-0.218, p 0.0003). On the ring |.| solves 2/8 and 1/8 against 8/8 at p_plus 0.5 and 0.75 (Fisher p 0.007 and 0.0014), and observation-step-free models give 0/8 against 8/8 on the ring and

on the torus (0.880 against 1.000). The signed median crosses the SOLVED threshold at epochs 57-480 in four batches, while most |.| runs flatten at a loss of 0.3-0.5 (Fig. 7).

The cost holds within 900 epochs at each step's best learning rate, without weight decay and at 2 layers d_model 256 (Table 4). With each step at its own best of three rates on the torus (App. H12) and of four on the ring (App. H1), |.| solves 0/8 and 2/8 against signed 8/8. Higher rates hurt the signed step first (on the torus it solves 8, 0 and 0 of 8 at 1e-3, 3e-3 and 1e-2, every run at the higher rates still descending) and do not help |.|.

Table 4. The cost of the |.| step by task, model and recipe (observation-step-free unless marked "full"; n = 8 per arm).

task and model	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	test
torus, 1 layer, full	default (lr 1e-3, weight decay 0.05)	0.821, 0/8	0.998, 8/8	perm p 0.0002; Fisher 0.0002
torus, 1 layer	default	0.880, 0/8	1.000, 8/8	perm p 0.0002; Fisher 0.0002
torus, 1 layer	each at its best of three rates (|.| 3e-3, signed 1e-3)	0.890, 0/8	1.000, 8/8	selection-aware p 0.0002 (SOLVED), 0.0095 (accuracy)
torus, 2 layers, d_model 256	default	0.967, 2/8	0.999, 8/8	perm p 0.0005; Fisher 0.0070
ring	default	0.911, 1/8	1.000, 8/8	perm p 0.0002; Fisher 0.0014
ring	weight decay 0	0.873, 0/8	1.000, 8/8	perm p 0.0002; Fisher 0.0002
ring	lr / 3 (3.3e-4)	0.841, 0/8	0.950, 6/8	Fisher 0.0070
ring	each at its best of four rates (|.| 3e-3, signed 1e-3)	0.963, 2/8	1.000, 8/8	selection-aware p 0.0070 (SOLVED), 0.0057 (accuracy)
ring, 1-layer diagonal complex SSM	lr 3e-3	0.821, 0/8	1.000, 8/8	perm p 0.0002; Fisher 0.0002
ring, 1-layer diagonal complex SSM	lr 1e-3	0.756, 0/8	0.994, 7/8	perm p 0.0002; Fisher 0.0014

The |.| solution exists, and training keeps or re-finds it. Proposition 3 constructs it from each solved signed observation-step-free model (angles unchanged mod 2 pi to 1.4e-6 rad, frozen accuracy equal to the source's on 8/8 seeds, weights within about 5x of the signed model's; App. D). On the ring the constructed model, trained 300 more epochs, stays solved on 8/8 seeds, keeping its cancelling code in the channels attention uses (98%, usage-weighted); on the torus it is perturbed early (peak loss 0.11-0.21) and re-found, ending solved on 8/8 seeds. From scratch the class solves 0/8 on both tasks (Fig. 6b-c).

Where the angles start matters too (ring, registered). Varying only the initial move angles, every other weight identical at a seed (Table 5, Fig. 8): from the |.| start, where both moves rotate the same way, signed still solves 7/8 and |.| 0/8 (Fisher p 0.0014); a |.| step started with opposite-move angle sums past pi solves 8/8 (against 0/8 from the small start, p 0.0002), while a large start whose sums stay below pi solves 2/8, so size alone does not rescue it. From past pi both classes cancel by wrapping; from small starts signed models cancel through zero (98-99% of channels end opposite-signed; analysis). On the ring the variable is the barrier parameterisation together with where the angle sums start relative to pi. On the torus the start lever does not transfer: a past-pi start leaves |.| at 0/8 and slows the signed step to 2/8, with 5/8 runs per large-start arm still descending (App. H3); exp's built-in past-pi start does not help there either (below).

Table 5. SOLVED (of 8) by start, move set and model (observation-step-free, rank 4; registered; App. H3, H4, H13, H14).

task, model	moves	initial move angles	signed	|.|	other
ring, attention	+1/-1	default (small, signs independent)	8/8	0/8	
ring, attention	+1/-1	the |.| start (both moves the same way)	7/8	0/8	
ring, attention	+1/-1	large, sums below pi	--	2/8	
ring, attention	+1/-1	large, sums past pi	8/8	**8/8**	
torus, attention	four moves	default	8/8	1/8	
torus, attention	four moves	large, sums past pi	2/8	0/8	
ring, attention	+1/+2 (no inverse pair)	default	5/8	2/8	UNMEASURED
ring, SSM, lr 1e-3	+1/-1	default (exp: 13 of 32 channels per head past pi)	7/8	0/8	softplus 0/8, exp **6/8**
torus, attention, best of lr 1e-3 / 3e-3	four moves	default (exp: 11 of 32 channels per head past pi)	8/8	0/8	softplus 0/8, exp 0/8

The signed escape is collective, and lives in the increment. From the |.| start, freezing the two move latents (W_out and omega still free) blocks the signed escape: 1/8 against 7/8 (Fisher p 0.010), ending with a |.|-like clock (1.18). With the latents free, one move's latent turns toward the negative of the other's and every channel cancels together, the antipodal end state MapFormer described (their Figs. 3d, 9b). A hybrid with the readout confined to the non-negative orthant, a_c(x) = omega_c |w_c| . z_x, but a signed increment solves 8/8 at both reference rates, against |.| 0/8 and 2/8 (Fisher p 0.0002 and 0.007; App. H8): a non-negative readout costs nothing; the barrier on the increment is what costs.

Losing the through-zero route does not account for the difficulty (registered). By Lemma 1 a |.| step represents, without wrapping or crossing zero, a map whose moves step the same way. With moves +1/+2 (Table 5; App. H4), which need integration (index RoPE 0.649, 0/8) and which signed models learn with a sign-coherent code (0.92 of channels end with both move angles of one sign, against 0.01 on +1/-1), |.| still solves only 2/8, no more than on +1/-1 (1/8 in the same batch; Fisher p 1.00), against signed 5/8. The registered prediction, that the cost disappears without an inverse pair, is not

confirmed; the comparison is unmeasured because +1/+2, where every revisit is a lap, is harder for signed steps too. Lemma 1 is about representation; what makes the barrier classes hard to train from a small start is not settled by it.

Barrier or route? (registered, half-read). A 2x2 on the ring from the |.| start (App. H11) gives |.| one learnable offset per move, a collective route to a point where every channel cancels by wrapping, with the barrier kept: it solves 0/8 against signed 8/8 (Fisher p 0.0002), while the same class started past pi solves 7/8 and ends on that point. The registered half-reading is that a route to the wrap does not carry |.| past pi from a small start; the cell without a barrier and without a route (signed, latents frozen) solves 4/8, so the 2x2 as registered is unmeasured.

The cost is not specific to softmax attention; exp escapes only in the SSM. Replacing softmax by the causal unnormalised kernel gives a one-layer diagonal complex SSM without decay or gate, with transition exp(-i omega Delta(x_t)). On the ring |.| solves 0/8 against signed 8/8 at learning rate 3e-3 and 0/8 against 7/8 at 1e-3, and softplus 1/8 and 0/8 (Table 4; App. H10, H13); the constructions are exact and the installed |.| solution is re-found. The exp step diverged at 3e-3 but shows no cost at 1e-3: 6/8 against 7/8 (Fisher p 1.0), at a rate where the signed anchor reached exact codes on only four seeds. It ends with every step positive, and 90 of its 95 cancelling channels are channels its initialisation, exp(0) = 1, already put past pi (analysis); softplus, which starts 9 of 32 channels per head past pi, does not solve. In attention exp pays. On the ring exp and softplus solve 1/8 to 3/8 against signed 8/8 at both reference rates (App. H9). On the T = 1024 torus (registered, App. H14), each at its best of two rates, exp solves 0/8 (0.916) and softplus 0/8 (0.908) against signed 8/8 (0.993; Holm p 0.0003 on SOLVED; budget-scoped), its cancelling channels again mostly past-pi-start ones cancelling by a wrap (95 of 107), without a map. So the escape is not exp's alone; between the two batches mixer, task and start (past pi on 41% of channels in the SSM, 34% on the torus) all change, and which matters is not separated. The cost is shown for |s|, softplus(s) and, in attention, exp(s).

Published content phases also miss the torus map. *Proposition 8 (proved):* faithful CARoPE advances every channel by (0, 1] per token, so the shortest return's phase lies in (0, 4], never 0 mod 2 pi (App. E5); it scores 0.667 on the torus (0/8), below index RoPE's 0.722 (p 0.040). Faithful CoPE can represent ring cancellation (App. I) yet solves 0/8 on the ring (0.887 against 1.000, p 0.0002). Faithful Selective RoPE, signed, builds a phase it relies on (+0.236 over index RoPE) but solves 0/8 against MapWM's 8/8 within 900 epochs, every run still descending; an observation steps its phase 0.86-0.96 as far as a move (MapWM: 0.027). Its state-tracking configuration keeps every step non-negative and fails like the barrier steps (0.769, 0/8).

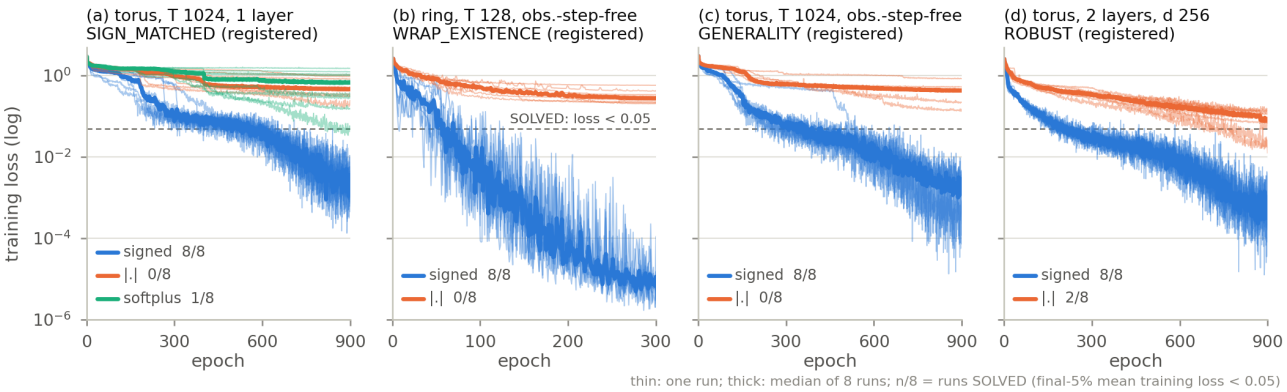


Figure 7 (`fig_loss_curves`). Training loss per epoch, signed (blue), |.| (orange) and softplus (green), every run of four batches (thick: median of 8; log scale). Dashed: the SOLVED threshold; n/8 = runs SOLVED. (a) Torus, T = 1024: signed 8/8, |.| 0/8, softplus 1/8. (b) Ring, observation-step-free: 8/8 against 0/8. (c) Torus, observation-step-free: 8/8 against 0/8. (d) As (c) at two layers and d_model 256: 8/8 against 2/8.

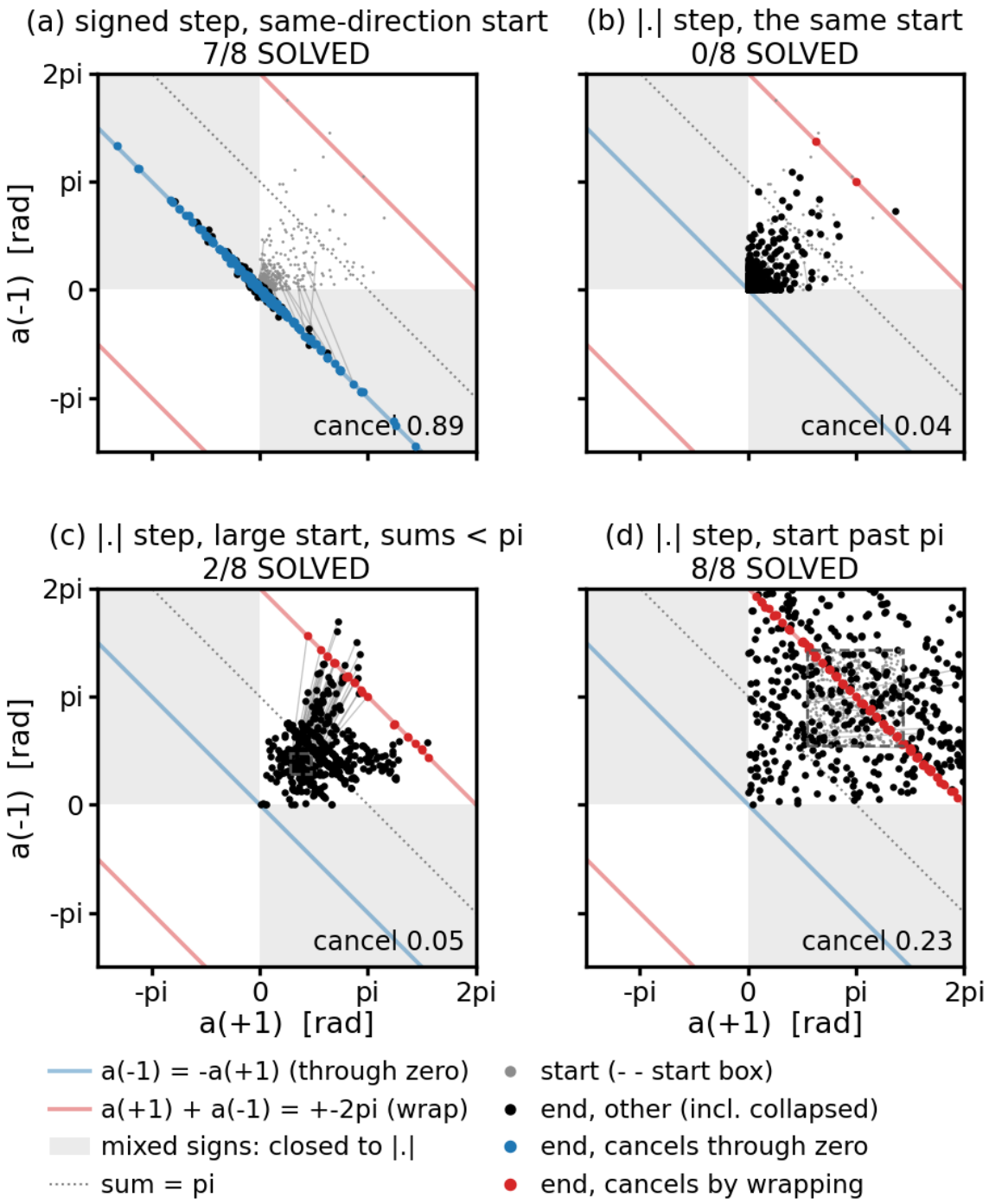


Figure 8 (`fig_mechanism`; ring, 8 runs per panel). Move angles (a(+1), a(-1)) of every channel at the start (grey) and after training (dots). A channel cancels opposite moves on the blue line (through zero, reached only with opposite signs) or the red line (by wrapping); grey shading: the mixed-sign quadrants the |.| step cannot reach; dotted: sums of pi. (a) Signed from the same-sign start: 7/8 solve, one move's angle crosses zero. (b) |.| from the same start: 0/8. (c) |.| from a large start below pi: 2/8, by wrapping. (d) |.| from a start past pi: 8/8, by wrapping (App. H).

7.4 Readers cannot repair a code

More computation reads a rank-2 code; it does not repair it. Looping the single block four times (identical parameters and initialisation) or using four real layers raises

rank-2 accuracy from 0.894 to 0.973 and 0.990 (permutation p 0.0034 and 0.0012; SOLVED 2/8 and 5/8 against 0/8; rank 4 0.998, 8/8). By Proposition 6 neither alters the code family. A one-pass, position-only reader fitted to each model's own phase recovers the one-layer models' accuracy but falls 0.32-0.44 short of the looped and deep models on long-gap and wrap-only revisits, and their codes stay far from rank 4's (p 0.0002); only 1 of the 16 runs finds a two-dimensional code (analysis; App. M). Downstream computation compensates for the rank-2 deficit and leaves it in place.

Where the angle comes from and how content uses it are separate choices. Path integration changes dphi; PoPE's polar score removes beta and, by Proposition 7, pins every kernel's peak at zero phase difference. On the short-walk torus PoPE's score raises the path model (MapPoPE 0.999 against MapWM 0.971) and lowers the index model (PoPE 0.679 against RoPE 0.805), an interaction of +0.154 (8/8); for the index model no kernel then prefers offset 3 over offset 1 (RoPE's is a comb at the retrace offsets; Fig. 9a; analysis). Freeing PoPE's phase offset (registered) raises the index model by +0.122 (8/8, sign-flip p 0.0078), most on multi-step reversals, and leaves the path model at ceiling (-0.018, p 0.31). At rank 2 on the long-walk torus the polar score repairs short-gap revisits through the readout term (target coherence 0.617 to 0.960) and leaves the code term low on long-gap (0.80 / 0.91 under the two scores) and wrap-only revisits (0.40 / 0.48, against 0.97 at rank 4; analysis; Fig. 9c): under either score rank 2 solves 3/12 (registered).

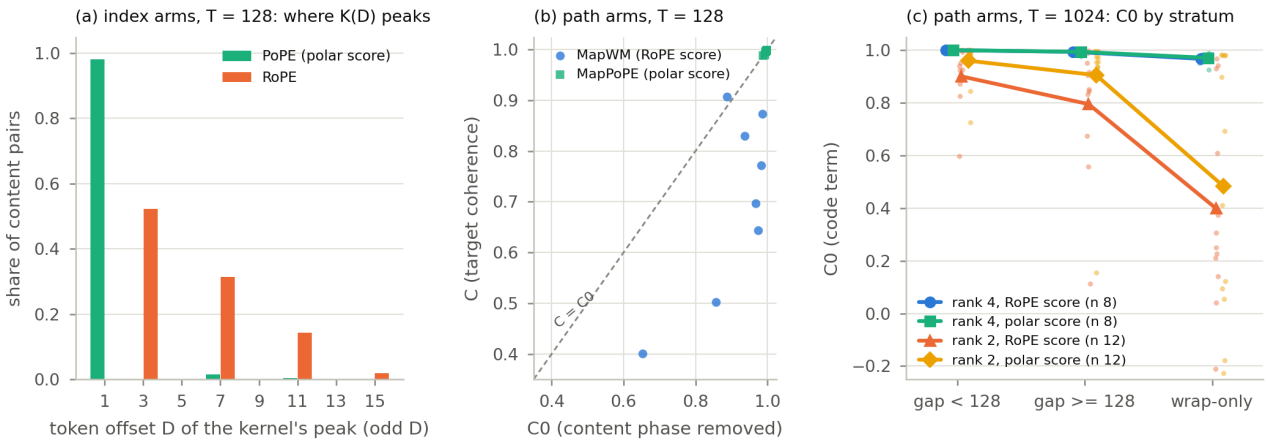


Figure 9 (`fig_kernel_peak`; analysis of registered batches). (a) Index arms, torus T = 128: the token offset at which each content pair's kernel peaks. PoPE peaks at offset 1 in 98.2% of pairs; RoPE's is a comb on the retrace offsets 3 / 7 / 11 / 15. (b) Path arms at T = 128: target coherence C against C0 (content phase removed), per seed: MapWM 0.703 against 0.905, MapPoPE 0.995 against 0.996. (c) C0 by stratum, torus T = 1024: rank 2 0.90 / 0.96 on short gaps, 0.80 / 0.91 on long gaps, 0.40 / 0.48 on wrap-only revisits (RoPE / polar score); rank 4 0.97-1.00 in every stratum.

8. Maps from words

Each move of the torus walk is written as a sentence: a direction word chosen from three synonyms, verbs that do not move the walker, optional adverbs and fillers, and, after some moves, an aside that names an object without visiting it ("she thought about a cat ."). Nothing in the input says which words move the walker (App. J). A walk of 1024

words is about 132 moves, so retrace scores 0.934 here (analysis), beside the constant (0.505).

One path layer learns navigation told in English. On fresh seeds (n = 6) one path layer scores 0.979 +/- 0.048 (6/6 SOLVED) against 0.505 for one-layer index RoPE (0/6), +0.474 (permutation and Fisher p 0.0022); two index layers reach 0.793 (0/6), below retrace on every seed. The decisive revisits are the 13.4% that retrace cannot reach (Table 6; analysis, all 200 evaluation walks): there the path layer scores 0.956 over all eight seeds (0.978 on the fresh seeds; re-scored with the attention-dropout scale, App. C9), while two index layers score 0.411 and one 0.454, both below the constant (0.506). On the 6.6% whose cell holds an object, the path layer scores 0.925 and two index layers 0.129, against a best constant of 0.088.

Table 6. Text world: accuracy by whether retrace can reach a revisit (analysis; 200 registered evaluation walks, 6,381 revisits; mean over 8 seeds, fresh seeds in brackets; path re-scored with the dropout scale unless marked).

model (n = 8)	all revisits	retrace applies (n 5,525)	retrace cannot reach (n 856)	... cell holds an object (n 423)
path, 1 layer	0.993	0.999	0.956 (0.978)	0.925 (0.967)
path, 1 layer, eval mode	0.969	0.975	0.934 (0.958)	0.884 (0.927)
index RoPE, 2 layers	0.772	0.828	0.411	0.129
index RoPE, 1 layer	0.505	0.513	0.454	0.004
best constant	0.505	--	0.506	0.088

What the step table holds: a word's offset depends on its role, not its identity. On every seed opposite direction words cancel once the component common to the four directions is removed (0.002-0.012), synonyms collapse to one step (cosine 1.000), and zeroing the direction steps drops accuracy to the constant floor (declared secondaries; Fig. 10). Adverbs and fillers get steps of 0.004-0.03 of a move; asides get word steps that cancel within the sentence, leaving the aside's noun 0.04-0.16 of a move off the place (analysis). That offset is why learned steps beat being told: a model whose steps are restricted to the direction words stops at 0.970-0.974, its errors naming objects told in an aside at that place, and offsetting its aside words by half a move lifts it to 0.999-1.000 (analysis). "Only actions move" is the wrong separation in language: what is mentioned must not be filed where it is not. The same rule covers the what-to-where leak with new objects in every episode: removing the objects' steps, or normalising the step input, lets every run converge (16/16 SOLVED against 0/8 for MapWM; App. K3).

Half the models also count moves, in channels they do not use. On four of eight registered seeds 31-38 of 64 channels drift between visits to the same cell (declared secondary), a per-move tick (Howard's Case II) beside the map. By Proposition 2(b) such channels cannot serve retrieval by place, and they do not: 79-99% of retrieval usage sits in channels whose tick is zero modulo pi, and removing the clock channels costs at most 0.009 on each clock model at ceiling, while removing as many map channels collapses accuracy (analysis, 24 models; App. J2-J3).

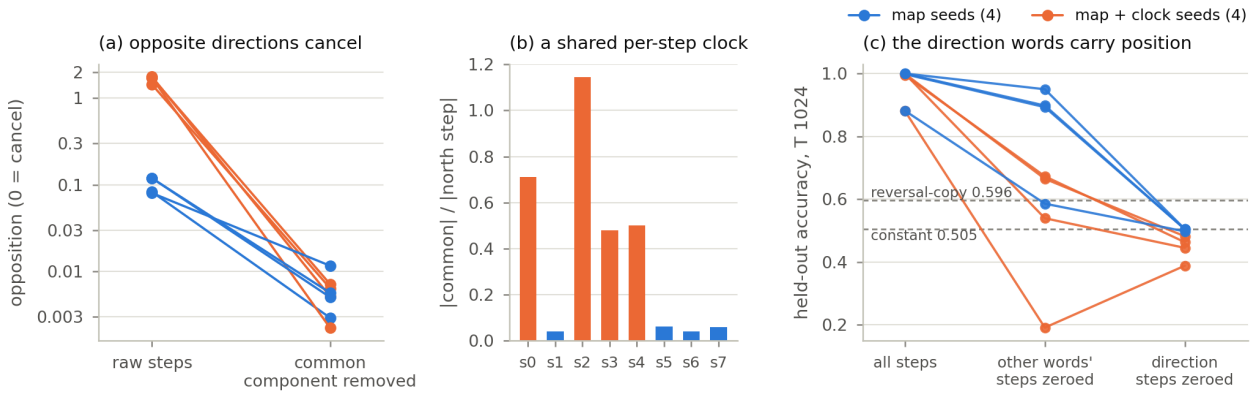


Figure 10 (`fig_textworld`). A one-layer path model on the torus walk told in English (n = 8). (a) Opposition of the direction-word steps, |A + B| / |A| (0 = cancel): after removing the common component, 0.002-0.012 on every seed. (b) The common component is small on four seeds (blue) and large on four (orange), where it points along the verbs' step: a per-move clock beside the map. (c) Accuracy with every step, with non-direction steps zeroed, and with direction steps zeroed: the last drops every seed to the constant floor (dashed). Eval mode.

Pixel observations

In the image torus (registered; App. K4) every observation of the 1024-move walk is a 28 x 28 image of its class (MNIST digits and six Fashion-MNIST classes), a fresh instance at every visit (test-split images at evaluation), read by a small CNN trained end to end. One path layer whose step the actions set scores 0.995 (8/8 solved, every seed at or above the perception ceiling of 0.993, a perfect map voting with a standalone classifier) against 0.701 for index RoPE (0/8, below retrace), +0.294 (Holm p 0.0003), and holds long-gap and wrap-only revisits at 0.994. Its phase never sees a pixel: what this shows is that a content-addressed CNN with an action-driven phase reaches the perception ceiling, and that the revisit loss alone trains the encoder (linear probe 0.985 against the standalone classifier's 0.980; index RoPE's encoder 0.965). The batch's registered primary compared the default MapWM, whose step reads every token, with index RoPE and reads unmeasured (0.668 against 0.701); after a pilot stop rule fired the path arm was changed to the action-driven step (Amendment 2; the headline is for that arm, partly pilot-informed). Why MapWM fails here is a design point (Sec. 9).

Table 7. Torus from pixels (registered; T = 1024, test-split images, held-out map, n = 8 per cell; App. K4).

model (1 layer)	observations	step set by	accuracy	solved	long gap / wrap-only
path	images	actions	**0.995**	8/8	0.994 / 0.994
path (MapWM), image embedding at gain 1	images	every token	0.994	8/8	0.993 / 0.991
path (MapWM), image embedding at token scale	images	every token	0.668	1/8 (5/8 collapsed)	0.589 / 0.544

model (1 layer)	observations	step set by	accuracy	solved	long gap / wrap-only
index RoPE	images	--	0.701	0/8	0.482 / 0.483
path	tokens	actions	1.000	8/8	1.000 / 0.997
floors			retrace 0.843, blank 0.508; ceiling 0.993		retrace 0.509 / 0.508

9. Discussion and related work

What the two failure axes share. For per-head rank 2 and for |.| steps a solution in the class exists, training keeps or re-finds it, and training from scratch does not reach it: in Remark 2's terms the initialisation sits outside the basin of a non-empty, stable solution set. Both failures end with a clock of the order of the map and strata that keep short-gap revisits and lose the rest, and four times the computation or parameters reads that code better without finding the rank-2 solution: the deficit is local to the content-to-angle map. For rank, the registered causal test names the obstacle: forbid the clock and rank 2 solves. For the step, the evidence points to the increment and its start: a signed increment under a non-negative readout learns, a |.| step started past pi learns on the ring, and the exp step learns in the SSM on the ring from a start with many channels past pi but not in attention on the torus; losing the through-zero route does not by itself account for the cost (+1/+2), and whether the barrier at pi or the missing route binds from a small start is half-read. MapEM's recency deficit is a contrast: its exact solution is found with four times the budget (0.578 to 0.928; not registered), while rank 2 stays at 0/8 with twice the budget (App. N).

Design guidance. Keep the phase increment signed; a non-negative readout is harmless. If a non-negative increment is needed, |.| and softplus from small starts are the forms that paid everywhere we looked; exp escaped in an SSM at one rate and paid in attention on the ring and the torus. Give each head rank above the map's dimension (D + 1), with weight decay; where the inverse pairs of actions are known, tying their steps to exact negatives removes the clock, and rank D then suffices. Keep observations from stepping the phase. With pixels this is decisive (Table 7): when the image embedding enters the step at token scale, 5 of 8 runs collapse, the encoder outputting a constant and the phase becoming a clock that ticks at every observation slot; this is not the token-level leak (no per-instance offsets) but a collapse of encoder and phase together. Scaling the image embedding down 30x, a total effect on the step input and the residual stream at once, restores learning (8/8); since the action-driven model is fine at token scale, the cost belongs to the step. Measure map use where retracing no longer suffices, by stratum, and read a phase's used-channel clock ratio, not its growth with the token index.

What and where. In the long-walk regime the model that learns the map separates where from what (analysis; App. K1): for the signed rank-4 model one shared position kernel per head times a per-pair content gain reproduces its accuracy (0.998), and removing token identity within type costs 0.032 (Holm over 24 contrasts 0.08), while the non-negative and index models lose 0.25-0.31. MapFormer showed the separation in the

embeddings (their Fig. 9); the models rely on it where a map is needed, in the sense TEM proposed.

The integrator view and text. A learned, signed rate can realise all three of Howard et al.'s cases, and which one a channel realises is set by what the readout keys on: for targets keyed to a place the used channels are clock-free (clock part over content part 0.064-0.079 on the torus, 0.06-0.23 in the text world; analysis), for k-back recency they are clocks (21.8-22.1). On ordinary text at small scale (n = 3 pilots) the used channels are clock-dominated (9.9-12.3 on TinyStories), MapWM matches index RoPE on TinyStories (1.5053 against 1.5050 nats per token), and at matched length path integration costs on code (+0.0056 bits per byte, 3/3 seeds). With egocentric turn-and-forward actions, recoding them allocentrically restores path integration's advantage (+0.050 to +0.488, n = 8, short walks; not registered; App. A5).

Related work. *Optimisation, not expressivity.* Shakerinava et al. (2026) find that input-dependent rotations in unit-modulus SSMs can represent non-Abelian state tracking but do not learn it from scratch. Howe (2026), also pre-registered, installs verified-exact solutions in Householder linear RNNs and finds that an additive input pathway pulls Adam off them, and that deleting it lets training find the exact automaton, as forbidding the clock lets rank 2 find the torus map here. Larson (2026) gives a rank law for matrix memories on group composition (Sec. 7.2). Grazzi et al. (2025) and Mamba-3 own the representability of rotations with non-negative angles; ours is the learnability cost of |.| and softplus steps in the rotary slot. A reparameterised weight w = |u| cannot change sign under gradient flow (Amid and Warmuth 2020, Prop. 2; Woodworth et al. 2020); scoRNN fixes the +/-1 diagonal of its Cayley transform (Helfrich et al. 2018); the LRU initialises its phases near zero (Orvieto et al. 2023). With fixed frequencies the parameter symmetries of rotary attention are per-plane complex scalars (Wang and Wang 2025; Tran et al. 2026); a learnable step adds the reflection of Lemma 1. *Depth and recurrence.* Depth can substitute for recurrence through shortcuts (Liu et al. 2023; Yao et al. 2021), error falls exponentially with depth in parallelizable models (Heo et al. 2026), shallow transformers learn sensitive functions unreliably (Hahn and Rofin 2024), and Selective RoPE and PaTH (Yang et al. 2025) solve parity with one layer; we give the rate at matched depth, the mechanism trained models use, and where it breaks.

10. Limitations

- The models are small (1-4 layers, 2 heads, d_model 128-256) and the tasks synthetic; natural text appears only as a three-seed pilot. Actions are allocentric translations. There is no LSTM or GRU baseline.
- "Not found" is relative to the stated budgets (300-1800 epochs), and many failing cells have runs still descending. Learning rates for the |.| contrast were swept on the ring (four rates) and the torus (three, decay coupled to the rate, no decay-matched control; 5 of 8 |.| runs at its best rate still descending). At 2 layers and d_model 256 the accuracy gap is a quarter of the one-layer gap, so at scale part of the cost may be speed. No other optimiser was tried.
- The parameterisation result is shown for |s| and softplus(s), and for exp(s) in attention (ring at two fixed rates; torus at the best of two, softplus's optimum not bracketed, most runs still descending, the exp batch partly pilot-informed). The exp step showed no cost in the SSM on the ring at one rate (1e-3; unstable at 3e-3),

where the signed anchor had partial codes on four seeds; whether the mixer, the task or the start explains this is not separated (no SSM on the torus, no exp started below pi). The start lever is registered on the ring and does not transfer to the torus within 900 epochs.

- Why the barrier classes fail from a small start is not settled. The registered test of Lemma 1's learnability reading (+1/+2) is unmeasured and did not confirm it; the route-versus-barrier 2x2 (ring, one start, one readable rate) has one registered half (a route to the wrap does not carry |.| past pi) and is unmeasured as a whole.
- Rank results use two heads and a latent shared across heads where MapFormer's is per head (per-head arms are reported). Within 900 epochs rank 4 learned only with weight decay 0.05 on all parameters (8/8 against 1/8 without; 5 of 7 failures still descending), credited to the optimiser setting. Rank 2 failed at three recipes, but neither alternative is a recipe at which rank 4 is known to learn, there is no lower-decay or combined arm, and 12 of 24 rank-2 runs were still descending; the claim is that rank 2 fails at each of three recipes, not that no recipe rescues it. The clock-exit test is PARTIAL with most unsolved runs descending.
- The torus exchange rate is a lower bound at n = 4, replicated on four fresh seeds, with 1 and 4 layers only.
- The pixel test uses class images on the discrete torus, one top-down image per visit, one layer, one length and one recipe. Its registered primary (default MapWM against index RoPE) was unmeasured; the reported path arm replaced it after a pilot stop rule, and the cost of an image-driven step was predicted from pilots and confirmed on fresh seeds, not blind; the gain effect is a total effect at two values.
- The text world is scripted (58 words, context-free direction words), and its walks are short enough that retrace scores 0.934; the map evidence there is the revisits retrace cannot reach and the step table.
- The theory of Sec. 4 was developed after the registered batches, except the clock-route batch, registered to test its rank account; the hiding feedback is a conjecture and the per-channel dynamics of |.| are not derived. Checks marked (analysis) are post hoc, on one batch per arm, with thresholds listed in App. O; one CPU analysis (the step-map basin) is not deterministic and is reported as exploratory.

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Appendix to "A map is a phase without a clock: what path-integrated attention learns, and what decides whether training finds it"

Draft 2 (2026-10-21; revised 2026-10-26: main-text Table 4 and Fig. 5 moved to G3 as Table G3a and Fig. G2, the revisit-class table is F1's, H4 and G9 scoped per the 2026-10-26 critical review; H14 added with EXPTORUS), for `docs/paper/MAIN_DRAFT2.md`. Every number is copied from the file named beside it; a CORRECTED or AUDIT block at the top of a results file supersedes its body. "Registered" = a pre-registered decision or declared secondary; "(analysis)" = computed on stored streams or checkpoints, eval-only, never changing a verdict; its script is listed in App. O. App. E states and proves every theory statement the main text uses; statements not used are in `docs/paper/APPENDIX_theory_v3.md`. Delete this paragraph before submission.

Contents. A: models, recipes, batches. B: floors. C: per-seed tables, SOLVED-threshold sensitivity, evaluation scale. D: constructions. E: theory. F: exchange-rate details. G: rank details. H: sign details. I: CARoPE, CoPE and Selective RoPE. J: text world. K: what and where; the leak. L: PoPE's score. M: loops and depth on a rank-2 code. N: EM's recency deficit, a contrast case. O: analysis scripts.

A. Models, recipes, batches

A1. Architecture

component	setting
layers	1 (Sec. 6: 1-4 real layers and one block looped x4; ROBUST torus: 2 layers, d_model 256)
heads, width	2 heads, d_model 128, d_head 64; feed-forward 4 x d_model with GELU; LayerNorm before attention and before the feed-forward; dropout 0.1 on attention weights, attention output and feed-forward output
rotary channels	32 two-dimensional blocks per head, 64 channels; every query and key coordinate is rotated
score	channel c contributes q_c^T R(theta_{t',c} - theta_{t,c}) k_c (content Q and K rotated by the path angle; not additive)
step map	Delta = W_out W_in x, W_in: 128 -> r, W_out: r -> 64, no bias; r = 4 shared across heads unless stated (MapFormer's latent is per head); per-head variants: one W_in / W_out block per head; block-diagonal rank 4: the shared rank-4 latent with the off-diagonal blocks of W_out held at zero (each head reads rank 2)
frequencies	omega_c = 2 pi (1 / Delta_max)^(i / 31) for block i = 0..31, the same ladder in each head; Delta_max = grid size (torus 64, ring 32, line 256); omega is a free parameter, its sign unconstrained
vocabulary	torus: 4 moves + 17 observation tokens (16 types + blank); ring and line: 2 moves + 17
parameters	about 204k for the one-layer torus MapWM; 798,741 for four index layers; 203,925 for the looped block (`FACTORIAL_RESULTS.md`); 1.59M for the ROBUST torus model

Step parameterisations (`model.py`, `model_sign.py`, `model_wrap_exist.py`): signed (`Signed_r4` = `Vanilla_r4`), |.| (`Abs_r4`; the absolute value is applied after W_out on each of the 64 channel steps, adding no parameters), softplus (`Pos_r4`; softplus(0) = 0.693, so every token starts

at about the same step), observation-step-free (`SignedAO_r4`, `AbsAO_r4` on the ring, `SignedAO4_r4`, `AbsAO4_r4` on the torus: every non-move token's step is 0 by construction). Index RoPE (`RoPE`): canonical ladder base^(-2c / d_head), base 10000.

A2. Recipes

	torus (Secs. 5-8)	**ring and line (Secs. 6, 7)**	**Dyck-2 (Sec. 6)**
optimiser	AdamW, weight decay 0.05	AdamW, weight decay 0.05	AdamW, weight decay 0.01
peak learning rate	1e-3	1e-3 (ROBUST and LRSWEEP vary it)	1e-4
schedule	5% linear warm-up, cosine to 10% of the peak	same	same
batch	16 walks of T = 1024 moves (2,048 tokens); T = 128 batches of 128	128 walks of T = 128	128 sequences
budget	98 batches per epoch, 900 epochs (1800 in Sec. 7.2; 300 at T = 128)	98 batches per epoch, 300 epochs	560,000 sequences (1x) or 1,680,000 (3x)
loss	cross-entropy on revisited observations	same	next-token cross-entropy on every position
training map	one per run	redrawn for every walk	--
evaluation	100 walks on one held-out map, eval mode	400 walks, maps redrawn	L32 D12 test set

Run classes (`stats_core.classify_run`, per-epoch training loss): SOLVED if the mean of the final 5% of epochs is below 0.05; otherwise STALLED if the final-10% mean is within 5% of the preceding 10% mean, else DESCENDING. Continuations of constructed models (App. D): ring 300 epochs at lr 1e-3 (schedule re-warmed, fresh optimiser state, fresh walk stream); torus 300 epochs at lr 1e-4, fixed before launch (about the rate at which the source solution was reached).

A3. Batches

"Code check" = an independent, results-blind code review, whose findings were amended into the pre-registration before any result was read (before launch, except TEXTWORLD and LEAK, checked while the batch trained). "Reproduced" = the analysis was re-run from stored checkpoints by an independent reproduction audit (`docs/paper/review_draft5/REPRO_AUDIT.md`, `REPRO_AUDIT_nonsign.md`, 2026-10-16) and matched (spot re-evaluations within 7e-4); batches that landed after that audit read "no". "Attention": standard, or `--fast-attn` (SDPA; agrees to ~1e-6 in logits, not bit-identical in training). Pilots ran on seeds outside the batch except where noted.

batch	**main text**	**task**	**arms**	**seeds**	**epochs**	**attention**	**code check**	**reprodu**
PAPER2X2	5, 7.4	torus T = 128	RoPE, PoPE, MapWM r2 / r4, MapPoPE r2 / r4	0-7	300	standard	no	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reprodu
SIGN_MATCHED	5, 7.3, 9	torus T = 1024	signed, |.|, softplus, RoPE	0-7	900	standard	no (authors' pre-launch gate)	yes
GENERALITY	5, 7.3	torus; ring p_plus 0.5	obs.-step-free signed / |.| and constructions; signed, CARoPE, CoPE, RoPE	700-707	900 (ring 300); +300	fast-attn (torus)	yes	yes
FACTORIAL	6	torus T = 1024	path 1 layer; index 1, 4 layers, looped x4	500-503 (n = 4)	900	fast-attn	yes	yes
FACTORIAL_N8 (F7)	6	torus T = 1024	FACTORIAL's four cells, its recipe asserted per run	2800-2803 (fresh, n = 4)	900	fast-attn	yes	no (anal re-run fr the run identica
DYCK_MDEPTH	6	Dyck-2 L32 D12	MapWM, MapPoPE (r2), RoPE, PoPE at 1-4 layers; 3x; mixture	0-7	1x / 3x	standard	no	yes
CANCEL	6	ring p_plus 0.5-1.0	path 1 layer; index 1-3 layers	0-7; main text 2-7 (0-1 were the pilot)	300	standard	no	yes
RANK_MI	7.2	torus	shared r2, per-head r2, shared r4	0-7 (per-head r2 s0-1 were a pilot)	900	standard	no	yes
RANK_SEP	7.2	torus	block-diagonal r4, per-head r4	0-7	900	standard	no	yes
RANK3	7.2	torus	per-head r3	0-7	900	standard	no	yes
LOOP_RANK_E1800 (part 1)	7.2	torus	shared r2, shared r4	0-7	1800	standard	no	yes
RANK_PROJ	7.2	torus	projected r2 (frozen, trained), r4 control, r2 continued	0-7	+900	standard	no	yes
RANK_ND	7.2	2D (32 per side), 3D (10 per side) tori	per-head rank D, D + 1	0-7	900	standard	yes	yes
RANK_WRAP	7.2		per-head rank D + 1	0-7	900	standard	yes	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reprodu
		2D and 3D tori of two sizes						
RANK_TESTS	7.2	ring; torus escape test	rank 1, 2 (+ variants); lifted / fresh rank 4	20-27; 0-7	300; 600	standard	yes	yes
SCORE_RANK	7.2, 7.4	torus T = 1024	MapWM / PoPE score at r2 and r4	r2 10-21, r4 10-17	900	standard	yes	no
RANKCLOCK (G7)	7.2	torus T = 1024	shared r2; r2 + per-axis / per-token clock slot; r2 drift-forbidden; r2 actions-only; shared r4 at weight decay 0.05 and 0	1900-1907	900	standard	yes	no
RANKRECIPE (G9)	7.2	torus T = 1024	shared r2 at lr 1e-3 / wd 0.05, lr 3e-3, wd 0.2; shared r4 at the default	2600-2607	900	standard	yes	no
LOOP_RANK	6, 7.4	torus T = 1024	r2 1 layer, r2 looped x4, r2 4 layers, r4	0-7 (A and C reused; A s0 reproduced bitwise)	900	standard	no	yes
RING_DIAL	7.3	ring p_plus 0.5 / 0.75 / 0.9	signed, |.|	30-37	300	standard	yes	yes
WRAP_EXISTENCE	7.3	ring p_plus 0.5	obs.-step-free signed / |.|, constructions	600-607	300 + 300	standard	yes	yes
LRSWEEP	7.3	ring	obs.-step-free signed / |.| at 1e-3 / 3e-3 / 1e-2 / 3e-2	1500-1507	300	standard	yes	no (window run class cross- checked
TORUS_LRSWEEP (H12)	7.3	torus T = 1024	obs.-step-free signed / |.| at 1e-3 / 3e-3 / 1e-2	2400-2407	900	fast-attn	yes	no (window run class cross- checked cell mea re-derive from the checkpo

batch	main text	task	arms	seeds	epochs	attention	code check	reprodu
ROBUST	7.3	ring (four recipes); torus 2 layers d 256	obs.-step-free signed / |.|	1400-1407	300; 900	fast-attn (torus)	yes	yes
ACCOUNTS	7.3	ring; line	T1 continuations; T2 frozen latents; T4 CoPE on a line	600-607; 1000-1007; 1100-1115	300	standard	yes	yes
MECHANISM	7.3	ring p_plus 0.5	eight start / step arms	800-807	300	standard	yes	yes
SIGNROUTE	7.3	ring, moves +1/-1	obs.-step-free signed, |.|, hybrid, abs_tied, exp, softplus at lr 1e-3 and 3e-3	1800-1807	300	standard	yes	no
G3_SSM (H10)	7.3	ring, moves +1/-1	one-layer diagonal complex SSM: signed, |.|, softplus, exp, index; constructions (H at 3e-3, H2 at 1e-3)	2100-2107	300; +100	none (linear kernel)	yes	no (anal re-run fr the run identica
G3_SSM block A2 (H13)	7.3	ring, moves +1/-1	the same SSM arms at lr 1e-3 from scratch	2100-2107	300	none (linear kernel)	yes (Amendment 3)	no (anal re-run fr the run identica
EXPTORUS (H14)	7.3	torus T = 1024	obs.-step-free signed, |.|, exp, softplus at lr 1e-3 / 3e-3	3400-3407	900	fast-attn	yes (Amendment 1; exp partly pilot-informed, Amendment 2)	no (window run class cross-checked cell mea and permuta p re-der from the evaluati files)
SIGNSEP (H11)	7.3	ring, moves +1/-1	obs.-step-free signed, |.|, |.| + wrap offset (W), signed with frozen move latents (F), W from past pi (P),	2300-2307	300	standard	yes	no

batch	main text	task	arms	seeds	epochs	attention	code check	reprodu
			one common start, at lr 1e-3 and 3e-3					
SROPE_TORUS	7.3	torus T = 1024	Selective RoPE (best LM config), without SiLU, state-tracking config, generator swap; MapWM; RoPE	1700-1707	900	fast-attn	yes	no
DELTA_FREE	7.4	torus T = 128, r2	PoPE / MapPoPE with delta pinned, free, frozen-random (path)	1600-1607	300	standard	yes	no
TEXTWORLD	8	torus in words	path 1 layer; index 1, 2 layers	0-7; main text 2-7 (0-1 were the pilot)	900	standard	yes	yes
TW_NORMSTEP	8	torus in words	MapWM, NormStep, DirOnly (direction words only step)	10-17	900	standard	yes	yes
LEAK	8	new-object torus	MapWM, ActOnly, NormStep	0-7	900	standard	yes	yes
IMAGE_TORUS (K4)	8	torus T = 1024, class images per observation	image ActOnly, MapWM (gain 30.37 and 1), RoPE; token MapWM, ActOnly	3100-3107	900	fast-attn	yes (twice: Amendments 1, 3)	no (anal re-run fr the run identica
GAIN_PHASE (K)	--	new-object torus	step x score 2x2	8-15	900	standard	yes	no
RING_BUDGET (H)	--	ring p_plus 0.5	|.| at 300 and 900 epochs	40-47	300 / 900	standard	yes	yes
TORUS_START (H)	--	torus	start arms	1200-1207	900	fast-attn	yes	yes
GAUGE (H)	--	ring, moves +1/-1 or +1/+2	obs.-step-free signed / |.|, RoPE	1300-1307	300	standard	yes	yes
WRAP_COPE (I)	--	ring	CoPE table starts	900-907	300	standard	yes	yes

batch	main text	task	arms	seeds	epochs	attention	code check	reprodu
MAPPOPE_PAIR (L)	--	torus T = 128	MapWM r2, MapPoPE r2 (32 / 64 angles)	10-25	300	standard	yes	no
RECENCY_EM, WARM, NOLEAK, SPREAD, SEARCH (N)	9	k-back recency	MapEM variants and installs	0-7	300 (SPREAD 1200)	standard	no	no
KNOB_SWEEP (A5)	9	torus T = 128, 16-epoch recipe	MapWM, RoPE under action variants	8 seeds	16	standard	not registered	no

ACCOUNTS T1's weight-decay-0.05 arms are bit-identical retrains of WRAP_EXISTENCE's constructed arms (a second readout of the same runs, not a replication). The shared rank-2 arm of every rank contrast is one stored set of eight runs. SIGN_MATCHED's signed arm is the source of the T = 1024 what/where analysis (App. K).

A4. Reproduction of MapFormer v4 Table 2 (2D)

With the paper's recipe (16 epochs x 98 batches, linear decay, lr 3e-4; `python3 -m mapformer.main --device cuda --epochs 16 --n-batches 98`), our MapWM scores 0.989 +/- 0.010 and our MapEM (single p_0) 0.987 +/- 0.012 on fresh maps at T = 128 (`PAPER_TASK_ACCURACY.md`), against 1.00 / 1.00 in v4 and 0.99 / 1.0 in v1 (MapWM-r2, MapEM-os, 2D IID).

A5. Egocentric actions (Sec. 9; `KNOB_SWEEP_n8.md`, not registered)

Torus at T = 128 with MapFormer's 16-epoch recipe, n = 8, held-out map. With allocentric moves (baseline) path integration's position effect over index RoPE is +0.438 (MapWM 0.967, RoPE 0.529). With egocentric turn-and-forward actions ("rotate") it is +0.050 (0.558 vs 0.508): a cumulative phase cannot integrate a heading-dependent step. Recoding the same walks into allocentric moves ("allocentric") restores it to +0.488 (0.996 vs 0.508). The index arm sits at the 0.508 floor in both action conditions, so these are boundary statements about the path model, not exchange rates.

B. Floors

A floor is the accuracy of the best map-free predictor on the same kind of evaluation walk. Rows marked (analysis) were computed after the batches.

task	predictor	value	stream	source
torus T = 128 (analysis)	blank / n-gram (orders 0-5) / retrace	0.507 / 0.598 / 0.934 (retrace per seed 0.928-0.946);	PAPER2X2 eval walks, seeds 0-7	`docs/audits/2026-10-16/paper2x2_T128_floors_out.txt`

task	predictor	value	stream	source
		wrap-only share 0.000		
torus T = 1024 (analysis)	blank / n-gram / retrace	0.506 / 0.575 / 0.843 (retrace per seed 0.837-0.849); wrap-only share 0.079	SIGN_MATCHED eval walks, seeds 0-7 (shared by RANK_MI, RANK_SEP, RANK3)	`docs/audits/2026-10-16/sign_floors_out.txt`
torus T = 1024 (analysis)	same	0.508 / 0.578 / 0.843; wrap-only 0.081	GENERALITY, seeds 700-707	`generality_floors_out.txt` (same directory)
torus T = 1024, 2 layers d 256 (analysis)	same	0.509 / 0.578 / 0.844; every run of both arms above retrace (lowest |.| 0.910)	ROBUST, seeds 1400-1407	`robust_floors_out.txt`
torus T = 1024 (analysis)	blank / retrace	0.509 / 0.844	FACTORIAL, seeds 500-503	`docs/paper/figs/fig_factorial_strata_out.txt`
ring p_plus 0.5 / 0.75 / 0.9 (analysis)	retrace	0.860 / 0.758 / 0.612 (blank 0.497-0.499; 0.864 at p_plus 0.5 on the registered eval stream)	2000 walks per bias, another stream of the same task	`docs/audits/2026-10-16/ring_floors_out.txt`
ring p_plus 0.5 / 0.75 / 0.9 / 1.0	better of best constant and n-gram (orders 1-5)	0.554 / 0.544 / 0.520 / 0.498	1000 held-out walks	`CANCEL_PREREG.md`, `CANCEL_RESULTS.md`
line (256 cells)	blank	0.503	ACCOUNTS T4	`ACCOUNTS_RESULTS.md`
ring, moves +1/+2	blank (retrace reduces to it)	~0.50	GAUGE	`GAUGE_RESULTS.md`
torus in words	best constant / reversal-copy	0.505 / 0.597	TEXTWORLD eval set	`TEXTWORLD_RESULTS.md`
Dyck-2 L32 D12	best stack-free A2f floor (n-grams of orders 1-6)	0.594	gate G4	`DYCK_MDEPTH_RESULTS.md`

Per-stratum floors (analysis). Torus, seeds 0-7 (`docs/paper/figs/torus_strata_floors_s0-7.json`): best constant 0.506 / 0.500 / 0.507 (short gap / long gap / wrap-only), retrace 0.911 on short gaps and equal to the constant on the others (walk runs are at most 10 moves, so a retrace never reaches a gap of 128 moves or a wrap-only

revisit). FACTORIAL's stream: 0.508 / 0.515 / 0.517, retrace 0.912. Ring (`ring_strata_floors_out.txt`): retrace 0.9998 at gaps 2-7, 0.943 at 8-31, 0.570 at 32-127, 0.493 wrap-only; constant 0.49-0.50 in every stratum.

Against its own seed's retrace floor (analysis, torus T = 1024): signed falls below on 0/8 runs, |.| on 3/8 (0.637, 0.690, 0.717), softplus on 4/8, index RoPE on 8/8.

Further floors (analysis unless marked).

task	predictor	value	stream	source
torus T = 1024	blank / n-gram / retrace	0.507 / 0.576 / 0.843	SCORE_RANK (registered in its results file)	`SCORE_RANK_RESULTS.md`
torus T = 1024	blank / n-gram / retrace	0.509 / 0.578 / 0.845	SROPE_TORUS, seeds 1700-1707	`SROPE_TORUS_RESULTS.md`; `docs/audits/2026-10-20/srope_torus_floors_out.txt`
torus T = 128	retrace	0.938	DELTA_FREE, seeds 1600-1607	`DELTA_FREE_RESULTS.md`; `docs/audits/2026-10-20/delta_free_floors_out.txt`
torus in words	always-'nothing' / one-step reversal-copy / retrace	0.5054 / 0.5965 / 0.9337	TEXTWORLD registered eval stream	`docs/paper/expansion/scripts/language/tw_retrace_floor_out.txt`
torus in words	same	0.5120 / 0.6024 / 0.9365	TW_NORMSTEP eval stream	same
torus in words, revisits retrace cannot reach	best constant	0.506 ('nothing'; 856 of 6,381 revisits, 13.4%)	all 200 walks of the TEXTWORLD stream	`docs/paper/expansion/scripts/language/tw_strata_200walks_out.txt`
torus in words, ... whose cell holds an object (the revisits retrace gets wrong)	best constant ('nothing' excluded by construction)	0.088 ('cat'; 423 of 6,381, 6.6%)	same	same
ring, moves +1/-1, observation-step-free	n-gram / retrace	0.554 / 0.860	SIGNROUTE reference	`SIGNROUTE_RESULTS.md`

C. Per-seed tables

Accuracy at the training length on held-out maps, eval mode. SOLV / STAL / DESC = SOLVED / STALLED / DESCENDING; a number in brackets is the final-5% training loss.

C1. Torus sign batch (Secs. 5, 7.3; `SIGN_MATCHED_ANALYSIS.txt`)

seed	signed	|.|	softplus	index RoPE
0	1.000 SOLV (0.003)	0.931 DESC (0.201)	0.590 STAL (1.250)	0.730 STAL (0.983)
1	0.997 SOLV (0.003)	0.917 DESC (0.314)	0.527 STAL (1.510)	0.729 STAL (0.939)
2	1.000 SOLV (0.005)	0.868 STAL (0.492)	0.630 STAL (1.084)	0.720 STAL (0.931)
3	1.000 SOLV (0.008)	0.637 STAL (1.043)	0.972 DESC (0.158)	0.739 STAL (0.910)
4	1.000 SOLV (0.004)	0.690 DESC (0.962)	0.674 STAL (1.009)	0.737 STAL (0.928)
5	0.990 SOLV (0.003)	0.888 STAL (0.446)	0.935 DESC (0.295)	0.744 STAL (0.885)
6	1.000 SOLV (0.004)	0.922 STAL (0.349)	0.922 DESC (0.345)	0.722 STAL (0.956)
7	1.000 SOLV (0.004)	0.717 STAL (0.834)	0.995 SOLV (0.049)	0.728 STAL (0.976)
mean	0.998	0.821	0.781	0.731

Registered: |.| - signed -0.177 (permutation p 0.0002; 0/8 vs 8/8, Fisher p 0.0002); softplus - signed -0.218 (p 0.0003; 1/8 vs 8/8, p 0.0014); signed - RoPE +0.267 (p 0.0002; 8/8 vs 0/8, p 0.0002). r(final loss, accuracy) -0.992 over 32 runs.

C2. Torus, index depth (Sec. 6; `FACTORIAL_{L1,L4,LOOP}.json`)

seed	path, 1 layer	index, 1 layer	index, 4 layers	index, looped x4
500	1.000	0.719	0.814	0.868
501	1.000	0.731	0.870	0.861
502	1.000	0.688	0.874	0.860
503	1.000	0.739	0.881	0.868
2800 (fresh)	0.996	0.730	0.840 (D)	0.845
2801 (fresh)	1.000	0.740	0.862	0.861
2802 (fresh)	1.000	0.713	0.811	0.866
2803 (fresh)	1.000	0.733	0.807	0.866

FACTORIAL (500-503): every path run SOLVED, every index run STALLED. Registered C0-C3: path beats index at one layer (+0.281), four layers (+0.141) and loop x4 (+0.136); depth and loop each help index (+0.140, +0.145); every firing a complete separation (p 0.029). FACTORIAL_N8 (2800-2803, analysed alone; App. F7): every path run SOLVED, every index run STALLED except one 4-layer run still DESCENDING (D); the same five labels (+0.270, +0.169, +0.140; +0.101, +0.130), each a complete separation (p 0.029): REPLICATES.

C3. Ring exchange rate, fresh seeds 2-7 (Sec. 6; `docs/audits/2026-09-27/fresh_seeds.txt`)

p_plus	path 1 layer	index 1 layer	gap to path at 1 / 2 / 3 index layers
0.5	1.000	0.710 +/- 0.020	+0.290 / +0.051 / +0.002
0.75	1.000	0.677 +/- 0.009	+0.323 / +0.038 / +0.002
0.9	1.000	0.824 +/- 0.007	+0.176 / +0.013 / +0.001
1.0	1.000	1.000	0 / 0 / 0

All eight seeds (`CANCEL_RESULTS.md`): index 1 / 2 / 3 layers 0.717 / 0.954 / 0.998 (p_plus 0.5), 0.676 / 0.969 / 0.998 (0.75), 0.825 / 0.988 / 0.999 (0.9), 1.000 (1.0); path 1.000 in every cell (32/32 SOLVED); registered exchange rate k = 3, 3, 3, 1 index layers.

C4. Rank (Sec. 7.2; `RANK_MI_ANALYSIS.txt`, `RANK_SEP_ANALYSIS.txt`, `RANK3_ANALYSIS.txt`, `RANK_SEP.json`, `RANK3.json`)

seed	shared r2	per-head r2	block-diag r4	per-head r3	shared r4	per-head r4
0	0.849 STAL (0.587)	0.656 STAL (1.049)	0.879 STAL (0.482)	0.942 DESC (0.243)	1.000 SOLV (0.004)	0.996 SOLV (0.004)
1	0.998 DESC (0.051)	1.000 SOLV (0.007)	0.936 DESC (0.302)	0.990 SOLV (0.003)	1.000 SOLV (0.003)	1.000 SOLV (0.010)
2	0.917 DESC (0.308)	1.000 SOLV (0.003)	1.000 SOLV (0.010)	1.000 SOLV (0.003)	1.000 SOLV (0.003)	1.000 SOLV (0.004)
3	0.917 DESC (0.321)	0.723 STAL (0.910)	0.945 DESC (0.234)	0.967 DESC (0.134)	1.000 SOLV (0.003)	1.000 SOLV (0.004)
4	0.947 DESC (0.177)	0.959 DESC (0.194)	0.964 DESC (0.202)	1.000 SOLV (0.003)	1.000 SOLV (0.002)	0.998 SOLV (0.002)
5	0.907 STAL (0.404)	0.949 DESC (0.211)	0.917 DESC (0.310)	1.000 SOLV (0.006)	0.994 SOLV (0.004)	1.000 SOLV (0.008)
6	0.773 STAL (0.675)	0.943 DESC (0.248)	1.000 SOLV (0.010)	1.000 SOLV (0.003)	1.000 SOLV (0.004)	1.000 SOLV (0.005)
7	0.842 STAL (0.605)	0.851 STAL (0.539)	0.947 DESC (0.166)	1.000 SOLV (0.005)	0.987 SOLV (0.010)	1.000 SOLV (0.003)
mean	0.894, 0/8	0.885, 2/8	0.948, 2/8	0.987, 6/8	0.998, 8/8	0.999, 8/8

Per-head r2 seeds 0-1 were a pilot run before registration (1/6 SOLVED on seeds 2-7). Registered: per-head rank 4 vs block-diagonal 8/8 vs 2/8 (Fisher and permutation p 0.0070); shared r4 vs block-diagonal the same (it also changes per-head rank); sharing (per-head r4 vs shared r4, +0.001) and W_out scale (block-diagonal vs per-head r2, +0.063) not detected; per-head r3 vs r2 +0.102 (permutation p 0.027; SOLVED 6/8 vs 2/8, Fisher p 0.13); r4 vs r3 +0.012, not detected. Shared r4 vs shared r2: 8/8 vs 0/8 (Fisher p 0.0002), +0.104 (p 0.0005).

Twice the budget (`LOOP_RANK_E1800_P1_ANALYSIS.txt`, 1800 epochs from scratch): shared r2 0/8 (final-5% losses 0.489, 0.383, 0.390, 0.340, 0.141, 0.231, 0.305, 0.571; 3 STALLED, 5 DESCENDING), accuracy 0.908; shared r4 7/8 (s5 STALLED at 0.180), 0.994.

C5. Sign, ring and torus (Sec. 7.3)

Ring dial (`RING_DIAL_VERDICTS.json`; full models, S = SOLVED):

seed	0.5 signed	0.5 |.|	0.75 signed	0.75 |.|	0.9 signed	0.9 |.|
30	1.000 S	1.000 S	1.000 S	0.994 S	1.000 S	1.000 S
31	1.000 S	0.977	1.000 S	0.958	1.000 S	0.999 S
32	1.000 S	0.987	1.000 S	0.957	1.000 S	0.885
33	1.000 S	0.964	1.000 S	0.954	1.000 S	1.000 S
34	1.000 S	1.000 S	1.000 S	0.964	1.000 S	1.000 S

seed	0.5 signed	0.5 |.|	0.75 signed	0.75 |.|	0.9 signed	0.9 |.|
35	1.000 S	0.944	1.000 S	0.967	1.000 S	0.825
36	1.000 S	0.977	1.000 S	0.993	1.000 S	0.960
37	1.000 S	0.931	1.000 S	0.958	1.000 S	0.964

Registered: p_plus 0.5 -0.028 (permutation p 0.007), 2/8 vs 8/8 (Fisher p 0.0070); 0.75 -0.032 (p 0.0002), 1/8 vs 8/8 (p 0.0014); at 0.9 the contrast did not reach the registered threshold (-0.046, below the 0.05 materiality; 4/8 vs 8/8, p 0.077).

Observation-step-free ring from scratch and constructed (`WRAP_EXISTENCE_ANALYSIS.txt`):

seed	signed	|.|	constructed |.|: frozen -> trained (max continuation loss)	constructed signed: frozen -> trained
600	1.000 S	0.905 STAL	1.000 -> 1.000 S (0.041)	1.000 -> 1.000 S (0.019)
601	1.000 S	0.949 STAL	1.000 -> 1.000 S (0.030)	1.000 -> 1.000 S (0.042)
602	1.000 S	0.853 STAL	1.000 -> 1.000 S (0.154)	1.000 -> 1.000 S (0.072)
603	1.000 S	0.951 STAL	1.000 -> 1.000 S (0.046)	1.000 -> 1.000 S (0.017)
604	1.000 S	0.907 STAL	1.000 -> 1.000 S (0.024)	1.000 -> 1.000 S (0.039)
605	1.000 S	0.930 STAL	1.000 -> 1.000 S (0.032)	1.000 -> 1.000 S (0.028)
606	1.000 S	0.950 STAL	1.000 -> 1.000 S (0.031)	1.000 -> 1.000 S (0.112)
607	1.000 S	0.922 STAL	1.000 -> 1.000 S (0.036)	1.000 -> 1.000 S (0.015)

Registered: from scratch 0.921 vs 1.000, -0.079 (permutation p 0.0002), 0/8 vs 8/8 (Fisher p 0.0002); constructed |.| HELD (8/8 SOLVED, 0/8 LOST).

Observation-step-free torus (`GENERALITY_ANALYSIS.txt`):

seed	signed	|.|	constructed signed (lr 1e-4)	constructed |.| (lr 1e-4): peak loss, epoch of peak / of re-entry
700	1.000 S	0.874 STAL	1.000 S	1.000 S; 0.168, 29 / 153
701	1.000 S	0.707 STAL	1.000 S	1.000 S; 0.110, 40 / 121
702	1.000 S	0.946 DESC	1.000 S	1.000 S; 0.173, 19 / 204
703	1.000 S	0.879 STAL	1.000 S	1.000 S; 0.143, 52 / 149
704	1.000 S	0.892 STAL	1.000 S	1.000 S; 0.131, 13 / 119
705	1.000 S	0.907 STAL	1.000 S	1.000 S; 0.155, 13 / 148

seed	signed	|.|	constructed signed (lr 1e-4)	constructed |.| (lr 1e-4): peak loss, epoch of peak / of re-entry
706	1.000 S	0.877 STAL	1.000 S	1.000 S; 0.208, 36 / 164
707	1.000 S	0.960 DESC	1.000 S	1.000 S; 0.117, 23 / 139

Registered: from scratch 0.880 vs 1.000, -0.120 (p 0.0002), 0/8 vs 8/8; the constructed |.| model leaves the solved region on 6/8 seeds (peak loss 0.11-0.21; signed control 0.004-0.017) and ends SOLVED on 8/8 at 1.000: by the reading fixed before launch, perturbed and re-found.

Recipe and scale (`ROBUST_ANALYSIS.txt`, seeds 1400-1407; accuracy at the training length, class):

```
ring  seed | def:SO | def:AO | wd0:SO | wd0:AO | lrx3:SO | lrx3:AO | lrd3:SO | lrd3:AO
1400 | 1.0000 SOLV | 0.8230 STAL | 1.0000 SOLV | 0.9078 DESC | 1.0000 SOLV | 1.0000 DESC |
1.0000 SOLV | 0.8893 STAL
1401 | 1.0000 SOLV | 0.9155 STAL | 1.0000 SOLV | 0.6239 STAL | 1.0000 SOLV | 0.9596 STAL |
0.6194 STAL | 0.6221 STAL
1402 | 1.0000 SOLV | 0.9237 STAL | 1.0000 SOLV | 0.8878 STAL | 1.0000 SOLV | 0.9702 STAL |
1.0000 SOLV | 0.8319 STAL
1403 | 1.0000 SOLV | 0.9136 STAL | 1.0000 SOLV | 0.8800 STAL | 1.0000 SOLV | 0.9995 SOLV |
1.0000 SOLV | 0.9167 STAL
1404 | 1.0000 SOLV | 0.9118 STAL | 1.0000 SOLV | 0.9119 STAL | 1.0000 SOLV | 1.0000 SOLV |
1.0000 SOLV | 0.8313 STAL
1405 | 1.0000 SOLV | 0.8791 STAL | 1.0000 SOLV | 0.8858 STAL | 1.0000 SOLV | 0.9650 STAL |
1.0000 SOLV | 0.8526 STAL
1406 | 1.0000 SOLV | 0.9211 DESC | 1.0000 SOLV | 0.9110 STAL | 1.0000 SOLV | 1.0000 SOLV |
0.9794 STAL | 0.9062 STAL
1407 | 1.0000 SOLV | 0.9997 SOLV | 1.0000 SOLV | 0.9735 DESC | 1.0000 SOLV | 0.9998 SOLV |
1.0000 SOLV | 0.8801 STAL
torus 1400 SO 1.0000 SOLV | AO 0.9941 DESC     1404 SO 1.0000 SOLV | AO 0.9935 SOLV
      1401 SO 1.0000 SOLV | AO 0.9981 SOLV     1405 SO 1.0000 SOLV | AO 0.9706 DESC
      1402 SO 0.9941 SOLV | AO 0.9705 DESC     1406 SO 1.0000 SOLV | AO 0.9098 DESC
      1403 SO 1.0000 SOLV | AO 0.9454 DESC     1407 SO 1.0000 SOLV | AO 0.9544 DESC
```

Learning-rate sweep (`LRSWEEP_RESULTS.md`, seeds 1500-1507): per-seed table in H2.

C6. Text world (Sec. 8; `TEXTWORLD_ANALYSIS.txt`)

seed	path, 1 layer	index, 1 layer	index, 2 layers
0 (pilot)	0.881	0.504	0.695
1 (pilot)	0.998	0.504	0.723
2	0.999	0.503	0.927
3	0.995	0.506	0.722
4	0.999	0.505	0.926
5	1.000	0.506	0.706
6	1.000	0.506	0.760
7	0.882	0.504	0.720

Fresh seeds 2-7 (`docs/audits/2026-09-27/fresh_seeds.txt`): path 0.979 +/- 0.048 (6/6 SOLVED), index 1 layer 0.505 (0/6), index 2 layers 0.793 (0/6); path - index one layer +0.474, permutation and Fisher p 0.0022. All eight seeds: path 0.969 +/- 0.054 (7/8; seed 0 still descending), +0.464 (p 0.0002; Fisher p 0.0014); path - index two layers +0.197 (p 0.0009).

C7. Dyck-2

Per-seed values are in `DYCK_MDEPTH_RESULTS.json` (cells `T12|{RoPE,PoPE,MapWM,MapPoPE}|{1-4}L|L32D12`); every one of the eight seeds has a positive position effect at every depth (sign-flip p 0.0078 per depth).

C8. SOLVED-threshold sensitivity

Sign batches (`docs/audits/2026-10-15/reanalyses/solved_thresholds_out.md`; predictions and decision rules written before the re-analysis; two-sided Fisher p in brackets):

batch	contrast	loss < 0.02	loss < 0.05 (registered)	loss < 0.1	accuracy >= 0.99
SIGN_MATCHED torus	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 7/8 (0.0014)
SIGN_MATCHED torus	softplus vs signed	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	1/8 vs 7/8 (0.0101)
RING_DIAL p 0.5	|.| vs signed	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)	4/8 vs 8/8 (0.0769)	2/8 vs 8/8 (0.0070)
RING_DIAL p 0.75	|.| vs signed	1/8 vs 8/8 (0.0014)	1/8 vs 8/8 (0.0014)	2/8 vs 8/8 (0.0070)	2/8 vs 8/8 (0.0070)
WRAP_EXISTENCE ring	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
MECHANISM ring	|.| vs signed, same-direction start	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)	0/8 vs 7/8 (0.0014)
ACCOUNTS T2 ring	frozen vs free move latents	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)	1/8 vs 7/8 (0.0101)
GENERALITY torus	|.| vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)
GENERALITY ring	CoPE vs signed	0/8 vs 8/8 (0.0002)	0/8 vs 8/8 (0.0002)	1/8 vs 8/8 (0.0014)	0/8 vs 8/8 (0.0002)

ROBUST (declared secondary, loss < 0.1): ring default 1/8 vs 8/8 (p 0.0014), weight decay 0 1/8 vs 8/8 (0.0014), lr / 3 0/8 vs 7/8 (0.0014), lr x3 4/8 vs 8/8 (0.077), torus 2 layers d 256 5/8 vs 8/8 (0.20); on that torus cell the accuracy contrast (0.967 vs 0.999, p 0.0005) carries the result. LRSWEEP (loss < 0.1, selection redone): 2/8 vs 8/8, same label.

Rank batches (analysis; read from the per-seed final-5% losses in C4): at loss < 0.02 the counts are 0 / 2 / 2 / 6 / 8 / 8 of 8 (shared r2, per-head r2, block-diagonal r4, per-head r3, shared r4, per-head r4), identical to the registered 0.05; at loss < 0.1 they are 1 / 2 / 2 / 6 / 8 / 8 (shared r2 seed 1 at 0.051 enters).

C9. Attention-dropout evaluation scale

Models trained with attention dropout expect the 1/(1 - p) attention scale; evaluated with it, below-ceiling one-layer runs score higher. The registered text-world evaluation re-run with the scale (`docs/audits/2026-10-04/DROPOUT_RESCORE.md`, all 24 runs) gives path 0.969 -> 0.993, index one layer 0.505 -> 0.506, index two layers 0.772 -> 0.772; path - index one layer +0.464 -> +0.488 (p 0.0002). The re-score was a declared secondary of RING_DIAL, RING_BUDGET, WRAP_EXISTENCE, GENERALITY, MECHANISM and WRAP_COPE and changed no label there; it was applied to SIGN_MATCHED by the same audit with no verdict changed. ACCOUNTS, TORUS_START, GAUGE and ROBUST were not re-scored (the correction does not apply at two layers). Main-text accuracies are eval mode without the scale except where marked re-scored (Sec. 8, Table 6, App. J).

C10. Sign route (Sec. 7.3; `SIGNROUTE_RESULTS.md`, registered; ring, observation-step-free, seeds 1800-1807)

Accuracy at T = 128 and run class (S SOLVED, T STALLED, D DESCENDING; none diverged):

```
lr     arm       1800   1801   1802   1803   1804   1805   1806   1807
1e-3   signed    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
1e-3   abs       0.912T 0.874T 0.902T 0.926T 0.918T 0.968T 0.939T 0.902T
1e-3   hybrid    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
1e-3   abs_tied  0.748T 0.748T 0.672T 0.723T 0.749T 0.751T 0.748T 0.770T
1e-3   exp       0.904T 0.973T 0.929T 0.957T 0.852T 0.958T 0.936D 0.972S
1e-3   softplus  0.841T 0.794T 1.000S 0.984T 0.734T 0.861T 0.909T 0.906D
3e-3   signed    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
3e-3   abs       0.921T 0.944T 0.935T 0.968T 0.949T 1.000S 0.945T 1.000S
3e-3   hybrid    1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S 1.000S
3e-3   abs_tied  0.743T 0.748T 0.652T 0.778T 0.775T 0.748T 0.774T 0.769T
3e-3   exp       0.897T 0.959D 1.000S 1.000D 0.902T 0.967D 0.960D 0.977D
3e-3   softplus  0.954T 0.959D 0.999S 1.000S 0.964T 0.975D 1.000S 0.880T
```

D. Constructions

D1. The rank-2 projection (Sec. 7.2; `RANK_PROJ_RESULTS.md`)

Each solved shared rank-4 model (seeds 0-7, a stored batch) is projected onto the top two directions of its action latent over the vocabulary (top-2 spectral energy 0.99952-0.99994), every other weight kept, and installed in the shared rank-2 architecture. Frozen, the projections score 0.9955 at T = 1024 against their sources' 0.9966 (at least 0.981 on every seed; per seed 0.9993, 1.0000, 0.9988, 1.0000, 0.9850, 0.9806, 1.0000, 1.0000). Trained 900 epochs from a learning-rate restart at 1e-3, 7/8 are SOLVED (one DESCENDING), as for the rank-4 control trained from its own solution (7/8); rank 2 continued from its own 900-epoch weights solves 1/8. Registered verdict: STABLE.

D2. Per channel (Lemma P1.0 in v3)

For a signed step Delta and omega != 0, Delta' = Delta mod (2 pi / |omega|) lies in [0, 2 pi / |omega|) and omega Delta' = omega Delta (mod 2 pi) for either sign of omega. The

model reads each channel's angle only through cos and sin, so the model with Delta' is the same function.

D3. The balanced |.| construction (Proposition 3(a), E2; the registered existence test)

Given a solved signed observation-step-free model with m stepping move tokens (m = 2 ring, 4 torus; r = 4):

1. Re-solve W_in so that the move latents are c e_1, ..., c e_m (c = the source's mean move-latent norm). W_in acts only on the move embeddings, so this is a free gauge.
2. Per channel set omega' = max(|omega|, 1) and the steps to remainder(omega Delta, 2 pi) / omega' (angles in [0, 2 pi) for either sign of omega).
3. Set W_out[:, j] = step_j / c for the m move columns and 0 elsewhere; copy every other weight.

The signed control is built by the same function with the unwrapped signed angle (steps omega Delta / omega'), so the two differ only in the wrap. On the torus m = 4 = r, exactly at Proposition 3(a)'s limit.

Exactness. Ring: frozen accuracy = source on 8/8; maximum angle error 1.2e-6 rad (|.|), 6.1e-7 (signed); maximum logit difference 5.1e-4 at T = 128. Torus: frozen = source on 8/8; maximum angle error 1.4e-6 rad (|.|), 5.8e-7 (signed). *Weight scale.* Ring: maximum |W_out| 0.31-0.38 in the signed sources, 1.07-1.70 built; |W_in| 0.15-0.21 -> 0.19-0.33. Torus: maximum |W_out| 0.48-0.58 -> 1.23-1.55, maximum |omega| 0.68-0.96 -> 1.00. Within about 5x of the trained scale.

Why the scale matters. A construction that keeps the source's W_in and solves W_out at minimum norm is also exact, but in a solved signed model the two move latents are nearly antiparallel (cosine -0.987 to -0.998), so W_out must be huge (up to 16,320 against 0.31-0.38). Adam's per-entry steps then lose that solution in the warm-up (8/8 LOST) before re-finding it (7/8 SOLVED). An existence test is informative only near a trained scale, which the balanced construction provides.

D4. The full |.| arms

To show that the task, not only a given signed table, is representable by a full |.| model (observations stepping), W_in is chosen to project onto two directions orthogonal to all 17 observation embeddings and W_out is solved per channel (`docs/audits/2026-10-12/ring_dial_audit.md` M1). In the real modules: |.| steps >= 0, observation-token |Delta| <= 1.7e-6, cos / sin equal to the signed code's to 6.3e-4 over a 512-step walk, circular mismatch <= 4.6e-4 on 2,000 same-cell revisit pairs.

D5. Softplus (Proposition 3(f), E2)

Softplus cannot reach 0, but observation steps need only be 0 mod 2 pi. With W_in annihilating every observation embedding, each observation steps softplus(0) = ln 2 in every channel, and omega'_c = 2 pi / ln 2 makes that exactly one turn; the move columns are set to softplus^{-1}((remainder(Theta, 2 pi) + 2 pi) / omega'). Built in the real softplus model from a solved ring model (`docs/paper/review_draft5/theory_check_extra.py` (2), CPU): minimum step 0.693 > 0, angle error mod 2 pi 8.4e-15 rad over all 19 tokens,

logits equal to the source's to 2.8e-11 (logit scale 21.6). On the torus m = 4 = r, so the same construction applies.

D6. Continuations and the LOST rule

A constructed model is trained further on a fresh walk stream and compared with its signed control at the same seed. *LOST* (registered): the run's maximum per-epoch training loss during the continuation exceeds 0.3, or exceeds 10x its control's maximum at that seed and 0.05. *HELD* = SOLVED on at least n - 1 seeds, mean accuracy at least control - 0.01, and LOST on at most one seed. Ring (lr 1e-3): HELD, 8/8 SOLVED, 0/8 LOST (maximum per-epoch loss 0.024-0.154 against the control's 0.015-0.112). Torus (lr 1e-4): LOST on 6/8 and SOLVED on 8/8 at the end, read by the rule fixed before launch as perturbed and re-found; at lr 1e-3 (descriptive) 6/8 SOLVED (0.988).

D7. What the held model keeps (ACCOUNTS T1, registered)

Weighted by usage (mean query norm at move tokens times mean key norm at observation tokens, per channel), the held ring |.| model keeps its cancelling code from 0.99 at construction to 0.98 after training; unweighted (every channel equal) the share falls from 0.87 to 0.18, because channels attention does not read leave the code. Weight decay 0 vs 0.05 changes nothing detected (-0.003).

E. Theory

Labels used here: **proved** (a proof is given), **cited** (a published construction applied to our setting), **conjecture** (a derivation whose assumptions or dynamics are not checked). Main-text numbering: Proposition 1 = E6.1; Proposition 2 = E7.1 + E7.2; Proposition 3 = E2 (a), (f), (g); Lemma 1 = E3; Proposition 4 = E8.1-E8.6; Proposition 5 and Conjecture 1 = E9.2-E9.3; Remark 2 = E10; Proposition 6 = E11; Proposition 7 = E12; Proposition 8 = E5; Remark 3 = E13; Remark 1 = E6.2. The check scripts named "(check)" are CPU-only and eval-only; those in `docs/paper/appendix_checks/` were refereed with the v3 theory appendix, those in `docs/paper/expansion/scripts/` are listed in App. O.

Notation. A rotary channel is c = (h, b), head h and block b, acting on query and key coordinates (2b, 2b + 1) of head h; R(phi) is the 2D rotation. The phase is theta_{t,c} = omega_c sum_{u <= t} S_{c, x_u}, with S = W_out W_in E the C x V step table (rank at most r) and Theta = diag(omega) S the angle table; omega is free in sign. For token x the latent is z_x = W_in e_x and channel c's readout row is w_c, so the pre-nonlinearity step is w_c . z_x and the angle a_c(x) = omega_c w_c . z_x (signed), omega_c |w_c . z_x| (|.|), omega_c softplus(w_c . z_x) or omega_c exp(w_c . z_x). For an inverse pair of tokens (x, y), such as opposite moves, write the clock k_c = a_c(x) + a_c(y), the map m_c = a_c(y) and delta_c = (a_c(y) - a_c(x)) / 2. Channel c *cancels through zero* if k_c = 0 with a_c(x) ! = 0, *cancels by wrapping* if k_c is a non-zero multiple of 2 pi, and is *blind* if both angles are 0 mod 2 pi. *Usage* q_c of a channel is its mean query norm at move tokens times its mean key norm at observation tokens (exact per token at layer 0).

E1. Score identity (proved; the identity is Mamba-3's Proposition 3)

With alpha piecewise constant, alpha = Delta_u on (u - 1, u], the homogeneous part of Howard et al.'s integrator, dF/dt = -s alpha(t) F, has propagator F(t) = exp(-s (Phi_t - Phi_{t'})) F(t'). With s = i omega, channel c of the rotary score is q_c^T R(theta_s - theta_t) k_c = Re(conj(q~*c) k~_c exp(-i omega_c (Phi_t - Phi_s))), with q~ = q*{2b} + i q_{2b+1}.

Proof. On each interval the ODE is linear with constant coefficient -s Delta_u; flows over tokens compose by adding exponents. R(theta) acts on q~ as multiplication by e^{i theta}, so (R(theta_t) q)^T (R(theta_s) k) = Re(conj(q~) k~ e^{-i (theta_t - theta_s)}). [] With input, the zero-order-hold solution is F(t) = e^{-s Phi_t} F(0) + sum_{u <= t} e^{-s (Phi_t - Phi_u)} ((1 - e^{-s Delta_u}) / s) f_u; a zero-rate token writes nothing there, whereas in attention a zero-step token still writes its key and value at full strength. *Check* (`a0_propagator.py`; `review_draft5/theory_check_extra.py` (1)): 3e-14 (propagator), 4e-15 (score), 1.1e-14 (input term).

E2. Representability by wrapping (proved)

(a) At most r stepping tokens. Let S be a signed table with at most m <= r non-zero columns (stepping tokens x_1, ..., x_m), whose embeddings are linearly independent modulo the span of the other embeddings (generic when V <= d_model, here 19 or 21 <= 128; automatic in observation-step-free models). Then a |.| model with the same embeddings and content weights has an angle table equal to Theta mod 2 pi, entrywise.

Proof. Pick W_in' with W_in' e_{x_j} = u_j (the j-th unit vector of R^r) and W_in' e_y = 0 for every other token. Set omega' = max(|omega|, 1) per channel and W_out'[:, j] = remainder(Theta_{., x_j}, 2 pi) / omega' >= 0, other columns zero. Then W_out' W_in' E is entrywise non-negative, so |.| leaves it unchanged, and it reproduces each stepping column mod 2 pi. [] *Full arms:* choosing W_in' to annihilate every observation embedding leaves only the moves stepping, so every |.| model in the paper can represent any code in which only the moves step. *Check* (`p1_wrap.py`): angle error <= 7e-14, logits <= 8e-14.

(f) Softplus. Under the same assumptions, send the stepping embeddings to u_1, ..., u_m and every other embedding to 0, set omega'*c = 2 pi / ln 2 for every c, and W_out'[c, j] = softplus^{-1}((remainder(Theta*{c,x_j}, 2 pi) + 2 pi) / omega'_c). Every non-stepping token then steps softplus(0) = ln 2, an angle of exactly 2 pi, and every stepping token's angle is Theta mod 2 pi. *Proof.* softplus is a bijection from R onto (0, inf); the targets lie in [ln 2, 2 ln 2); omega' ln 2 = 2 pi. [] *Check* (`theory_check_extra.py` (2)): built in the real softplus model, logits equal to the source's to 2.8e-11.

(g) Exp. The same construction with omega'*c = 2 pi and W_out'[c, j] = ln((remainder(Theta*{c,x_j}, 2 pi) + 2 pi) / 2 pi) makes every non-stepping token step exp(0) = 1, an angle of exactly 2 pi, and every stepping token's angle Theta mod 2 pi. *Proof.* exp is a bijection from R onto (0, inf); the targets (remainder + 2 pi) / 2 pi lie in [1, 2). [] Built in trained one-layer SSMs (App. H10): frozen accuracy equal to the source's on 8/8, move-angle error <= 2.3e-6 rad.

What representability does not cover. For a rank-1 |.| model and two stepping tokens with steps of strictly opposite sign, exact representation mod 2 pi fails for almost every signed

table (v3 P1(c), proved); rank-1 non-negative codes approximate any table mod 2 pi at a large angle scale (P1(d), Kronecker). Every claim in the paper uses (a), (f) or the full-arm construction, which are exact.

E3. The relative-sign gauge (Lemma 1; proved)

Take a MapWM model with any number of layers and a set F of channels. Form a second model by (a) negating every token's step in each channel of F (by negating omega_c; in the signed class equivalently row c of W_out, which is not available in the |.| class since |.| is applied after W_out) and (b) negating output coordinate h d_head + 2b + 1 of the query and key projections (weight row and bias entry, every layer) for each c = (h, b) in F. The two models have identical attention scores at every layer, hence identical logits.

Proof. With J = diag(1, -1), J R(phi) J = R(-phi) and J is orthogonal. (a) replaces theta_{t,c} by -theta_{t,c}; (b) replaces q_c, k_c by J q_c, J k_c, exactly including biases. The channel term becomes (J q_c)^T R(-(theta_s - theta_t)) (J k_c) = q_c^T R(theta_s - theta_t) k_c. Every layer uses the same cos / sin, so the scores agree layer by layer. [] *Check* (`t1_gauge.py`): 35 channels flipped and reflected give bitwise-identical logits (control without the reflection: 6.6).

Corollaries. (i) The |.| class realises a signed angle table Theta exactly as real numbers, with the same latent, iff every row of Theta is sign-coherent. (ii) It therefore excludes no map *function* for which a code with sign-coherent rows exists, and that code needs no wrapping: one stepping token (parity, Z_m counting) or two moves that step the same way (+1 and +2: a(+2) = 2 a(+1)). (iii) Up to the gauge, the excluded tables are exactly those with a row containing entries of strictly opposite sign. For an inverse pair every through-zero cancelling channel is such a row; cancellation by wrapping, (a, 2 pi - a), is allowed whenever E2 applies. The constraint removes the through-zero branch of the inverse-pair problem, not cancellation.

E4. One channel's geometry (proved)

Take one channel and an inverse pair (x, y) with angles (a_x, a_y), everything else fixed; the loss is a function on the angle torus. Write sigma = a_x + a_y.

E4.1 (cut circle). (a) The closed quadrant [0, inf)^2 maps onto the torus, so the signed and |.| classes attain the same loss values in this channel. (b) The non-blind cancelling set in the quadrant consists of the segments sigma = 2 pi k, k >= 1; the through-zero line meets the quadrant only at the blind origin. (c) From a start in the open quadrant with sigma < 2 pi, the infimum distance (in angle coordinates) to a non-blind cancelling point is min(sigma, 2 pi - sigma) / sqrt 2 in the signed class and (2 pi - sigma) / sqrt 2 in the |.| class; they coincide iff sigma >= pi. (d) Any continuous |.| path from |sigma| < pi to a non-blind cancelling point passes through |sigma| = pi.

Proof. (a), (b) from the quotient map. (c) The distance to sigma = 2 pi k is |sigma - 2 pi k| / sqrt 2; the |.| class has only k >= 1. (d) |sigma| is continuous along a |.| path and a non-blind cancelling point has |sigma| >= 2 pi. [] *Check* (`p3_p5_geometry.py`): a brute-force grid matches both formulas to 2e-4.

E4.2 (sticky wall). Let s be the pre-|.| step of one token in one channel and a = omega |s|, with L(s) = l(|s|) and l continuously differentiable near 0. (a) For s != 0, dL/ds = omega

sign(s) dL/da. (b) If l'(0) > 0, s = 0 is a strict local minimum of L along s with every other parameter fixed. (c) Gradient descent on s with step eta, l' in [mu, M] on [0, rho], mu > 0, eta M <= rho and |s_0| <= rho, gives |s_{k+1}| <= max(|s_k| - eta mu, eta M). *Proof.* Chain rule; first-order expansion; case split. [] In parameter space s = w_c . z_x is shared across tokens and channels, so (b) is not a local minimum of the loss. In trained |.| models the wall is the cone boundary of E8.2 (App. H7).

E5. CARoPE cannot cancel (Proposition 8; proved)

CARoPE (faithful, `model_mono_baselines.MapFormerWM_CARoPEF`) advances channel i of head h by f_h(x)^i per token, with f in (0, 1), so every per-token advance lies in (0, 1]; channel 0 advances by exactly 1, an index clock. (a) The phase of any interval of k >= 1 tokens lies in (0, k]. The shortest return spans 3 tokens between the move query and the observation key, and 4 between two observation tokens, so its phase lies in (0, 4], a subset of (0, 2 pi): no channel, at any parameter value, reads every return at phase 0. (b) Fix eps in (0, pi - 2). If a channel reads each return of an excursion of J moves out and J back within eps of 0 mod 2 pi, then the move-driven phase of any interval with at most 2J moves lies in [0, 2 eps]: every channel is either a clock or blind over that range.

Proof. (a) Sum k numbers in (0, 1]. (b) The depth-j return phase P_j satisfies P_1 in (0, 3] and P_{j+1} - P_j in (0, 4]; the arc (eps, 2 pi - eps) is longer than 4 when eps < pi - 2, so a sequence starting in (0, eps] with increments at most 4 cannot jump it; induction gives P_j in (0, eps], hence J (c_+ + c_-) <= eps. [] *Check* (`p6_carope.py`): eight trained ring CARoPE models; 10,331 random draws satisfy the bound.

E6. Path integration supplies the prefix sum; index attention needs a count

E6.1 (Proposition 1; proved). Tokens interleave moves and observations, [a_1, o_1, a_2, o_2, ...]; o_t is predicted at the token a_t. Let P_t be the unwrapped displacement after move t. The map is redrawn per evaluation, so the only source of o_t is an earlier observation o_s with P_s = P_t (mod N per axis). In MapWM, with observation steps zero and a fixed step vector per move, theta_t = omega P_t and score(t, s) = q^T R(omega (P_t - P_s)) k: the predicate is computed outside attention, exactly (a float32 cumsum), for every gap and every T. In a channel whose frequency times the move step is a multiple of 2 pi / N, the score is N-periodic in P, and wrap-around costs nothing. [] In an index-RoPE layer score(t, s) = q(h_t)^T R((t - s) Omega) k(h_s): position enters only through t - s, so the predicate "the moves in (s, t] sum to zero" must be written into the residual stream h.

E6.2 (Remark 1; cited). *Route A, a global count.* Layer 1: one head with constant query and key attends uniformly to the prefix; its value carries the move vector on move tokens, so its output is P_t / n_t. A feature proportional to 1/n_t is available in the same layer (for example a head with a fixed self-attention bonus). The MLP rescales to P_t and forms periodic features cos(2 pi k P_t / N), sin(2 pi k P_t / N); layer 2 matches on them. This is Liu et al.'s (2023) Lemma 6 followed by a match, and Yao et al.'s (2021) depth-2 Dyck construction (as summarised in Liu et al., App. A.5). In RASP terms the task has depth 2 in 1D and 2D alike. Its costs: a normaliser (Lemma 6 uses a padding token and an absolute channel log(2T - t); our torus and ring sequences have neither, Dyck has a BOS token); precision 1/n_t (1/256 at the end of a ring walk, 1/2048 per axis at the end of a torus walk); MLP width 4nT and weights up to 4nT + 2 in Lemma 6's mod gate (32 x 256

for the ring, 64 x 2048 per axis for the torus; our MLP has width 512); and sensitivity growing with the gap (Hahn and Rofin 2024, Cor. 5 and Thm. 6, Boolean inputs; used for direction only). *Route B, local return rules.* Retrace (while the current run reverses the previous one, the cell is the one seen 2j moves, 4j tokens, earlier: a bounded-window rule) and the one-lap clock (when the walk never reverses, the cell N moves back) need no count. A return that spans k direction runs is a function of those k runs; selecting its offset needs a summary of k runs, and stacking layers extends the reach. The reach is measured (App. F), not derived.

E7. Clock, map, the position-factor bound and the tick rule (proved)

E7.0 (phase decomposition). Over an interval with n_x moves x and n_y moves y, displacement d = n_y - n_x, the phase of channel c is a_c(y) n_y + a_c(x) n_x = m_c d + k_c n_x, exactly.

E7.1 (Proposition 2(a), a bound on the position factor). Let a revisit at displacement 0 have n backward moves; its phase in channel c is k_c n. For usage weights q_c (sum 1), the same-cell position factor F(n) = sum_c q_c cos(k_c n) equals 1 at n = 0; if a usage share s of the channels has kappa <= |k_c| <= K (unwrapped), then F(n) <= 1 - s (1 - cos(kappa n)) for n <= pi / K.

Proof. phi = m d + k n with d = 0. For n <= pi / K, kappa n <= |k_c| n <= pi, and cos is decreasing on [0, pi], so cos(k_c n) <= cos(kappa n) on the share s and <= 1 elsewhere. [] The bound covers n <= pi / K only. It says how fast a clock erodes the position factor of a revisit; it is not a theorem that a clocked code loses long-gap revisits, and near |k_c| n = pi it says nothing: two signed torus models started past pi solve with 0.43 and 0.99 of their usage on channels whose per-pair clock is near pi (App. H7). Index RoPE is the extreme case: every token, observations included, advances every channel, so the whole code is clock. Wrap-only revisits also need the lattice condition N m_c = 0 mod 2 pi.

E7.1' (the measured clock / map ratio). Every clock / map number in the paper (Table G3a, G3, H7) is rho_cm = sqrt(sum_{pairs} sum_c q_c wrap(k_c)^2 / sum_{pairs} sum_c q_c (wrap(a_c(x))^2 + wrap(a_c(y))^2) / 2), with the usage q normalised over the channels of both heads (one value per run), or over one head's channels (per head). The symmetric denominator makes it invariant to which move of a pair is called y (rule 8); for unwrapped |.| channels |k_c| >= |m_c| (E8.1), so rho_cm >= 1 there by construction. Generator: `scripts/sign/clock_table.py`.

E7.2 (Proposition 2(b), tick rule). Split each move's contribution in channel c into the map part and a tick tau_c, the phase every move adds whatever its direction. If the target at t is the key at the last visit to the same cell, matched pairs come at lags L drawn from a distribution, and with the map part cancelling over the closed loop the matched phase difference is L tau_c. A channel scores matched pairs consistently only if L tau_c is constant mod 2 pi over the lags that occur, i.e. tau_c = 0 mod 2 pi / g, g the greatest common divisor of the return lags (g = 2 on a bipartite lattice such as the even torus). Otherwise its matched score decorrelates once |wrap(tau_c)| sd(L) exceeds about 1 rad. [] A pi tick is therefore invisible to revisits on the even torus: it encodes the parity of the cell.

E8. Route inventory (Proposition 4; proved)

E8.1 (the |.| clock floor). In the |.| class, for an inverse pair in channel c (one omega_c of either sign): (a) a_c(x) and a_c(y) have the same sign, so |k_c| = |a_c(x)| + |a_c(y)| >= max(|m_c|, 2 |delta_c|), with |k_c| = |m_c| iff a_c(x) = 0 and |k_c| = 2 |delta_c| iff one of the two angles is 0. (b) As real numbers k_c = 0 iff the channel is blind, so the non-blind zero-clock points are exactly the wrap sheets |k_c| in 2 pi Z_{>=1}. (c) On a zero-displacement revisit with n = g/2 backward moves the phase is k_c g / 2 >= |m_c| g / 2; if |k_c| < 2 pi it is within eps of 0 mod 2 pi only for g <= 2 eps / |k_c| <= 2 eps / |m_c| (or g near a multiple of 4 pi / |k_c|).

Proof. (a) same sign of omega_c times non-negative numbers; triangle equality cases. (b) from (a). (c) from E7.0. [] An unwrapped |.| channel is at least as much clock as map, and the better it resolves neighbouring cells, the shorter the backtracks it can read. E4.1 is the geometric form of (b). *Check* (`sign/sign_clock.py`): zero violations of the inequality in 4,059 sub-pi |.| channels.

E8.2 (the collective route of the signed class). Signed class, inverse pair, readout W in R^{C x r} of full column rank, omega_c != 0. (a) The clock vector is k = diag(omega) W u with u = z_x + z_y. The latent pairs at which every channel has zero clock form the r-dimensional linear subspace {u = 0}; the map part (z_y - z_x)/2 is free on it, so it contains non-blind points. (b) Along z_y(tau) = (1 - tau) z_y - tau z_x (W, omega fixed), u(tau) = (1 - tau) u(0), and every channel's clock is (1 - tau) k_c(0). (c) For the quadratic surrogate R(z) = (1/2) sum_c q_c k_c^2, gradient flow on (z_x, z_y) gives du/dt = -2 H u with H = W^T diag(omega) diag(q) diag(omega) W, positive definite when the used channels' rows span R^r, and the map part is constant: u(t) = exp(-2 H t) u(0) -> 0 from every start, and {u = 0} is the only stationary set. (d) For almost every (W, omega), {u = 0} is the only latent set at which every channel cancels mod 2 pi when C > r.

Proof. (a)-(c) linear algebra: k_c = omega_c (w_c . z_x + w_c . z_y) and dR/dz_x = dR/dz_y = H u. (d) diag(omega) W u is in 2 pi Z^C iff (2 pi n_c / omega_c)_c lies in col(W); for fixed n != 0 the map omega -> (2 pi n_c / omega_c) is a diffeomorphism of (R \ 0)^C onto itself, so the preimage of the r-dimensional subspace col(W) is null when r < C; take the countable union over n. [] The surrogate guarantee concerns the clock term alone; with the content frozen, gradient descent on the step map rarely takes this route from the same-direction start (App. H7).

E8.3 (no collective route for |.| on an inverse pair). |.| class, C channels, rank r. (a) The latent route of E8.2(b) ends, under |.|, at a(y) = a(x) in every channel, since |w . (-z_x)| = |w . z_x|: a clock. (b) For almost every (W, omega), at that fixed (W, omega) no latent pair (z_x, z_y) makes more than r channels of each sign-coherence class (pre-|.| steps of equal or of opposite sign), hence more than 2r channels, cancel non-blindly. (c) Hence every |.| map code with more than 2r cancelling channels has its readout tuned channel by channel: in step units the channel-c condition is |w_c . z_x| + |w_c . z_y| = 2 pi n_c / |omega_c|, an offset that differs across the frequency ladder (omega_max / omega_min = 32 on the ring, 64 on the torus).

Proof. (a) evenness of |.|. (b) Fix a set S of 2r + 1 channels, a sign pattern per channel and non-zero integers n_c (a non-blind |.| channel cancels only by wrapping, E8.1(b)). On the sign region k_c = omega_c (e_c(x) w_c . z_x + e_c(y) w_c . z_y) is linear in zeta = (z_x,

z_y) in R^{2r}, k = diag(omega) N zeta, and the conditions k_c = 2 pi n_c for c in S hold for some zeta iff (2 pi n_c / omega_c)_{c in S} lies in col(N), a subspace of dimension <= 2r < 2r + 1 that does not depend on omega. As in E8.2(d) the set of omega_S for which it holds is null; Fubini and a countable union finish. Within one coherence class the sum depends on zeta only through z_x + z_y or z_x - z_y, an r-vector, giving the per-class bound. [] *Check* (`sign/prop_checks.py` (1)): at the default readouts of eight seeds, least-squares latents never cancel 9 channels exactly (smallest maximum error 0.027 rad); 8 channels cancel exactly (1e-14) only when they split 4 / 4 between the coherence classes.

E8.4 (same-direction moves). For a relation n_x a(x) + n_y a(y) = 0 (+1/-1: n = (1, 1); +1/+2: n = (-2, 1)): signed has one linear zero set; |.| with n_x n_y < 0 (no inverse pair) has two collective zero sets, {n_x z_x + n_y z_y = 0} and its mirror under z_y -> -z_y, both with non-blind points; with n_x n_y > 0 (an inverse pair) neither is non-blind. In the quadratic surrogate restricted to a sign region in which both coherence classes contain channels whose rows span R^r, the only latent stationary point is the blind origin, and the latent flow contracts toward it until some step reaches zero. *Proof.* As E8.2-E8.3; on the region the gradients are n_x (H_+ u_+ + H_- u_-) and n_y (H_+ u_+ - H_- u_-) with H_+- positive definite, vanishing iff z_x = z_y = 0. [] For +1/-1 every channel's residual is the clock k_c >= 0, so the |.| latent gradient that removes the clock is a contraction toward the wall, while the signed one projects onto u = 0 and leaves the map alone (`prop_checks.py` (2): signed ends at |u| / |z| < 1e-3 with the map part unchanged; |.| ends at the blind origin).

E8.5 (rank; the clock shares every channel at rank D). One head, latent dimension r, move latents in R^r; for a D-dimensional map write the map frame U = [d_1 ... d_D] in R^{r x D} (half-differences of inverse pairs), rank U = D, and the common clock c (the latent step shared by every move, plus the mean observation step). Channel i reads a phase per unit displacement k_i = U^T f_i in R^D (f_i its readout row in diag(omega) W_out) and a drift tau_i = per move. (a) r = D, U invertible: c = U beta for a unique phantom displacement beta (cells per move), and tau_i = in every channel. A channel is drift-free iff its wavevector is orthogonal to beta, i.e. iff it does not resolve the drift direction. So while c != 0 no channel is both drift-free and a full D-dimensional map, and the only way to a clean code is c = 0 (the collective route). (b) r >= D + 1: split c = c_par + c_perp with c_perp orthogonal to range(U). If c_perp != 0, the readout change delta = -tau_i c_perp / |c_perp|^2 cancels channel i's drift with U^T delta = 0 exactly: the map is untouched to all orders (*parking*). Independent clock directions can be parked only up to r - D of them: on the torus a single common clock needs r >= 3, two independent pair clocks r >= 4. (c) At r = D the minimum-norm cancelling readout change sets k_i <- k_i - (k_i . bhat) bhat: it removes the channel's resolution along the drift direction (*hiding*). (d) At D = 1 hiding sets k_i = 0 (the channel dies), and the drift a channel sees in cells, tau_i / k_i = beta, is the same for every channel and every readout: no readout change alters the drift-to-map ratio; only shrinking c or growing U does.

Proof. (a) c in R^D = range(U) when U is invertible; = . (b) U^T c_perp = 0 by definition; for q independent clock directions solve = -tau_j, a linear system with independent coefficient vectors. (c) = ; Cauchy-Schwarz; the minimiser has U^T delta parallel to beta. (d) D = 1: k_i = U f_i, tau_i = c f_i, ratio c / U. [] Per head, because each head's channels read only that head's r-dimensional image: a block-diagonal rank-4 latent gives each head

rank 2 and no c_perp. E8.5 and E8.2 use different clocks (the common step c, and the per-pair sums u); both vanish on a map code.

E8.6 (route inventory, stated once). For an inverse-pair map task, a class reaches the zero-clock set from a generic small start through the collective route iff its clock is a linear function of one latent vector whose zero set contains non-blind points, and through the per-channel route iff each channel has a clock-free, full-map readout reachable without crossing a fold. Then: signed, r >= D + 1: both (E8.2, E8.5(b)); signed, r = D: collective only (E8.5(a)); signed with frozen move latents: per-channel only (each channel can still cross zero through W_out); |.|: neither from a small start (E8.3: the latent route ends at a clock and no latent choice wraps more than 2r channels; E8.1 + E4.1: the per-channel zero-clock points lie at |k| = 2 pi n, across |k| = pi). A hybrid with readout |w_c| and signed latent, a_c(x) = omega_c |w_c| . z_x, has the signed class's collective route (E8.2 with W replaced by |W|) and its near-side cancelling point k = 0. |.| with one learnable offset per move, a_c(x) = omega_c |w_c . z_x + tau_x pi / |omega_c||, keeps E8.1's barrier (both angles keep one sign) and gains a collective route to the wrap: where every pre-|.| step is non-negative, k_c = omega_c w_c . u + pi sign(omega_c)(tau_x + tau_y), and u = 0, tau_x + tau_y = 2 cancels every channel at once for any readout (`SIGNSEP_PREREG.md` W1, W2; tested in App. H11).

E9. Hiding at rank D (Proposition 5; Conjecture 1)

Assumptions. (A1) Small drift: for the gaps that carry gradient, g |tau_i| << 1, so the drift loss is L_d = (1/2) sum_i a_i tau_i^2 with a_i > 0. (A2) The map loss depends on the frame only through the spectrum of the head's positional precision P = sum_i A_i k_i k_i^T, L_m = sum_j f(lambda_j(P)) with f decreasing and convex, plus a norm cost on readout rows. (A3) U near-isometric on the readout's span, so rotating k_i preserves |f_i|. Write r_beta = (bhat^T P bhat) / (tr P / D), the head's resolution along its own drift direction relative to its mean.

E9.1 (two routes). A change of parameters lowers tau_i only by changing c (shared by every channel of the head; the collective route) or by changing f_i with = -tau_i (the local route); E8.5 says what the local route costs.

E9.2 (Proposition 5; proved under A1-A3). (i) D >= 2, P isotropic: rotating any channel's k_i by a small angle eps from bhat toward a direction v orthogonal to bhat changes L_m by 0 at first order (a rotation leaves tr P unchanged) and changes L_d by -eps a_i |k_i|^2 |beta|^2 sin(2 phi_i) / 2 < 0 for every channel not already aligned with or orthogonal to bhat. So gradient flow on the readout moves resolution off the drift axis at once, for any beta != 0. (ii) D = 1: the only readout move that lowers L_d is shrinking |k_i|, whose first-order map cost is -f'(lambda) A_i > 0; it does not change beta (E8.5(d)). (iii) r >= D + 1 with c_perp != 0: parking lowers L_d at rate a_i |c_perp|^2 per channel with zero change in L_m at all orders. *Derivation*. k_i = |k_i| (cos phi bhat + sin phi v), tau_i = |k_i| |beta| cos phi, and d/deps (a_i / 2) tau_i^2 = -a_i |k_i|^2 |beta|^2 cos phi sin phi. []

E9.3 (Conjecture 1). Under A1 the gradient of L_d with respect to beta is P_a beta (P_a = sum_i a_i k_i k_i^T), so the collective route removes the drift along bhat at a rate proportional to bhat^T P_a bhat, the quantity hiding lowers. At rank D >= 2 this is a positive feedback (drift, then hiding, then slower removal, then persisting drift) that

leaves the drift in place or produces a long plateau; the clean state (beta = 0, P isotropic) remains a stable fixed point, which is the projection that the rank-2 existence test installs and training holds. Two refinements: long gaps carry the loss but not the gradient (once g |tau_i| > 1 the cosine averages out), so drift removal is driven by short gaps while the failure shows at long gaps and wrap-only revisits; and the CLOCK and COLLAPSE end states of rank-D heads (partial and complete hiding) are points on one curve. *Status.* The first-order statement is proved; the feedback is a conjecture. A CPU pilot at T = 128 (n = 3) saw a failing rank-2 seed whose weak axis was present at initialisation, and at initialisation on the real task the drift is too large for A1 (App. G6). A registered batch (App. G7) shows that on the torus removing the clock by construction lets rank 2 solve (8/8 against 1/8 with only the observation steps zeroed); that supports the premise that a remaining clock blocks rank D, and does not test the feedback.

E10. Exists, kept, not found (Remark 2; definitional)

Let Phi_B be the stochastic training map for budget B, S_eps = {theta : L(theta) <= eps} the solution set of a class and P_0 the initialisation distribution. Define the basin Bas = {theta : P(Phi_B(theta) in S_eps) >= 1/2}. *Exists:* S_eps is non-empty in the class (E2, constructed). *Kept:* the constructed points lie in Bas (ring 8/8, torus re-found 8/8, rank-2 projection 7/8). *Not found:* P_0(Bas) is small (SOLVED fractions 0-2/8 estimate it). Together they place the obstruction in the position of supp(P_0) relative to Bas, not in the emptiness or instability of S_eps. Two measurements of that position are in App. H7: the loss along straight paths from the start to a constructed solution, and the basin's extent along such a segment.

E11. Readers and codes (Proposition 6; proved)

E11.1 (one layer is a position-only reader). In a one-layer MapWM the pre-softmax score between the query at i and the key at j, head h, is s_h(i, j) = sum_c |q_c(x_i)| | k_c(x_j)| cos(theta_c(i) - theta_c(j) + psi_c(x_i, x_j)) / sqrt(d_head), where q_c, k_c are 2D blocks of W_q LN(e(x_i)), W_k LN(e(x_j)) and psi_c the angle between them. Both depend only on the token identities; position enters only through theta(i) - theta(j); the values depend on identity alone. So for scored revisit queries a one-layer model is a channel-weighted cosine kernel with token-type-dependent weights and offsets, a soft vote over the observations it lands on, and an MLP on the vote.

E11.2 (extra passes widen the reader, not the code). `MapFormerWM.forward` computes delta = action_to_lie(token_emb(x)) and cos, sin = path_integrator(delta) once and hands them to every layer; `MapFormerWM_Looped.forward` does the same and applies one block n_loops times. From the second pass on, q and k depend on the residual stream, so the weights |q_c||k_c| and offsets psi_c become context-dependent and values can carry retrieved content; theta is unchanged. A loop with the same parameters and initialisation is therefore the same code family with a larger reader function class. []

E11.3 (a score rule acts on the readout term). Any score of the form sum_c A_c cos(dphi_c + beta_c), with A_c >= 0, sees position only through dphi. Write the target coherence (the A-weighted cos(dphi + beta) on the target key) as C = C0 + (C - C0), with C0 the A-weighted cos(dphi) on the target, the code term. A score rule changes A and beta, so it changes C - C0 and can reweight channels within C0, but cannot change dphi. []

E12. Peak pinning (Proposition 7; proved given delta = 0)

RoPE / MapWM score: s_ij = sum_c Re(q_c conj(k_c) e^{i dphi_c}), so A_c = |q_c||k_c| and beta_c = arg q_c - arg k_c, both content-dependent. PoPE / MapPoPE (`model_pope.py`, cos(dphi - delta)): A_c = softplus(q_c) softplus(k_c), beta_c = -delta_c, content-free. If beta_c = 0 for all c, then for any content pair s_ij <= sum_c A_c, with equality iff dphi_c = 0 mod 2 pi on every channel with A_c > 0; as a function of displacement the kernel is a non-negative combination of cosines, positive-definite by Bochner, so content can reweight the frequencies but cannot move or invert the peak. *Corollary (index phase).* With PoPE's frequencies theta_c = 10000^(-c/64) <= 1 rad per token, each cos(theta_c D) decreases on D in [0, pi], so with beta = 0 every content pair's kernel strictly decreases over offsets 0-3: a one-layer pinned-peak index head cannot prefer the observation 3 tokens back to the one 1 token back, while the one-layer torus solution needs peaks at the retrace offsets 4j - 1. *Assumption check:* delta is in the clamp's dead zone on 98-100% of entries in every PoPE run (App. L).

E13. The used-channel ratio (Remark 3; light theory)

With token frequencies f_w, split every step into a common part and a content part: Delta_c(w) = mu_c + d_c(w), mu_c = sum_w f_w Delta_c(w), so sum_w f_w d_c(w) = 0. Over an interval of L tokens, Phi_c(t) - Phi_c(t') = mu_c L + (sum of d_c over the interval). The first term is a clock (Case I if carried by every token, Case II if only by selected tokens); the second can be a map (Case III) only if it cancels over closed loops. Define rho_c = |mu_c| L / range(content_c, L) at the training length L. E7.2 constrains mu_c only in channels the readout uses. Prediction: in used channels rho_c << 1 when targets are keyed to a latent state reached at variable lags, and rho_c >> 1 when they are keyed to a token or event count. The pooled growth exponent of the summed step weights channels by their raw range, so a large tick in unused channels dominates it; the used-channel median of rho does not have this problem.

E14. Status

#	statement	status	check
E1	rotary position factor = Howard's propagator with s = i omega	proved (identity: Mamba-3 Prop. 3)	`a0_propagator.py`
E2 (a), (f), (g)	exact non-negative (	.	, softplus, exp) representation mod 2 pi with <= r stepping tokens
E3	relative-sign gauge; excluded codes are opposite-sign rows	proved	`t1_gauge.py`
E4	cut circle; sticky wall along s	proved	`p3_p5_geometry.py`
E5	CARoPE: every channel a clock or blind	proved	`p6_carope.py`
E6.1	prefix sum in the phase; one layer at any gap	proved	same-cell attention (App. F2)
E6.2			probes (App. F3)

#	statement	status	check
	two-layer global count (normaliser, precision, width)	cited (Liu et al. Lemma 6; Yao et al.)	
E7	phase decomposition; position-factor bound; tick rule	proved	bound consistent with the strata, not a test of them; text-world drift (App. J)
E8.1-E8.4		.	clock floor; signed collective route; no
E8.5	rank D: no clock-free full-map channel; rank > D: parking	proved	end states (App. G3-G4); that parking is why rank > D helps: conjecture, registered test PARTIAL (App. G7)
E8.6	route inventory	proved (inventory)	Table 4; SIGNROUTE; SIGNSEP (App. H11: route class fails, 2x2 unmeasured); exp from its past-pi start learns in the SSM at 1e-3 by wrapping (App. H13), not in attention on the torus (App. H14)
E9.2	hiding free to first order at D >= 2	proved under A1-A3	drift on the least-resolved axis (App. G)
E9.3	hiding starves the collective route	conjecture	feedback not tested; its premise, that a remaining clock blocks rank D, has registered support on the torus (App. G7)
E10	basins	definitional	App. H7
E11	readers and codes	proved (architecture, algebra)	one-pass reader (App. M)
E12	peak pinning	proved given delta = 0 (measured)	DELTA_FREE (App. L)
E13	used-channel ratio	light theory	App. J4

Statements of the theory appendix v3 not used here: P1(c)-(e), P2 (rescaling sensitivity; its drift account is withdrawn), P5 (the pi-mirror) and P7 (the CoPE lattice); they are in `docs/paper/APPENDIX_theory_v3.md` with their check scripts.

F. Exchange-rate details

Dyck-2 (`DYCK_MDEPTH_RESULTS.md`). Task and metric as MapFormer v4 App. B.4, trained and scored at length L = 32 and nesting depth D = 12; primary metric A2f, Hewitt's distance-averaged closing accuracy over prefixes where a closer can be emitted (CE-optimal predictor 1.000, chance 0.500, best stack-free floor 0.594). 2 heads, d_model 128, 1-4 layers; path-integrated arms MapWM and MapPoPE at rank 2; index arms RoPE and PoPE at base 10000 (and base 32 in the 4-layer 3x cell); n = 8 paired seeds. Position main effect = ((MapWM - RoPE) + (MapPoPE - PoPE)) / 2.

condition	layers	RoPE	PoPE	MapWM	MapPoPE	position main effect
1x budget	1	0.648	0.640	0.996	0.998	+0.353
1x budget	2	0.889	0.850	1.000	1.000	+0.130
1x budget	3	0.964	0.946	1.000	1.000	+0.045

condition	layers	RoPE	PoPE	MapWM	MapPoPE	position main effect
1x budget	4	0.982	0.971	1.000	1.000	+0.024
3x budget	4	0.998	0.997	1.000	1.000	+0.002
mixture of depths 4-12	4	0.898	0.881	0.999	0.999	+0.110

Every effect is positive on 8/8 seeds (sign-flip p 0.0078). Index base 32 minus base 10000 at four layers, 3x budget: +0.001 (RoPE), -0.001 (PoPE), within the MDE. Three times the budget raises the index arms by +0.021 at four layers (8/8).

Ring (C3). The index-one-layer accuracies are not monotone in p_plus (0.710, 0.677, 0.824, 1.000 on fresh seeds; every pair differs, p 0.0022-0.0108), so the exchange rate, not the one-layer accuracy, is the quantity that tracks whether steps are undone.

Torus (`FACTORIAL_RESULTS.md`, Fig. F1). Strata (short gap / long gap / wrap-only; mean and run range, n = 4): path one layer 1.000 / 1.000 / 1.000; index one layer 0.769 (0.735-0.787) / 0.480 (0.456-0.508) / 0.482 (0.470-0.495); four layers 0.943 (0.900-0.958) / 0.458 (0.367-0.506) / 0.457 (0.381-0.494); looped x4 0.938 (0.937-0.940) / 0.507 (0.496-0.531) / 0.509 (0.496-0.521). Floors on this stream (analysis): constant 0.508 / 0.515 / 0.517, retrace 0.912 on short gaps. Memorisation check: four index layers 0.897 on their own training map against 0.860 held out; looped 0.862 against 0.864; path 1.000 on both. The fresh-seed replication and the retrace-floor comparison are in F7.

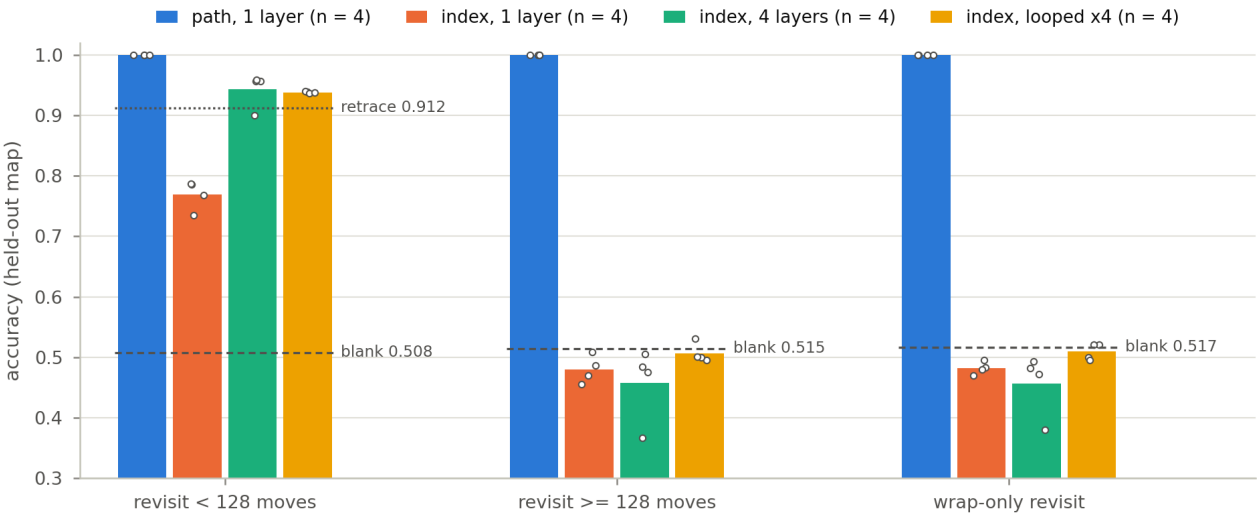


Figure F1 (`fig_factorial_strata`). Revisit accuracy by stratum on the torus at T = 1024 (trained and tested at T = 1024; bars: means of n = 4 runs, dots: runs). Dashed: always-blank within each stratum, 0.508 / 0.515 / 0.517; dotted: retrace within the short-gap stratum, 0.912 (elsewhere it equals the constant); floors on this batch's own evaluation walks. Path integration with one layer is at 1.000 in every stratum. Four index layers or a loop of four lift short-gap revisits to 0.94, just above retrace; at long gaps and wrap-only revisits every index model stays at or below the constant (0.46-0.51).

F1. Which revisits depth buys on the torus (analysis; `scripts/depth/torus_depth.py`, `summarize_torus.py` -> `out/torus_depth_out.txt`, `summarize_torus.py --segments` -> `out/torus_segments_out.txt`)

The FACTORIAL evaluation stream was regenerated (env seed 10000, np seed 1234 + s, 100 walks); overall accuracy per run equals the stored strata files' "all" to 4 decimals for all 16 runs, and the re-implemented forward equals the model's logits (max |diff| 0.0). Per revisit: s = the last visit to the cell; *retrace* = the current run reverses the previous one and has not overshot it; *runs* = maximal single-action runs among moves s+1..t. Means of per-seed accuracies (n = 4; range in brackets).

revisit class	share	index 1L	index 4L	loop x4	path 1L
short gap, retrace	0.680	0.833 (0.79-0.86)	0.996 (0.99-1.00)	1.000	1.000
short gap, not retrace	0.148	0.475	0.699 (0.51-0.77)	0.655	1.000
... 3-4 runs	0.031	0.477	0.926 (0.74-0.99)	0.977	1.000
... 5-8 runs	0.056	0.476	0.735 (0.52-0.81)	0.645	1.000
... 9-16 runs	0.052	0.474	0.561 (0.38-0.64)	0.499	1.000
... 17+ runs	0.009	0.472	0.493	0.502	1.000
long gap (>= 128)	0.090	0.480	0.458	0.507	1.000
wrap-only	0.083	0.482	0.457	0.509	1.000

Gap in moves (plain revisits): index 4L / loop 1.000 / 1.000 at 2-7, 0.978 / 0.986 at 8-31, 0.614 / 0.551 at 32-127, 0.458 / 0.506 at 128-511; one layer 0.935 at 2-7, 0.681 at 8-31, floor beyond. Runs at a fixed gap of 32-127 moves (plain revisits, retraces included; mean of per-seed accuracies, n pooled over the 4 seeds; the 2-run cells are all retraces of merged runs; pooled over seeds instead, the small cells read 4L 0.896 at 2 runs and 0.785 at 3-4):

runs spanned	n	index 1L	index 4L	loop x4
2	240	0.412	0.874	0.996
3-4	503	0.483	0.832	0.911
5-8	4565	0.464	0.688	0.575
9-16	6569	0.474	0.560	0.499
17+	1202	0.472	0.493	0.502

At gaps of 8-31 moves: 4L 0.993 / 0.940 / 0.818, loop 1.000 / 0.987 / 0.768 at 2 / 3-4 / 5-8 runs. Long-gap revisits span 17 or more runs in 98.5% of cases (median 37, minimum 11), wrap-only revisits in 99% (median 65). The deep models score below the constant floor there with high loss (2.1-4.8 nats): they predict a specific wrong object rather than abstaining to blank. *Decomposition* (an identity, since short = retrace + not retrace): the short-gap retrace floor is (0.680 x 1 + 0.148 x ~0.5) / 0.828 = 0.911 (measured 0.912); four index layers give (0.680 x 0.996 + 0.148 x 0.699) / 0.828 = 0.943. The class contrasts at n = 4 (e.g. 4L against the loop at 3-4 runs; 4L range 0.74-0.99) carry no test.

F2. Where index attention looks (analysis; same script)

Attention at the revisit query (token a_t), final pass, per head; *same* = mass on earlier observation tokens of the same cell (uniform chance 0.001-0.003); *last* = mass on the last visit's observation and arrival tokens.

model, final pass	retrace: same / last	short, not retrace: same	long gap: same	wrap-only: same
index 1L	0.05 / 0.10 (h0), 0.06 / 0.12 (h1)	0.00	0.00	0.00
index 4L, layer 4	0.55 / 0.56, 0.58 / 0.58	0.22 / 0.22	0.01 / 0.01	0.00 / 0.00
loop x4, pass 4	0.50 / 0.52, 0.49 / 0.50	0.20 / 0.20	0.00 / 0.00	0.00 / 0.00
path 1L	0.83 / 0.70, 0.76 / 0.65	0.82 / 0.76	0.81 / 0.74	0.66 / 0.58

In the deep index models retrieval happens in the last pass only, and on retraces all of it is on the last visit: they find the mirror point, not the cell.

F3. No global count on the torus (analysis; `scripts/depth/torus_probe.py` -> `out/torus_probe_out.txt`)

Ridge probes from the residual stream at move tokens (train walks 0-29, test 30-39, t >= 16; out-of-sample R^2, mean of 4 seeds, x and y averaged; best pass):

target	index 1L	index 4L	loop x4	ring index 3L, p_plus 0.5 (t >= 8; see F5)
unwrapped displacement P_t	-0.06	0.10	-0.01	0.60
normalised count P_t / (t+1)	-0.03	0.21	0.06	0.69
1 / (t+1)	-0.01	0.78	0.74	0.89
displacement over the last 4 moves	0.90	0.96	0.95	0.97
... last 16	0.47	0.93	0.86	0.86
... last 64	0.09	0.77	0.56	0.70
... last 256	-0.01	0.37	0.14	--

The deep torus models know elapsed time (1/t) but carry no normalised count; they carry a recent-displacement code whose error grows with the window (0.5 cells at 4 moves, 1.8 at 16, 7.1 at 64, 25 at 256 for 4L; the d_w targets at t < w are the displacement since the start, so the rmse at w = 256 is approximate). Linear probes bound only what is linearly readable; the attention readout of F2 does not rely on them. The ring column's count-like R^2 is a window code (F5).

F4. Run length moved at evaluation (analysis, distribution-shifted; `scripts/depth/torus_runlen.py`)

The stored FACTORIAL models re-evaluated on walks whose run length is fixed at 2, 5 or 10 moves. Two-run returns are gap-invariant for deep index models (retraces at gaps of 16-63 moves with runs of 10: 0.978 / 0.995 for 4L / loop). Down a column of fixed gap

(16-63 moves) 4L falls from 0.98 (2 runs) to 0.87-0.93 (3-4), 0.72-0.80 (5-8), 0.64-0.70 (9-16) and 0.58 (17+); within a run bin across gaps it moves by 0.02-0.2. Runs set the level and the gap trims it. Path 1.000 in every cell of every condition.

F5. The ring: rules, runs and a count (analysis; `scripts/depth/ring_depth.py`, `summarize_ring.py`)

CANCEL evaluation stream (overall accuracy equals each run's stored value for all 128 runs); fresh seeds 2-7. Classes: *retrace*; *lap* = the displacement over the last 32 moves is 0 mod 32 (a fixed one-lap offset lands on the cell); *neither*, split by runs. p_plus 0.5, mean (range):

class	share	index 1L	index 2L	index 3L	path 1L
retrace	0.716	0.797	0.987	1.000	1.000
lap, not retrace	0.035	0.497	0.977	1.000	1.000
neither, 3-4 runs (wrap-only included)	0.127	0.487	0.891 (0.72-0.94)	0.997	1.000
neither, 5-8	0.097	0.489	0.802 (0.55-0.87)	0.993	1.000
neither, 9-16	0.025	0.485	0.678 (0.48-0.75)	0.972	1.000
wrap-only	0.118	0.503	0.819	0.989	1.000

The "neither" rows include wrap-only revisits (29% / 47% / 56% of the three bins); on plain revisits only, 2L scores 0.910 / 0.828 / 0.742 and 3L 0.999 / 0.996 / 0.988 (`ring_plain_bins.py`; the conclusions, 2L decays and 3L is flat, are the same). p_plus 0.75 / 0.9 (index 1L): retrace 0.692 / 0.677, lap 0.987 / 0.994, neither 0.487 / 0.592, against best-constant floors of 0.493 / 0.491 / 0.502 for the "neither" class at p_plus 0.5 / 0.75 / 0.9 (`ring_class_floors.py`): at 0.9 one layer sits an unexplained +0.09 above it. At p_plus 1.0 every revisit is a lap revisit, 1.000 at every depth. *No global count on the ring either* (`ring_probe_window.py` -> `out/ring_probe_window_out.txt`). Probes read at t >= 8 on 128-move walks (P_t R^2 0.60, P_t / t 0.69, 1/t 0.89 for three layers; one layer 0.09 / 0.18 / 0.14; two 0.40 / 0.60 / 0.67) cannot separate a prefix count from a recent-window code, since half of P_t / t's variance sits at t <= 32 and for t < 64 the 64-move window equals P_t. Read late, three layers give P_t R^2 0.59 at t >= 64 (two layers 0.32), and at t >= 96 the displacement before the last 64 moves has R^2 0.01 (before the last 32: 0.31), while the windows are well coded (last 64 moves 0.72); the true 64-move displacement alone predicts P_t there with R^2 0.51, against the model's 0.53. Three ring layers carry a displacement code reaching 32-64 moves back, which on 128-move walks covers most revisits; their last-layer attention reaches the cell on wrap-only revisits (same-cell mass 0.36 / 0.38). *One-layer accuracy across p_plus (descriptive):* a p-invariant mixture does not reproduce it: per-class accuracies pooled over p predict 0.678 / 0.683 / 0.749 against the observed 0.710 / 0.677 / 0.824 at p_plus 0.5 / 0.75 / 0.9, because the class accuracies themselves move with p (retrace 0.797 -> 0.692 from 0.5 to 0.75, as large as the net dip of 0.033). No mechanism is claimed for the dip at 0.75.

F6. Dyck-2: blocks to skip (analysis; `scripts/depth/dyck_distance.py`, `dyck_crosstab.py`)

Closing accuracy by the number of complete balanced blocks between the top opener and the prefix (mean of 8 seeds):

blocks to skip (n prefixes)	RoPE 1L	2L	3L	4L	MapWM 1L
0 (3682)	0.998	0.999	0.999	0.999	0.999
1 (5839)	0.640	0.925	0.984	0.991	0.997
2 (572)	0.641	0.778	0.885	0.934	0.993
3-4 (161)	0.613	0.760	0.860	0.925	0.993

Dyck has a BOS token, so the count route's normaliser is available, and both blocks and distance matter for the index models; the path model is flat in both (0.987-0.998). 93% of scored prefixes need at most one block, which is why three index layers already reach 0.964 in aggregate.

F7. Fresh-seed replication of the torus factorial (FACTORIAL_N8; registered, `FACTORIAL_N8_RESULTS.md`, `FACTORIAL_N8_PREREG.md` + Amendment 1)

FACTORIAL repeated exactly (cells, recipe and flags asserted per run against FACTORIAL's checkpoints, evaluation and decision functions imported unchanged) on fresh seeds 2800-2803, scored on new evaluation walks; kalman GPU 1, 2026-10-24/25, code `ab872fb`; independent code check before launch. Registered primary = the fresh seeds alone (n = 4 per cell; minimum p 0.029); pooling with FACTORIAL is cross-batch and descriptive (two guarded training modules changed additively since FACTORIAL's commit).

cell	acc T = 1024 (fresh)	SOLVED	acc - retrace, same walks (seeds above)	short gap	long gap	wrap-only	own map	pooled n = 8 (SOLVED)
path, 1 layer	0.999	4/4	+0.153 (4/4)	1.000	0.997	0.987	0.999	0.999 (8/8)
index, looped x4	0.859	0/4	+0.014 (3/4)	0.931	0.494	0.493	0.869	0.862 (0/8)
index, 4 layers	0.830	0/4	**-0.016 (1/4)**	0.913	0.413	0.407	0.919	0.845 (0/8)
index, 1 layer	0.729	0/4	-0.117 (0/4)	0.779	0.474	0.476	0.790	0.724 (0/8)
floors (these walks)	blank 0.511, retrace 0.846			retrace 0.911	0.514	0.514		

Registered (fresh alone): C0 PATH WINS AT ONE LAYER (+0.270); C1 and C2 INDEX DOES NOT CATCH UP (+0.169 and +0.140; 4/4 vs 0/4, Fisher p 0.029; one 4-layer run still descending); C3 both HELP (+0.101, +0.130); replication summary REPLICATES. Declared secondaries: on FACTORIAL's own walks the deep cells sat at +0.015 (4 layers, 3/4 above) and +0.020 (loop, 4/4) over retrace, so over both batches four index layers are at the retrace floor (-0.000, 4/8 above) and the loop +0.017 (7/8). Batch shift per cell <= 0.029 (perm p >= 0.17). Four layers learn more of their own map (0.919) than they transfer (0.830). Out of distribution, T = 512 / 2048: path 1.000 / 0.968; 4 layers 0.881 / 0.573; loop 0.900 / 0.632; 1 layer 0.759 / 0.545. r(final loss, acc) -0.909 over the 16 runs. Predictions held except "no 4-layer run descending" (1/4).

G. Rank details

G1. Arms, curves and dimension

Arms (A1). Shared r2 (`Vanilla`), per-head r2 (`Vanilla_r2ph`), block-diagonal r4 (`Vanilla_r4mibd`: the shared rank-4 latent with W_out's off-diagonal blocks held at zero), per-head r3 (`Vanilla_r3ph`), shared r4 (`Vanilla_r4mi`), per-head r4 (`Vanilla_r4ph`), all built from the shared rank-2 model's initial weights at each seed. Initial angle-increment scale (std): 0.208 for block-diagonal r4 against 0.33-0.34 for the other arms, which fail (rank 2) or solve (rank 4) from the same scale. Per-seed results: C4.

Training curves (Fig. G1; `fig_loss_curves_rank_out.txt`). Shared r2 0/8 SOLVED (4 STALLED, 4 DESCENDING; median final loss 0.35) against shared r4 8/8 (2.8e-3); per-head r2 2/8 (median 0.22), r3 6/8 (3.5e-3), r4 8/8 (3.1e-3). The rank-3 and rank-4 medians cross the threshold at epochs 478-532; the rank-2 medians never do.

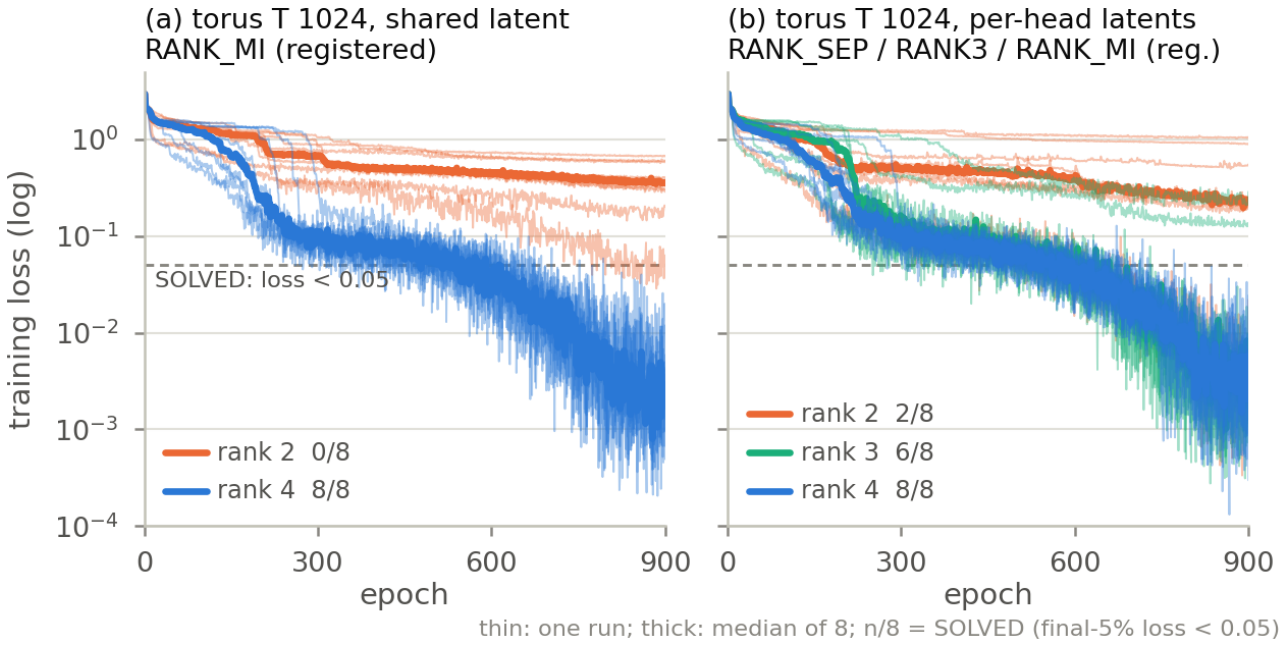


Figure G1 (`fig_loss_curves_rank`). Training loss per epoch on the torus (T = 1024) by the per-head rank of the content-to-angle map; thin lines are runs, thick lines medians of 8; dashed: the SOLVED threshold. (a) One latent shared by the heads: rank 2 solves 0/8, rank 4 8/8. (b) Per-head latents: rank 2 2/8, rank 3 6/8, rank 4 8/8.

Restart curves (`RANK_PROJ_ANALYSIS.txt`). Projected rank 2: peak training loss 0.37-0.46 after the restart, back below 0.05 at epochs 553-654 on 7/8 seeds. Rank-4 control: peak 0.26-0.30, back at 497-669 on 7/8. Rank 2 continued: peak 0.55-0.82, back on 1/8.

Rank against dimension (`RANK_ND_RESULTS.md`, `RANK_WRAP_RESULTS.md`, `RANK_TESTS_RESULTS.md`). Per-head rank, every arm from the same base, trained and tested at T = 1024, 900 epochs:

torus	D	per-head rank	wrap-only share of revisits	SOLVED	held-out accuracy
32 per side	2	2	0.37	1/8	0.769
32 per side	2	3	0.37	8/8	0.999
10 per side	3	3	0.71	1/8	0.800
10 per side	3	4	0.71	4/8	0.847
18 per side	3	4	0.35	8/8	1.000

2D rank 3 vs rank 2: 8/8 vs 1/8 (Fisher p 0.0014), +0.230 (permutation p 0.0014). 3D, 18 vs 10 per side at rank 4: +0.153 (p 0.0085). On a 100-cell 2D torus (10 per side) rank 3 memorises its training map (0.986 on it, 0.273 on an unseen map), so that cell does not measure map learning. 1D ring (32 cells, T = 128, 300 epochs): rank 1 7/8 (0.993), rank 2 8/8 (1.000); with the component common to the move steps removed by construction, rank 1 solves 8/8 in a median of 24 epochs (58 without).

Escape test (`RANK_TESTS_RESULTS.md`). Lifting the eight stalled shared-rank-2 runs to rank 4 (function-preserving) and training 600 epochs solves 3/8, against 1/8 for rank 2 continued and 2/8 for a fresh rank-4 step map on the same networks; at this budget the contrasts are not resolved.

G2. PoPE's score does not rescue rank 2 (`SCORE_RANK_RESULTS.md`, registered NO RESCUE)

Torus at T = 1024, 900 epochs, 32 angles per head in every arm; rank-2 arms on seeds 10-21, rank-4 arms 10-17:

arm	accuracy	SOLVED
MapWM r2	0.848 +/- 0.178	3/12
PoPE score r2	0.946 +/- 0.040	3/12
MapWM r4	0.9998	8/8
PoPE score r4	0.9995	8/8

Primary: 3/12 vs 3/12 (Fisher p 1.00); the rank-2 deficit replicates on these fresh seeds under both scores (8/8 vs 3/12, p 0.0014 each). The stratum readouts are secondaries (short gaps +0.115, p 0.036; gap >= 128 +0.043; wrap-only -0.015).

G3. The clock at the end of training, per head (analysis; `scripts/sign/clock_table.py -> clock_table_out.txt`; latent column `scripts/sign/rank_clock.py -> rank_clock_out.txt`)

Both axes (moved from main-text Sec. 7.1, 2026-10-26). Training from a small start ends in one of two places on either axis. The models that succeed end with clock-free heads: the signed and rank-3/4 arms at arm means 0.001-0.051, with no head above 0.14, and every solved rank-2 run with at least one clean head (0.03-0.07; two of the four keep a drifting second head, 0.54 and 0.68). The models that fail end with a clock of the order of their map: the non-negative arms at 1.25-1.57 (arm means; for an unwrapped |.| channel a ratio of at least 1 holds by construction, so the informative part is that the channels did not wrap) and the unsolved rank-2 runs at 0.54-0.69 (medians 0.58-0.79). Four of the 20 unsolved rank-2 runs end with clock-free heads whose two map axes have collapsed onto

one direction, and one more sits just above the SOLVED threshold (final loss 0.051). Weighted by usage, signed steps cancel opposite moves mod 2 pi in 70-93% of channels and |.| steps in 2-10% (H6).

Table G3a. Clock at the end of training (analysis; clock / map ratio of Sec. 4 and E7.1', one value per run, arm means unless marked; SOLVED of 8). For unwrapped |.| channels the ratio is at least 1 by construction; the informative rows are the frozen-latent signed arm and the rank arms.

arm	SOLVED	clock / map
ring, signed (three batches)	8/8, 7/8, 8/8	0.015-0.018
ring, |.| from small starts (three batches)	0/8, 0/8, 1/8	1.47-1.57
ring, signed with move latents frozen, |.| start	1/8	1.18 (the solved run 0.06)
ring, |.| from a start past pi	8/8	0.32 (wrapped; 0.82 of usage on cancelling channels)
torus, signed / |.|, observation-step-free	8/8 / 1/8	0.001 / 1.25
torus, rank 2, unsolved runs (three arms)	0/8, 2/8, 2/8	0.54-0.69 (medians 0.58-0.79); 4 of 20 runs collapsed, clock-free
torus, rank 2, solved runs (4): six clean heads / two drifting heads (per head)	4 runs	0.03-0.07 / 0.54, 0.68
torus, rank 3 and rank 4, solved runs	6/8, 8/8, 8/8	0.043-0.051

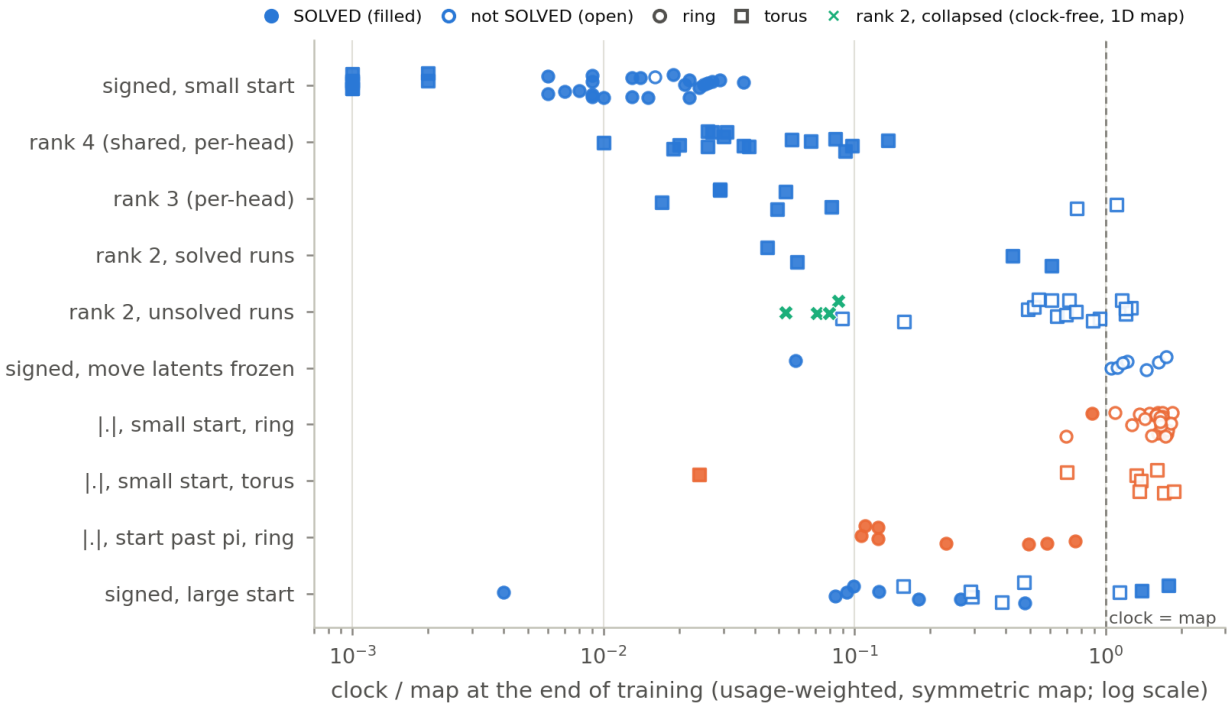


Figure G2 (`fig_clock_endstate`; moved from the main text; analysis of registered batches). Clock / map ratio at the end of training, one dot per run (circles: ring; squares: torus; filled: SOLVED; log scale), on the one definition of Table G3a. Successes from small starts are clock-free: signed 0.001-0.036 (31 of 32 runs SOLVED; the unsolved one at 0.016),

rank 4 0.010-0.136 (16/16), solved rank 3 0.017-0.081. Failures carry a clock of the order of the map: |.| from small starts 0.70-1.85 on the ring (1 of 24 SOLVED, at 0.88) and 0.70-1.87 for the unsolved torus runs (the solved one at 0.024), signed with move latents frozen 1.055-1.741 (the solved run 0.058), unsolved rank 3 0.767-1.105 and unsolved rank 2 up to 1.26 (median 0.62). Four unsolved rank-2 runs are clock-free (0.053-0.086) with their two map axes collapsed onto one direction (crosses). The solved rank-2 runs span 0.045-0.611 (two keep a drifting second head). Starts past pi cancel by wrapping and solve at larger ratios: |.| on the ring 0.11-0.76 (8/8); signed 0.004-1.78 (10 of 16, ring and torus), the two solved torus runs at 1.39 and 1.78 with 0.43 and 0.99 of their usage on channels whose per-pair clock is near pi.

Per head.

Clock / map ratio of Sec. 4 (E7.1') per run (usage pooled over both heads) and per head (usage normalised within the head); full models (observations step). The latent clock (|z_x + z_y| over the map frame, raw latent coordinates, mean over unsolved runs' heads) is a different quantity and not invariant to the GL(r) gauge of the latent (in the W_out image the unsolved values become 1.4-2.6).

arm	**SOLVED**	**solved runs: run mean (heads median, range)**	**unsolved runs: run mean, median (heads median, range)**	**latent clock, unsolved**
shared r2	0/8	--	0.595, 0.579 (0.619, 0.07-1.26)	1.72
per-head r2	2/8	0.236 (0.045, 0.03-0.54)	0.694, 0.792 (0.756, 0.02-1.98)	3.19
block-diagonal r4 (rank 2 per head)	2/8	0.335 (0.067, 0.04-0.68)	0.541, 0.575 (0.084, 0.01-1.41)	1.63
per-head r3	6/8	0.043 (0.038, 0.02-0.08)	0.936 (0.678, 0.12-1.25)	3.68
shared r4	8/8	0.049 (0.036, 0.01-0.10)	--	--
per-head r4	8/8	0.051 (0.030, 0.02-0.14)	--	--

The solved rank-2 run means hide a split: in each of the four solved rank-2 runs at least one head is clean (per-head r2 s1 h1 0.03, s2 0.05 / 0.05; block-diagonal s2 0.04 / 0.06, s6 h0 0.07), and two runs keep a drifting second head (per-head s1 h0 0.54, block-diagonal s6 h1 0.68). Not every failure is clock-coupled: 5 of the 20 unsolved rank-2 runs have both heads below 0.15 (shared s1; per-head s5, s6; block-diagonal s1, s7), and 4 of them (all but shared s1) are in the collapsed class of `rank/sweep_final_out.txt`: visible drift <= 0.01 per move with the two map axes near-collinear (frame conditioning 0.00-0.03), a map that has lost a dimension; shared s1 ends just above the SOLVED threshold (final loss 0.051). Heads below 0.15 among unsolved runs: shared 4/16, per-head 4/12, block-diagonal 7/12. The rank-3 unsolved runs are clock-coupled. The observation clock (mean observation step over the map; `rank_clock.py`) is 0.015-0.018 in solved rank-3/4 heads and 0.18-0.52 in unsolved heads. At r = 2 = D, rank_clock's "in-span" share of the clock is 1 by construction before wrapping, so it is not reported as a measurement.

G4. The leftover latent clock and hiding (analysis; `scripts/rank/`; a latent quantity, not the ratio of G3)

Here the clock is c = a + o, the mean move latent a plus the mean observation latent o (the drift a walk accumulates per move and observation), measured in the native latent of each head's read subspace against the map frame U (lam = |c| / |U|); "off-plane share" = |c_perp| / |c|; "readgain" = how strongly the readout sees the leftover clock relative to the map (1 = read like the map, 0 = invisible). This latent metric and G3's channel ratio agree that what is still read in solved heads is small; they are not numerically comparable. The native latent metric is not invariant to the GL(r) gauge of the latent; the independent audit found the balanced gauge gives the same picture (solved-set medians 0.94-1.00).

What the leftover clock is made of (`clock_decomp.py` -> `clock_decomp_out.txt`; medians over solved heads): a and o are each 0.017-0.037 of a map step and anti-parallel (cosine -0.79 to -0.999), so they cancel; at T = 1024 they lie mostly in the map plane (off-plane share 0.05-0.35 for a, 0.02-0.08 for o). Only their small residual c is off-plane. SEARCH's per-pair clocks z(-e_j) + z(+e_j), which leave the observation step out, are 0.034-0.057 of a map step and in-plane (off-plane 0.14-0.21 on the 2D sets, below the random-direction median); with the two observation steps of one out-and-back added, 0.73-0.99 off-plane. The per-pair clock therefore overstates the drift a walk accumulates; the two definitions agree on what is read.

set (solved heads)	|a| / |U|	|o| / |U|	cos(a, o)	off-plane share of a / o / c
torus G64 per-head r3	0.021	0.020	-0.988	0.15 / 0.07 / 0.98
torus G64 shared r4 (RANK_MI)	0.017	0.017	-0.980	0.19 / 0.08 / 1.00
3D G18 r4	0.021	0.022	-0.999	0.05 / 0.06 / 0.81
torus T = 128 r4	0.034	0.029	-0.897	0.48 / 0.40 / 0.99
ring rank 2	0.037	0.032	-0.793	0.47 / 0.17 / 0.99

Where the residual lies in solved heads of rank above D (`sweep_final.py`, `summarise_final.py`; consistent with parking, but removal through the latent, which takes the read components first, leaves the same end state, so this does not discriminate the two):

set (solved heads)	r - D	lam	off-plane share (random direction, median)	readgain
torus G64 per-head r3	1	0.004	0.978 (0.50)	0.15
torus G64 shared r4 (RANK_MI)	2	0.004	0.996 (0.71)	0.11
torus G64 per-head r4	2	0.003	0.988 (0.71)	0.16
torus G64 shared r4 (SCORE_RANK)	2	0.003	0.998 (0.71)	0.09
2D G32 r3	1	0.003	0.987 (0.50)	0.09
3D G10 r4	1	0.008	0.964 (0.40)	0.23
3D G18 r4	1	0.001	0.810 (0.40)	0.57
ring rank 2 (1D), 300 epochs	1	0.025	0.989 (0.71)	0.21
torus T = 128 r4, 300 epochs	2	0.021	0.988 (0.71)	0.27

set (solved heads)	r - D	lam	off-plane share (random direction, median)	readgain
escape, fresh r4 step map, 600 epochs	2	0.003-0.004	0.98 (0.71)	0.18

In the clean head of solved rank-D runs (ring rank 1, torus rank 2) the clock is removed (lam <= 0.0005 on the T = 1024 2D sets, <= 0.0013 on the ring), since it cannot lie off the plane; the other head of such runs often drifts (lam 0.15-1.75). The leftover is about 6x larger after 300 epochs than after 600-900, of the size decoupled weight decay alone gives an unread direction over a cosine schedule (consistent, not a test). The escape arms (below) show the latent clock shrinking, not held: in the fresh rank-4 step maps its absolute size falls 14-85x (relative to the map step 270-1300x), the off-plane part 15-72x in 7/8 runs (1.6x in s1, which started at 0.005), and in the three solved lifted runs 23-490x (`clock_decomp_out.txt`). Unsolved heads of rank above D read their clock in all three sets and hold it in the map plane in two (torus per-head r3: off-plane 0.27 against a random-direction median of 0.50, readgain 0.61; escape lifted runs: 0.08 against 0.71, readgain 0.76; 3D G10 r4: off-plane 0.53 against 0.40, readgain 0.78).

Drifting rank-D heads drift along their least-resolved axis (`c1_runlevel.py`): over 116 drifting heads in nine sets the median r_beta (resolution along the drift direction relative to the head's mean) is 0.087, and 85% are below 0.5 against 29% for a random direction given each head's own frame; at the run level 57 of 62 runs against 28.7 expected (P < 5e-5, 20,000 simulations); escape C continues rank_mi's own networks, so counting each network once gives 50 of 55 against 25.1 (`c1_runlevel_once.py`). Clean heads of rank above D sit at r_beta 0.79-1.10 (medians). This geometry is what E9 leads to and also what pruning of a visible drift after any collapse would leave; it fixes the geometry, not the order of events.

The escape test, re-read (`escape_start.py`; RANK_TESTS escape arms, 600 epochs on the same eight trained networks):

arm	runs with a drift-free head at the end	heads drift-free	off-plane share at end
fresh rank-4 step map (start drift 0.28-5.86 cells per move)	8/8	16/16	0.91-1.00
stalled rank 2 lifted to rank 4	3/8	6/16	1.00 in the 3 solved
rank 2 continued	1/8	2/16	--

Within the lifted arm the clock's off-plane share at the start (the lift adds two latent rows at default scale beside large trained rows) is 0.09 / 0.14 / 0.11 in the three runs that solve and 0.01-0.06 in the five that do not (exact one-sided rank-sum p 0.018; post hoc). Two caveats make this suggestive only. The statistic was chosen after the fact: with the off-plane clock relative to the map step instead (|c_perp| / |U|), the solved runs rank 1st, 2nd and 8th (p 0.29). And s1's source also solves under plain continuation (arm C s1 SOLVED) and starts nearly clock-free (|c| / |U| 0.05), so its large share is that of a near-zero clock; without s1, p 0.048 (`escape_start_out.txt`). A seeded escape test (RANK_THEORY's G2) would test it. The fresh step maps remove the drift everywhere, and the residual geometry does not track SOLVED within that arm (the 12 unsolved heads show the same

off-plane share, 0.985, and readgain, 0.19, as the 4 solved ones; audit); learning the rest of the map within 600 epochs (2/8 SOLVED) is a separate matter.

G5. The 1D exemption (registered + analysis)

On the ring (`RANK_TESTS_RESULTS.md`), rank 1 solves 7/8 (0.993) and rank 2 8/8 (1.000); with the drift forbidden by construction (opposite move latents tied, observation steps zero) rank 1 solves 8/8 in a median of 24 epochs (58 without). Every channel of a rank-1 head sees the same phantom drift in cells (E8.5(d)); over the 8 rank-1 runs the 11 clean heads (visible drift <= 0.01 per move) end at lam <= 0.0013, and the 5 drifting heads (visible drift 0.04-0.50 per move; two of them in the unsolved run s22) have the largest map steps (|U| 2.8-5.0 against 1.6-2.6), the only other global way to shrink the drift in cells.

G6. Dynamics (analysis, CPU pilot; `scripts/rank/train_snap.py`, `analyze_snap.py`, `pilot_t128/`)

Paper torus T = 128 (grid 64), batch 32, 40 epochs cosine, lr 1e-3, seeds 3000-3002 (outside every batch); arms shared r2, shared r4 (from r2's base) and R2Clock (shared r2 plus a fixed extra latent coordinate that every token steps by 1, giving each channel a learned clock rate; identical to r2 at initialisation). Snapshots every 49 steps.

arm	seed	final loss	accuracy T = 128 / 512	latent clock
r2	3000	0.005	1.000 / 0.937	removed in lockstep with the visible drift (lam 0.63 -> 0.004)
r2	3001	0.230	0.922 / 0.836	kept, on the unresolved axis (r_beta 0.00); frame near-collinear from step 0
r2	3002	0.003	1.000 / 0.981	removed in lockstep
r4	3000	0.005	1.000 / 0.954	off-plane 0.62 -> 0.96, then removed (lam falls 66x)
r4	3001	0.028	0.990 / 0.959	off-plane 0.55 -> 0.98, then removed (180x)
r4	3002	0.005	1.000 / 0.946	off-plane 0.81 -> 1.00, then removed (5.5x)
R2Clock	3000-3002	0.003-0.006	1.000 / 0.941-0.959	the slot holds a fixed unit clock by construction (off-plane ~1 from step 0)

At rank 4 the visible drift fell tenfold one or two snapshots before the latent clock did in 4/6 heads, and the latent clock was then removed too (5-180x within 40 epochs); in the rank-2 control the same order appears in 2/6 heads (n = 3). R2Clock is not counted: its slot clock cannot shrink tenfold by construction. At the same initial weights as the failing rank-2 seed 3001, R2Clock solved (loss 0.006 against 0.230) and repaired its frame (one matched pair). The failing seed's weak axis was present at initialisation; the readout gradient at initialisation on the real T = 1024 task gives no consistent hiding direction (drift 0.4-2.7 cells per move, outside assumption A1; one stochastic sample of two batches, dropout on; `grad_hiding.py`). Initial frame conditioning does not separate solved from unsolved rank-D runs at T = 1024 (one-sided p 0.19; `init_frame.py`); initial drift is lower in the runs that solve (as a test only the independent SCORE_RANK seeds count: one-sided p 0.073; pooled over sets whose seeds share a base, 0.012, descriptive). MapFormer's 5D setting (T = 16, grid 5) solved at every rank under our recipe (rank 5, 6 and 5 + clock slot 3/3 each; index RoPE stalled), so it does not test rank here (`pilot_5d/`).

G7. Clock routes at rank 2 (`RANKCLOCK_RESULTS.md`, registered; `RANKCLOCK_PREREG.md` with Amendment 1)

Design. Torus 64 x 64, T = 1024, the RANK_MI recipe (900 epochs, lr 1e-3, 98 batches of 16, cosine, AdamW weight decay 0.05, 1 layer, 2 heads, d_model 128, standard attention), fresh seeds 1900-1907, 7 arms x 8 = 56 runs in one batch on one GPU from one code version (`a33d6d0`, md5-guarded). r2, both slot arms, drift-forbidden and actions-only share every initial weight at a seed (bitwise; `test_rankclock.py`); r4 is built from r2's base as in RANK_MI. An independent results-blind code check before launch (Amendment 1: 3 MAJOR, 5 MINOR, 2 NOTE) was folded in; its first MAJOR added the per-axis slot as the primary. Pilots on seeds 1990-1992 are not pooled. Evaluation `eval_noise_refine` on fresh maps (env seed 10000), 100 trials per length. Analysis `analyze_rankclock.py` -> `RANKCLOCK_ANALYSIS.txt`, `RANKCLOCK_VERDICTS.json`.

Two clock exits. Per inverse pair the latent clock is u_x = z(+x) + z(-x) (likewise u_y). A walk's drift per channel is (t / 2)(u_x + u_y) / 4, a common part growing with the step count, plus (n_x - n_y)(u_x - u_y) / 4, a differential part growing with the net axis imbalance; at initialisation the second is as large as the first (|u_x - u_y| / |u_x + u_y| median 1.22 for r2, 1.01 for r4; reviewer's measurement). Two independent pair clocks need r >= 4 to be parked (E8.5(b)), so rank 4 has both exits. The per-axis slot gives rank 2 both without extra map rank; the per-token slot only the common one (on this stream it is equivalent to a learned index-RoPE rate). The slots add parameters but not map rank, so the over-parameterisation account predicts r2-like behaviour for both. (G6's R2Clock pilot is a different arm: a fixed extra latent coordinate that every token steps by 1, at T = 128.)

arm	step map	parameters	initialisation against r2 at the seed
r2	shared rank 2, every token steps	204,373	--
slotxy (primary)	r2 + per-channel clock_x on the two x-moves and clock_y on the two y-moves; observation steps unchanged	204,501 (+128)	shared tensors bitwise equal; slots 0; forward bitwise equal
slot (P2)	r2 + one per-channel clock on every token	204,437 (+64)	as slotxy
DF, drift-forbidden	pair (+x, -x) steps +-W_out W_in (e(+x) - e(-x)) / sqrt 2, likewise y; every other step 0	204,373	state dict bitwise equal
AO, actions-only	the four move steps W_out W_in e(move) free; every other step 0	204,373	state dict bitwise equal
r4	shared rank 4 built from r2's base	204,757	differs only in W_in, W_out
r4wd0	as r4, weight decay 0	204,757	as r4

Measured on the constructed models: DF's non-move steps are exactly 0, its pairs exact negatives and its move-sum clock exactly 0; AO's non-move steps are exactly 0 and its initial move clock is 0.743. DF's initial pair step is sqrt 2 times r2's half-difference (as in the ring precedent). Shared, not per-head, rank throughout.

Existence before the batch (prereg Sec. 3; `runs/rankclock_pilot/construct_rankclock.json`). Source: a trained shared r4 (pilot seed 1990, outside the batch; frozen accuracy 1.0000, final-5% loss 0.0021; step-table singular values 12.61, 10.51,

0.028, 0.009, so effectively rank 2). Each class is scored frozen by the best construction it can represent and then warm-started for 60 cosine epochs at peak lr 1e-3 (HELD = still SOLVED and frozen accuracy within 0.02):

class	frozen, best construction	other constructions	after 60 epochs	
r2	rank-2 projection 1.0000	DF 1.0000, AO 0.6045	0.9998, peak loss 0.289, tail 0.0073	HELD
slot	rank-2 projection 1.0000	slot 1.0000, DF 1.0000, AO 0.6045	1.0000, peak 0.210, tail 0.0025	HELD
slotxy	rank-2 projection 1.0000	xy 1.0000, DF 1.0000, AO 0.6044	1.0000, peak 0.197, tail 0.0052	HELD
AO	DF 1.0000	AO 0.6045	1.0000, peak 0.135, tail 0.0002	HELD
DF	DF 1.0000	--	1.0000, peak 0.017, tail 0.0001	HELD

Every class contains a solution and holds it, so a failure in the batch is a search failure. The slot classes are knocked off their solution (peak loss about 0.2; one Adam step at lr 1e-3 moves a slot's top-channel phase by 12.9 rad over 2,048 tokens) and re-find it. Constructed weight scales are 2-3x the source's. Descriptive, one source: the trained r4 with its observation steps zeroed (AO) scores 0.60 and with its moves antisymmetrised (DF) 1.00, so it cancels part of its move clock with its observation steps.

Results (SOLVED = final-5% training loss < 0.05; S / T / D = SOLVED / STALLED / DESCENDING; no run diverged; accuracy mean +/- sd over 8 seeds; `RANKCLOCK.md`, `RANKCLOCK_ANALYSIS.txt`):

arm	SOLVED	S / T / D	T = 512	T = 1024	T = 2048
r2	1/8	1 / 2 / 5	0.954 +/- 0.042	0.918 +/- 0.053	0.851 +/- 0.056
slotxy (primary)	3/8	3 / 1 / 4	0.984 +/- 0.037	0.972 +/- 0.046	0.927 +/- 0.059
slot (P2)	3/8	3 / 2 / 3	0.979 +/- 0.023	0.951 +/- 0.045	0.891 +/- 0.075
DF	8/8	8 / 0 / 0	1.000 +/- 0.000	1.000 +/- 0.000	1.000 +/- 0.001
AO	1/8	1 / 0 / 7	0.990 +/- 0.011	0.962 +/- 0.029	0.890 +/- 0.072
r4	8/8	8 / 0 / 0	1.000 +/- 0.000	1.000 +/- 0.001	0.986 +/- 0.018
r4wd0	1/8	1 / 6 / 1	0.766 +/- 0.128	0.760 +/- 0.130	0.748 +/- 0.130

Per seed, accuracy at T = 1024 and run class (final-5% training loss):

seed	r2	slotxy	slot	DF	AO	r4	r4wd0
1900	0.883 STAL (0.377)	0.965 DESC (0.172)	0.916 STAL (0.264)	1.000 SOLV (0.000)	0.916 DESC (0.326)	1.000 SOLV (0.002)	0.735 STAL (0.641)
1901	0.896 DESC (0.329)	0.981 DESC (0.121)	0.962 DESC (0.167)	1.000 SOLV (0.000)	0.993 DESC (0.075)	1.000 SOLV (0.002)	0.715 STAL (0.616)
1902	0.923 DESC (0.354)	0.997 SOLV (0.044)	1.000 SOLV (0.003)	1.000 SOLV (0.000)	0.948 DESC (0.263)	1.000 SOLV (0.002)	1.000 SOLV (0.003)
1903	0.981 DESC (0.089)	0.995 DESC (0.062)	0.900 STAL (0.417)	1.000 SOLV (0.000)	0.968 DESC (0.148)	0.998 SOLV (0.003)	0.783 STAL (0.496)

seed	r2	slotxy	slot	DF	AO	r4	r4wd0
1904	0.939 DESC (0.226)	0.978 DESC (0.159)	0.930 DESC (0.274)	1.000 SOLV (0.000)	0.935 DESC (0.264)	1.000 SOLV (0.009)	0.692 STAL (0.872)
1905	0.882 STAL (0.488)	0.861 STAL (0.585)	0.897 DESC (0.302)	1.000 SOLV (0.000)	0.981 DESC (0.157)	1.000 SOLV (0.003)	0.890 DESC (0.249)
1906	0.998 SOLV (0.002)	1.000 SOLV (0.003)	1.000 SOLV (0.009)	1.000 SOLV (0.000)	1.000 SOLV (0.001)	1.000 SOLV (0.003)	0.590 STAL (1.077)
1907	0.843 DESC (0.494)	1.000 SOLV (0.003)	0.999 SOLV (0.011)	1.000 SOLV (0.000)	0.959 DESC (0.147)	1.000 SOLV (0.003)	0.673 STAL (0.831)

Gates: G0 r4 8/8 (>= 6 required; RANK_MI 8/8); G1 r2 1/8 (<= 2 required; RANK_MI 0/8). Both pass.

Registered readings.

readout	rule	label	numbers
P (primary): slotxy vs r2	>= 6/8 and Fisher < .05: CLOCK EXITS HELP; <= 2/8: DO NOT HELP; else PARTIAL	**PARTIAL**	3/8 vs 1/8, Fisher p 0.5692; accuracy 0.972 vs 0.918, permutation p 0.0500 (printed beside, not a decision rule)
P2: slot vs r2	same thresholds (common-mode exit)	PARTIAL	3/8 vs 1/8, Fisher p 0.5692; accuracy 0.951 vs 0.918, permutation p 0.2068
E2: DF, AO, r2	AO <= 2/8, DF >= 6/8 and Fisher(DF, AO) < .05: THE CLOCK IS THE OBSTACLE (first match of five branches)	**THE CLOCK IS THE OBSTACLE**	DF 8/8 vs AO 1/8, Fisher p 0.0014; AO vs r2 1/8 vs 1/8, p 1.0000; DF vs r4 8/8 vs 8/8 (descriptive)
X2: r4wd0, park or remove	needs >= 4 SOLVED r4wd0 runs	X2 UNREADABLE	r4wd0 SOLVED 1/8
X2b: order of read and latent clock decline	two-sided sign test over solved runs	NO ORDER	r4wd0: no usable runs (1 solved); r4: read clock first in 2 of 4 usable runs, sign p 1.0000

Power (prereg Sec. 5): with r2 at 1/8 the primary could read HELPS only at >= 6/8 (Fisher p 0.041); a slot at 3-5/8 reads PARTIAL by design. The slots were used, not left at zero: median max |omega x slot| 0.142 / 0.124 rad (clock_x / clock_y) and 0.068 rad (per-token slot).

Budget (rules 3-4; read before quoting the primary). The slot arms are not stalled at rank 2's level: of their 5 unsolved runs each, slotxy has 4 DESCENDING and 1 STALLED, slot 3 DESCENDING and 2 STALLED; rank 2 has 5 of 7 unsolved runs DESCENDING. The registered claim is "within 900 epochs"; a slot that solves later than rank 4, or a rank 2 that would solve with more epochs, fits this table, so PARTIAL is a speed-or-capability question the batch does not separate. Rank 2 did not solve in 1800 epochs from scratch (0/8 vs rank 4 7/8; C4), so a longer budget is not certain to rescue the slot arms either.

Checks. Rule 2: r(log10 final loss, accuracy) = -0.537 / -0.655 / -0.824 at T = 512 / 1024 / 2048 over 56 runs; the readings are on SOLVED and accuracy is descriptive.

Not registered (post hoc). - *Weight decay looks close to necessary for rank 4 here.* r4wd0 solves 1/8 against r4's 8/8 (Fisher p 0.0014), with 6/8 STALLED, and is the only

arm far from ceiling at T = 512 (0.766). No r4 vs r4wd0 contrast was registered (the arm was there to read X2), so this is an observation to confirm in a registered batch. It is also why X2 could not be read. - *The solved wd-0.05 rank-4 runs end with little clock:* median shrink |u_end| / |u_init| 0.134, readgain 0.026 of its initial value, u_ratio 0.022 (8/8 solved). With decay on, removal and decay of a parked clock cannot be told apart, so this says only that the end state carries little clock; it is not evidence for parking or for removal. - *DF holds its solution past the training length:* 1.000 +/- 0.001 at T = 2048 against rank 4's 0.986 +/- 0.018. - *Geometry at the end of training* (medians over all runs, descriptive): u_ratio r2 0.169, slotxy 0.243, slot 0.056, AO 0.481, r4 0.022, r4wd0 0.105; shrink r2 1.86, slotxy 3.15, slot 0.41, AO 7.30, r4 0.13, r4wd0 1.56. In r2, slotxy and AO the latent clock grew over training (shrink > 1), most in AO; in the per-token slot arm it shrank, as in rank 4.

G8. Weight decay and rank 4 (RANKWD; registered, `RANKWD_RESULTS.md`, `RANKWD_PREREG.md` + Amendment 1)

Shared rank 4 (`Vanilla_r4mi`) at AdamW weight decay 0.05 vs 0, fresh seeds 2100-2107, torus 64 x 64, T = 1024, 900 epochs, lr 1e-3 cosine (the RANK_MI / RANKCLOCK recipe); independent check before launch. Existence: a solved rank-4 model holds its solution through 60 epochs at peak lr with weight decay 0. Registered reading: DECAY NEEDED, scoped to "AdamW decay 0.05 on all parameters, at lr 1e-3 cosine, within 900 epochs" (not credited to the step map or the clock); 5 of the 7 unsolved weight-decay-0 runs were still DESCENDING at 900 epochs. The park-vs-remove secondary was unreadable by design (1 solved run).

Primary

arm	SOLVED	classes (S / T / D)	acc T = 512	acc T = 1024	acc T = 2048
wd 0.05	**8/8**	8 / 0 / 0	1.000 +/- 0.000	0.9995 (1.000 +/- 0.001)	0.976 +/- 0.023
wd 0	**1/8**	1 / 2 / 5	0.865 +/- 0.123	0.860 +/- 0.124	0.846 +/- 0.124

G0 passes (wd 8/8 >= 6). P: k_wd0 = 1 <= 2 and Fisher p 0.0014 < .05 -> DECAY NEEDED. Accuracy at T = 1024: 0.9995 vs 0.8603, unpaired permutation p 0.0014, paired sign-flip p 0.0156 (exact; the smallest p the n = 8 design allows is 0.0078).

Per seed, accuracy at T = 1024 (class): wd 0.998S 1.000S 0.999S 1.000S 1.000S 1.000S 1.000S 1.000S; wd 0 0.853D 0.866D 0.837T 0.952D 0.870D 1.000S 0.920D 0.585T.

G9. A recipe search for rank 2 (RANKRECIPE; registered, `RANKRECIPE_RESULTS.md`, `RANKRECIPE_PREREG.md` + Amendment 1)

Shared rank 2 (`Vanilla`) at three recipes and shared rank 4 (`Vanilla_r4mi`) at the default, fresh seeds 2600-2607, torus 64 x 64, T = 1024, 900 epochs, cosine (5% warm-up), AdamW, the RANKWD recipe otherwise (standard attention); 32 runs on one GPU and one code version; independent check before launch (2 MAJOR, 8 MINOR, folded in). At a seed the rank-2 cells share every initial weight, and rank 4 shares all non-bottleneck weights. Existence: a rank-2 solution exists and is held (App. D1, G7), and the construction held a 60-epoch warm start at both new recipes. The pilot (seeds 2692-2693) kept the higher

rate at 3e-3 (not UNSTABLE). Selection of rank 2's best recipe on training loss only (most SOLVED, then lowest mean final-5% loss); decision p = max(selection-aware, naive). Gates: rank 4 >= 6/8; rank 2 at the default <= 2/8.

cell	recipe	SOLVED	classes (S / T / D)	acc T = 512	acc T = 1024	acc T = 2048	per-move angle growth (end / init)
rank 2	lr 1e-3, weight decay 0.05 (default)	0/8	0 / 3 / 5	0.888	0.860	0.803	0.566
rank 2	lr 3e-3, weight decay 0.05	**0/8**	0 / 3 / 5	0.954	**0.910**	0.838	0.555
rank 2	lr 1e-3, weight decay 0.2	0/8	0 / 6 / 2	0.932	0.885	0.802	0.296
rank 4	lr 1e-3, weight decay 0.05 (default)	**8/8**	8 / 0 / 0	1.000	**1.000**	0.997	0.504

Gates pass (rank 4 8/8; rank 2 at the default 0/8, replicating App. C4 and G7). Selected: lr 3e-3 (SOLVED tied at 0; lowest mean final-5% loss 0.370; also rank 2's best on accuracy). Primary, rank 2 at its best recipe against rank 4: SOLVED 0/8 vs 8/8, naive Fisher and selection-aware p 0.0002; accuracy at T = 1024 0.9095 vs 0.9999 (d -0.090), naive permutation and selection-aware p 0.0002. Registered reading: RANK D+1 NEEDED SURVIVES A THREE-RECIPE SEARCH, scoped to "the best of these three recipes": the decay exponents are 2.41 / 7.24 / 9.66, so both alternatives raise the total decay; there is no lower-decay or combined arm; 12 of the 24 rank-2 runs were still descending. No run UNSTABLE or diverged. r(log10 final loss, accuracy) -0.629 / -0.793 / -0.925 at T = 512 / 1024 / 2048 over 32 runs. Every cell keeps its effective per-move angle bounded (maximum 2.09-3.98 rad per move), unlike rank 4 without decay (~9x growth, App. G8).

Scope of the search (2026-10-26, after the critical review B-M5). Selecting rank 2's best of three cannot hurt rank 2, but neither alternative is a recipe at which rank 4 is known to learn: at lr 3e-3 the related observation-step-free signed rank-4 model solves 0/8 on this torus within 900 epochs (H12, a different model, cited as an observation), and the higher decay shrinks the angles most and leaves the most rank-2 runs STALLED. The search is therefore limited, and the main text states the result as "rank 2 fails at each of three recipes" (within 900 epochs).

Per seed, accuracy at T = 1024 (class): rank 2 default 0.731D 0.883D 0.720T 0.922D 0.983D 0.792T 0.984D 0.862T; lr 3e-3 0.923D 0.902D 0.892T 0.885T 0.907D 0.910D 0.878T 0.978D; weight decay 0.2 0.864T 0.881T 0.882D 0.951D 0.871T 0.882T 0.877T 0.870T; rank 4 1.000S x 4, 0.999S, 1.000S x 3.

H. Sign details

H1. Learning rates (`LRSWEEP_RESULTS.md`; ring, observation-step-free, seeds 1500-1507)

learning rate	signed SOLVED, accuracy	|.| SOLVED, accuracy
1e-3	8/8, 1.000	1/8, 0.917
3e-3	8/8, 1.000	2/8, 0.963
1e-2	4/8, 0.972	1/8, 0.872
3e-2	1/8, 0.559	0/8, 0.584

Registered B1 (each arm at its best rate, selection redone in the null over all 12,870 relabellings): |.| 3e-3 2/8 vs signed 1e-3 8/8, -0.037, decision p 0.0070 (SOLVED) / 0.0057 (accuracy). The |.| optimum is interior to the grid. Three |.| runs at 3e-3 were still descending at 300 epochs. At 3e-2 both arms sit near the 0.50 blank floor. The torus counterpart (three rates) is H12. Per seed:

```
seed | 1e-3:SO | 1e-3:AO | 3e-3:SO | 3e-3:AO | 1e-2:SO | 1e-2:AO | 3e-2:SO | 3e-2:AO
1500 | 1.0000 SOLV | 0.9362 STAL | 1.0000 SOLV | 0.9111 STAL | 0.9460 DESC | 0.9936 DESC |
0.4960 STAL | 0.4958 STAL
1501 | 1.0000 SOLV | 0.8759 STAL | 1.0000 SOLV | 0.9662 DESC | 0.9460 STAL | 0.4960 STAL |
0.4960 STAL | 0.4960 STAL
1502 | 1.0000 SOLV | 0.8995 STAL | 1.0000 SOLV | 0.9428 STAL | 0.9407 STAL | 0.9703 DESC |
0.4960 STAL | 0.4960 STAL
1503 | 1.0000 SOLV | 0.9116 STAL | 1.0000 SOLV | 1.0000 SOLV | 1.0000 SOLV | 0.6000 STAL |
0.4960 STAL | 0.4960 STAL
1504 | 1.0000 SOLV | 0.9415 STAL | 1.0000 SOLV | 0.9686 STAL | 0.9460 STAL | 1.0000 DESC |
0.4960 STAL | 0.8159 DESC
1505 | 1.0000 SOLV | 1.0000 SOLV | 1.0000 SOLV | 0.9993 SOLV | 1.0000 SOLV | 0.9950 DESC |
0.9998 SOLV | 0.4961 STAL
1506 | 1.0000 SOLV | 0.8736 STAL | 1.0000 SOLV | 0.9595 DESC | 1.0000 SOLV | 0.9996 SOLV |
0.4960 STAL | 0.4960 STAL
1507 | 1.0000 SOLV | 0.8971 STAL | 1.0000 SOLV | 0.9587 DESC | 1.0000 SOLV | 0.9184 DESC |
0.4959 DESC | 0.8799 DESC
```

H2. Recipe cells (`ROBUST_RESULTS.md`, seeds 1400-1407)

cell	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	d [95% CI]	perm p	Fisher p
ring	lr 1e-3, weight decay 0.05	0.911, 1/8	1.000, 8/8	-0.089 [-0.125, -0.055]	0.0002	0.0014
ring	lr 1e-3, weight decay 0	0.873, 0/8	1.000, 8/8	-0.127 [-0.200, -0.075]	0.0002	0.0002
ring	lr 3e-3, weight decay 0.05	0.987, 4/8	1.000, 8/8	-0.013 [-0.025, -0.005]	0.026	0.077
ring	lr 3.3e-4, weight decay 0.05	0.841, 0/8	0.950, 6/8	-0.109 [-0.215, +0.010]	0.089	0.0070
torus	2 layers, d_model 256	0.967, 2/8	0.999, 8/8	-0.032 [-0.055, -0.015]	0.0005	0.0070

At lr 3e-3 the registered contrast was below its decision threshold in this batch (4/8 vs 8/8); on fresh seeds in the sweep the same rate gives 2/8 vs 8/8 (H1). Declared secondaries: cancelling share (unweighted) |.| 0.04-0.13 in every cell, signed 0.43-1.00; torus strata (short / long / wrap-only) signed 1.000 / 0.999 / 0.992, |.| 0.989 / 0.911 / 0.799; on the torus 6/8 |.| runs were still descending at 900 epochs (final losses 0.018-0.144).

H3. Starts (`MECHANISM_RESULTS.md`, ring p_plus 0.5, seeds 800-807; `TORUS_START_RESULTS.md`, torus, seeds 1200-1207)

Observation-step-free models differing only in the initial move angles (every other weight identical at a seed):

initial move angles	signed (ring)	|.| (ring)	signed (torus)	|.| (torus)
default (small, signs independent)	8/8	0/8	8/8	1/8
the |.| start: both moves rotating the same way	7/8	0/8	--	--
large, opposite-move sums below pi, U(0.9, 1.5)	--	2/8	--	0/8
large, sums above pi, U(1.7, 4.5)	8/8	8/8	2/8	0/8

Ring, registered: same start, sign matters (0/8 vs 7/8, Fisher p 0.0014; -0.092 [-0.135, -0.045]); a start past pi rescues |.| (8/8 vs 0/8, +0.109 [+0.085, +0.135], p 0.0002); a large start with sums below pi does not (2/8; vs past pi Fisher p 0.0070); from past pi the sign makes no difference (8/8 vs 8/8). The large start also moves the angles 13-22x faster per Adam step (the sub-pi control about 5x), so sum and size are only partly separated. Analysis: signed models from the small starts end with 98-99% of channels opposite-signed and cancel through zero; from past pi both signs cancel by wrapping (signed 88%, |.| 100% of cancelling channels); the rescued |.| models cancel in 23% of channels. Torus, registered: the past-pi start does not rescue |.| (0/8 vs 1/8, +0.005 [-0.060, +0.070]), and it slows the signed model (2/8; 5/8 runs per large-start arm still descending at 900 epochs); the default-start sign cost replicates (1/8 vs 8/8, Fisher p 0.0014). The start lever is a ring result.

H4. Moves +1/-1 vs +1/+2 (`GAUGE_RESULTS.md`, ring, seeds 1300-1307)

By Lemma 1 (ii) a |.| step represents, without wrapping or crossing zero, a map whose two moves step the same way. With moves +1/-1 (an inverse pair) the cost replicates: |.| 0.931, 1/8 vs signed 1.000, 8/8 (-0.069 [-0.095, -0.045], Fisher p 0.0014). Moves +1/+2 need integration (index RoPE 0.649, 0/8) and every revisit is a lap of at least 16 moves; there |.| scores 0.883, 2/8 against signed 0.947, 5/8 (-0.064 [-0.200, +0.075], Fisher p 0.31), a registered comparison that is not resolved because the task is harder for both arms. The step code is learned in both classes (analysis; `scripts/sign/latent_geometry.py`, `lattice_p.py`): in the channels attention uses the relation a(+2) = 2 a(+1) holds on 0.98-1.00 of usage in 8/8 |.| runs (signed 0.99-1.00 in 7/8; angle tolerance 0.05 rad, a post hoc choice: at 0.3 rad the lattice shares change a lot), and the latent relation in 5/8 |.| runs, one of them through the mirror copy z_2 = -2 z_1 of E8.4 (cosine -0.98), against 7/8 signed. The unsolved |.| runs and 2 of the 3 unsolved signed runs have the relation in place (signed s1300: 0.68); what they miss lies outside the step code. The registered comparison is unmeasured: there is no evidence either way of a sign cost without an inverse pair, and |.| solves +1/+2 no more often than +1/-1 (2/8 against 1/8, Fisher p

1.00), so losing the through-zero route does not account for |.|'s difficulty (main text Sec. 7.3). Signed models on +1/+2 end with 0.92 of channels having both move angles of one sign (0.01 on +1/-1), the sign-coherent code Lemma 1 allows.

H5. Three times the ring budget (`RING_BUDGET_RESULTS.md`, seeds 40-47)

The |.| step at 900 epochs solves 3/8, as at 300 epochs on the same seeds (3/8); the solved sets differ.

H6. Cancelling share and ring strata

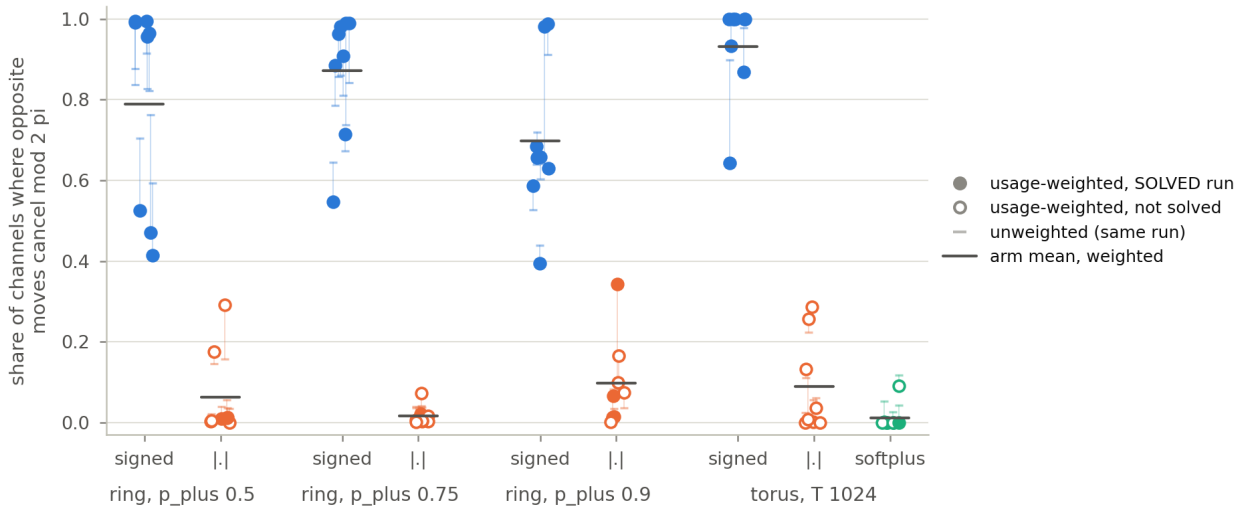


Figure H1 (`fig_wrap_share`; analysis). Share of rotary channels in which opposite moves cancel mod 2 pi, per run, weighted by how much attention uses each channel (dots; filled = SOLVED) and unweighted (faint dashes, joined to the same run). A channel cancels when |wrap(a(move) + a(opposite move))| < 0.1 rad; channels whose move angle has collapsed to within 0.05 rad of 0 mod 2 pi are excluded from the denominator. Usage weight: mean query norm of the channel at move tokens times mean key norm at observation tokens. Ring of 32 cells at three move biases (n = 8 each) and the torus at T = 1024 (n = 8). Bars: arm means of the weighted share. Signed steps 0.70-0.93; |.| steps 0.02-0.10; softplus 0.01. Seven of the eight non-negative runs that solve stay at or below 0.07.

cell	arm	SOLVED	weighted mean (range)	unweighted mean (range)
ring p_plus 0.5	signed	8/8	0.79 (0.41-0.99)	0.79 (0.59-0.92)
ring p_plus 0.5	|.|	2/8	0.06 (0.00-0.29)	0.06 (0.02-0.16)
ring p_plus 0.75	signed	8/8	0.87 (0.55-0.99)	0.78 (0.64-0.86)
ring p_plus 0.75	|.|	1/8	0.02 (0.00-0.07)	0.03 (0.02-0.04)
ring p_plus 0.9	signed	8/8	0.70 (0.39-0.99)	0.64 (0.44-0.91)
ring p_plus 0.9	|.|	4/8	0.10 (0.00-0.34)	0.06 (0.02-0.10)
torus T 1024	signed	8/8	0.93 (0.64-1.00)	0.98 (0.90-1.00)
torus T 1024	|.|	0/8	0.09 (0.00-0.29)	0.11 (0.00-0.27)
torus T 1024	softplus	1/8	0.01 (0.00-0.09)	0.03 (0.00-0.12)

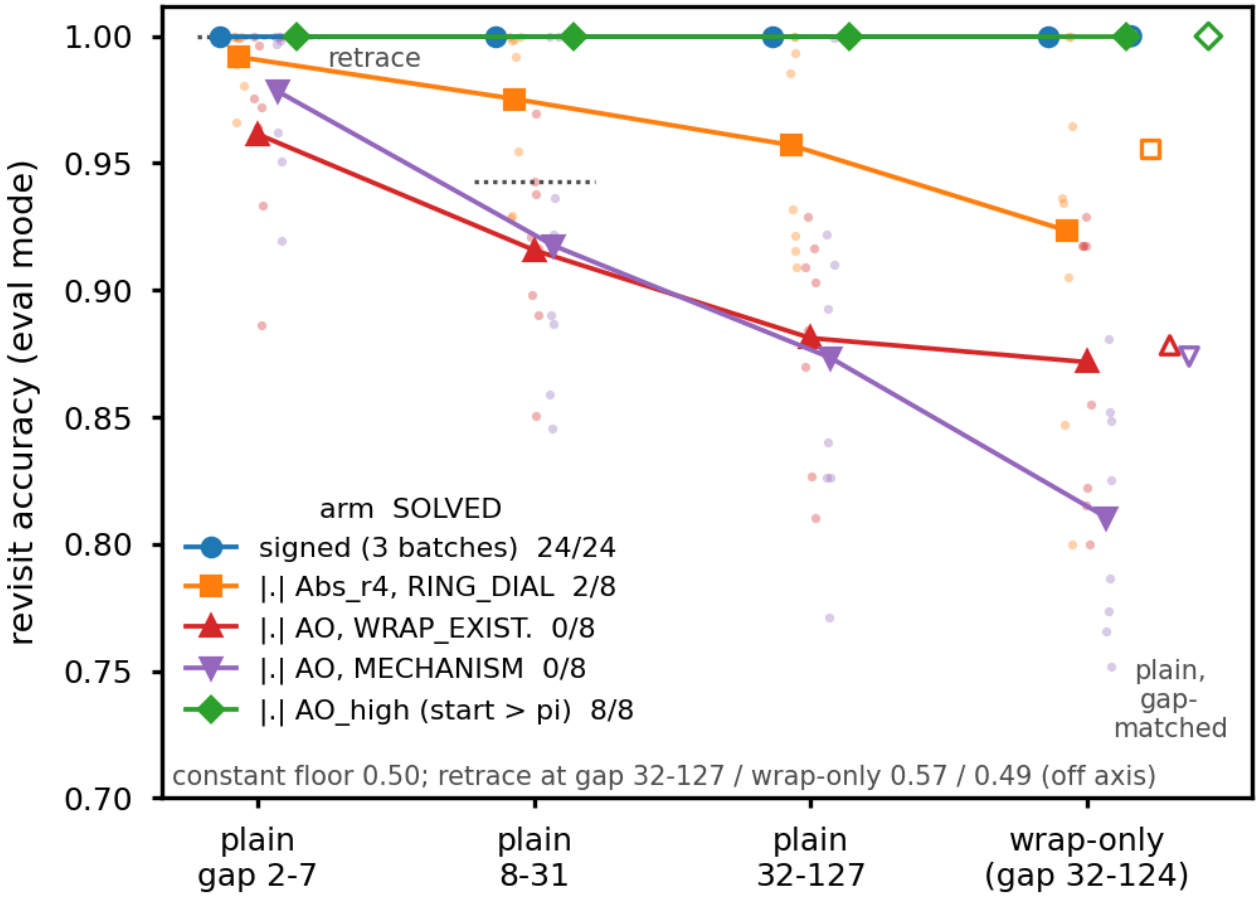


Figure H2 (`fig_ring_strata`; analysis). Revisit accuracy on the 1D ring by stratum (eval mode; 400 trajectories, T = 128; maps redrawn). "Plain" revisits have an earlier visit at the same unwrapped position; "wrap-only" revisits were seen only a lap away. Lines: arm means; dots: runs; hollow markers: plain revisits re-weighted to the wrap-only gap distribution. Signed steps (24 runs over three batches) and |.| steps started past pi are at 1.000 everywhere. |.| steps from the default start lose accuracy with the gap (0.96-0.99 at gaps 2-7, 0.87-0.96 at gaps 32-127). Dotted: retrace within each stratum, 1.000 at gaps 2-7 and 0.943 at gaps 8-31 (0.57 and 0.49 in the long-gap and wrap-only strata, off axis); constant floor 0.50.

H7. Routes, measured (analysis; `docs/paper/expansion/scripts/sign/`)

End-state clocks (`clock_table.py` -> `clock_table_out.txt`: the clock / map ratio of E7.1', one value per run; the usage share on cancelling channels from `sign_clock.py`; `sign_clock.py`'s one-sided map wrap(a_y), the old column, gave 1.683 / 1.632 / 1.549 / 1.394 / 1.520 / 1.219 for the |.| and frozen rows, and is reproduced by `clock_table.py` as a check):

batch, arm	SOLVED	clock / map, arm mean	usage on cancelling channels
ring MECHANISM, signed (default start)	8/8	0.018	0.96
ring MECHANISM, signed from the |.| start	7/8	0.015	0.96
ring GAUGE, signed (+1/-1)	8/8	0.016	0.95
ring MECHANISM, |.| (default start)	0/8	1.471	0.08

batch, arm	SOLVED	clock / map, arm mean	usage on cancelling channels
ring MECHANISM, |.| (|.| start)	0/8	1.571	0.02
ring GAUGE, |.| (+1/-1)	1/8	1.559	0.06
ring ACCOUNTS T2, signed, move latents frozen, |.| start	1/8	1.181 (the solved run 0.058)	0.15
ring ACCOUNTS T2, signed, move latents frozen, default start	7/8	0.167	0.87
ring MECHANISM, |.| from a large sub-pi start	2/8	1.423 (solved runs 0.10, 0.59)	0.20
ring MECHANISM, |.| from a start past pi	8/8	0.317 (wrapped; runs 0.11-0.76)	0.82
ring MECHANISM, signed from a start past pi	8/8	0.166 (runs 0.004-0.48)	--
torus TORUS_START, signed	8/8	0.001	0.77
torus TORUS_START, |.|	1/8	1.248 (the solved run 0.024)	0.16
torus TORUS_START, signed from a start past pi	2/8	0.737 (solved runs 1.39, 1.78)	--

"Clock-free at the end" holds for runs that succeed from a small start (arm means 0.001-0.018 here, 0.043-0.051 for the rank arms of G3); runs that succeed from a large start can end with a large ratio. In the two solved signed torus runs from a start past pi, 0.43 and 0.99 of the usage sits on channels whose per-pair clock is near pi (|wrap(k)| > 2.6 rad; `clock_table_out.txt`), where E7.1's bound says nothing; the independent audit found the second of them (s1206) at revisit accuracy 1.000 in all four parity strata of (n_-x, n_-y) on 16 fresh walks although the correct key's score flips sign with the parity (mechanism not identified). For unwrapped |.| channels the ratio is at least 1 by construction (E8.1).

The |.| cone inequality |k| >= |m| holds with zero violations on every sub-pi channel (an identity for |.|, listed as a check of the code). Trained signed ring models end on the collective zero set: |z_x + z_y| / (|z_x| + |z_y|) 0.044-0.053, cosine -0.994 to -0.995 (`latent_geometry.py`), MapFormer's antipodal end state (their Figs. 3d and 9b).

The cone boundary. Sub-pi |.| ring channels within 10% of the cone boundary (|delta| >= 0.9 |k| / 2, one move's step near zero): 0.30 (MECHANISM default start), 0.36 (|.| start), 0.25 (GAUGE), against 0.056 at initialisation and 0.00-0.02 for the large starts. |.| models end with 18.8-22.2 one-sided and 13.2-16.0 blind channels of 64 (signed 2.1-4.1 and 8.1-8.9; initialisation 15.6 and 5.0); these counts are unweighted over all 64 channels, unused ones included. The MECHANISM initialisation values (0.056; 15.6 / 5.0) were recomputed with the script's functions by the independent audit; the script itself prints GAUGE's (0.078; 19.8 / 3.1). At fixed clock the largest map a |.| channel can carry sits on that boundary, so collapse onto the wall is the boundary optimum of a clock-map trade-off as well as the sticky kink of E4.2.

Latent-reachable share at initialisation (`latent_reach.py`, MECHANISM seeds 800-807). Share of the needed change of the angle sums that a latent move can produce: signed (to zero) 1.000 on every seed (E8.2(a)); |.| (to the nearest wrap) 0.339 unweighted and 0.561

usage-weighted (random-subspace baseline 0.125). The rest must come from per-channel readout changes. In the held constructed |.| models the wrap targets are carried by the readout (`aobal_geometry.py`): they cancel in 9-13 channels carrying 92-98% of usage, and a readout drawn at initialisation contains 0.3-11% of that target (random-subspace baseline 6%).

Straight paths with the content of a solved model fixed (`path_profiles.py`; ring, MECHANISM signed seeds 800-807; balanced gauge, so a straight line in W_out is a straight line in angle space; 64 fixed walks):

path	**loss at tau = 0, 0.2, ..., 1 (mean of 8)**	**accuracy at tau = 0 -> 1**	**largest rise above the running minimum, per seed**
signed, |default| start -> signed solution	9.46 9.18 8.65 7.50 6.41 0.00	0.30 -> 1.00	0.27 0.04 0.30 0.40 0.48 0.38 0.05 0.36
|.|, same start -> wrapped solution	9.46 8.23 7.87 7.96 7.44 0.00	0.30 -> 1.00	1.92 0.87 1.41 0.60 1.09 1.01 1.14 0.74
signed, collective latent route only	9.46 9.39 9.49 9.25 8.35 2.98	0.30 -> 0.72	0.03 0.00 0.00 1.36 2.15 1.44 0.21 0.02
|.|, the same latent route	9.46 9.41 9.66 9.90 9.68 9.21	0.30 -> 0.30	0.17 1.11 0.18 2.40 2.23 1.68 1.33 0.51

Every |.| seed's largest rise exceeds every signed seed's (complete separation, exact two-sided permutation p 1.6e-4).

The step map's basin with the content frozen (exploratory; `step_basin.py`, `basin_summary.py`; ring, seeds 800-807; content of the solved signed model frozen, only W_in, W_out and omega trained for 2000 steps of batch 32, about 2% of a full run; fresh-stream accuracy). These CPU runs are not deterministic and the step-map search is bistable: re-runs of single cells in the independent audit gave S_abs s801 0.944 -> 0.509 and 0.939, A_abs s803 0.548 -> 0.450 and A_high s804 0.921 -> 0.839, so the per-seed values and the "runs >= 0.9" column do not reproduce; only contrasts much larger than these swings are read:

start	**class**	**accuracy start -> end (mean)**	**runs >= 0.9**
default angles	signed	0.33 -> 0.83	3/8
|default| (the |.| start)	signed	0.29 -> 0.47	1/8
|default|, move latents frozen	signed	0.29 -> 0.40	0/8
|default|	|.|	0.29 -> 0.36	0/8
past pi, U(1.7, 4.5)	|.|	0.36 -> 0.64	1/8
halfway along the straight segment to the wrapped solution	|.|	0.32 -> 0.61	0/8

With the content fixed the start decides the step map's search (default vs |.| start, signed: +0.365, p 0.005; past pi vs small start, |.|: +0.280, p 0.0007; both survive Holm over the eight exploratory tests, two others at p 0.012-0.013 do not). The signed escape from the same-direction start that full training shows (7/8) is not reproduced by the step map alone (signed vs |.| from that start +0.104, p 0.32): in full training the content co-adapts. No run from halfway along the straight segment to the wrapped solution reached 0.9

(end accuracies 0.42-0.82), so within this budget the wrapped solution's basin did not extend that far; a re-run could change single runs.

H8. The sign route (`SIGNROUTE_RESULTS.md`, registered SUPPORTED; per seed in C10)

Six observation-step-free rank-4 arms on the ring at lr 1e-3 and 3e-3 (LRSWEEP's recipe), fresh seeds 1800-1807. The *hybrid* confines the readout to the non-negative orthant, a_c(x) = omega_c |w_c| . z_x, with the latent and hence the increment signed. *abs_tied* is |.| with the move latents tied antipodal, z_y := -z_x, so both moves step omega |w . z| and its code is a pure clock.

lr	signed	|.|	hybrid	abs_tied	primary (hybrid vs |.|, gated on signed vs |.|)
1e-3	8/8, 1.000	0/8, 0.918	8/8, 1.000	0/8, 0.739	Fisher p 0.0002; d +0.082
3e-3	8/8, 1.000	2/8, 0.958	8/8, 1.000	0/8, 0.748	Fisher p 0.007; d +0.042

Hybrid vs signed: d 0.000 at both rates. Descriptive end states (medians): hybrid u-ratio | z_x + z_y| / (|z_x| + |z_y|) 0.043 / 0.002 and cosine(z_x, z_y) -0.996 / -1.000, as signed (0.055 / 0.001; -0.994 / -1.000); |.| 0.661 / 0.664 and -0.133 / -0.165; the hybrid's effective readout is 0% negative and fully row-sign-coherent. Both the absence of the collective route (E8.3) and the far-side cancelling point (E8.1, E4.1) predict this outcome, since the hybrid restores both; the batch localises the cost to the increment and shows that non-negativity in the readout costs nothing.

H9. Exp and softplus increments (`SIGNROUTE_RESULTS.md`, registered breadth cells at fixed rates)

lr	arm	SOLVED	accuracy	d vs signed	label (Holm per test, 4 cells)
1e-3	exp	1/8	0.935	-0.065	signed learns better (Fisher 0.0056, perm 0.0006)
1e-3	softplus	1/8	0.879	-0.122	signed learns better (Fisher 0.0056, perm 0.0042)
3e-3	exp	1/8	0.958	-0.042	signed learns better (Fisher 0.0056; SOLVED test only; 5/7 unsolved runs still descending)
3e-3	softplus	3/8	0.966	-0.034	signed learns better (Fisher 0.0256; SOLVED test only; 2/5 unsolved still descending)

Both start every token at a positive step (mean move step at initialisation exp +1.051, softplus +0.706, signed +0.003), i.e. as index RoPE with a rescaled omega; no best-rate sweep was run for them.

H10. The sign test in a state-space model (G-3; `G3_SSM_RESULTS.md`, registered, `G3_SSM_PREREG.md` + Amendment 1)

Model. Per head (2 heads, 32 complex channels, value width 64), H_t = diag(exp(-i omega Delta(x_t))) H_{t-1} + k_t v_t^T and y_t = Re(conj(q_t)^T H_t), with k_t, q_t the (even, odd) pairs of the real key and query projections; y is RMS-normalised per head, then the attention model's output projection, residual, feed-forward and head. It is computed in the cumsum-of-angles form (E1; Mamba-3 Prop. 3) as causal *unnormalised* rotary linear attention, equal to the recurrence to 1.1e-12 relative in float64 and 8.7e-5 in float32 at

4T. Only the mixer differs from the attention ring arms (softmax with attention dropout replaced by the linear kernel, +128 norm parameters); at a seed all 24 shared tensors start bitwise equal to the attention arm's. No decay (|a| = 1), no output gate, Delta enters only the phase (declared, not Mamba-faithful). Task and recipe: the ring anchors' (32 cells, moves +1/-1, only the moves step, T = 128, 300 epochs, cosine, AdamW 0.05, rank 4), lr 3e-3; fresh seeds 2100-2107; configuration `minimal_h2` (104 runs); independent check before launch (no BLOCKER, 6 MAJOR, 11 MINOR, folded in). Floors: blank 0.496, n-gram 0.554, retrace 0.864.

From scratch (block A).

arm	SOLVED	classes (S / T / D)	accuracy T = 128	T = 512 (descriptive)	RUNAWAY	vs signed
signed	8/8	8 / 0 / 0	1.000	0.999	0	--
|.|	0/8	0 / 7 / 1	0.821	0.457	0	signed learns better (Fisher 0.0002, perm 0.0002; d -0.179)
softplus	1/8	1 / 2 / 5	0.935	0.617	0	signed learns better (Holm: Fisher 0.0014, perm 0.0003)
exp	0/8	0 / 8 / 0	0.473	0.421	8	signed learns better (8 RUNAWAY): no learnability statement
index	0/8	0 / 5 / 3	0.755	0.445	0	integration gate passes

Registered reading: SIGN COST IN THE SSM, worded "not specific to softmax attention (it appears in unnormalised rotary linear attention = this SSM)". RUNAWAY (Amendment 1.4) = final move-step magnitude > 10^3 or a post-warm-up training loss above the memoryless 2.08; 7 exp runs by magnitude (up to 3.6e24), one by loss. Exp's failure at 3e-3 is instability at that rate; the second rate from scratch (1e-3) was run afterwards as block A2 (App. H13: exp 6/8 vs signed 7/8, no sign cost; attention at 1e-3: exp 1/8, softplus 1/8, |.| 1/8, signed 8/8; App. H1, H9). 5 of 7 unsolved softplus runs were still descending (budget-scoped). r(log10 final loss, accuracy) -0.674 at T = 128 and -0.927 at T = 512 over the 40 runs. Mechanism (descriptive, medians): signed cos(z_+, z_-) -1.000, u-ratio 0.001, every non-blind channel cancelling through zero; |.| -0.334, 0.592, 7% cancelling; softplus +0.738, 0.933, 12%.

Existence (blocks H, H2). Each solved signed source (8/8) is converted by E2 (a), (f), (g) with D3's balancing (max |W_in| = max |W_out|; |.| <= 3.7x, softplus <= 1.5x, exp <= 1.2x the source's) and a signed control differing only in the wrap. Frozen: accuracy = source on 8/8 for every class and the control; move-angle error <= 2.3e-6 rad. (f) and (g) fix omega' at 2 pi / ln 2 and 2 pi against the sources' 0.59-0.84 (declared). Move-angle sensitivity, median: |.| 108, softplus 180, exp 224, control 27 (attention AO_bal 87-96). Continuations: 100 epochs on a fresh stream, LOST and HELD as in D6.

lr	class	SOLVED	LOST (of which > 0.3)	max loss (control)	E2 (registered)	E2' (> 0.3 only; secondary)
3e-3	|.|	8/8	8 (8)	0.320-0.678 (0.006-0.039)	perturbed and re-found	perturbed and re-found

lr	class	SOLVED	LOST (of which > 0.3)	max loss (control)	E2 (registered)	E2' (> 0.3 only; secondary)
3e-3	softplus (f)	3/8	8 (8)	1.215-1.481	not held [construction (f)]	not held
3e-3	exp (g)	1/8	8 (8)	1.491-1.811	not held [construction (g)]	not held
1e-3	|.|	8/8	6 (0)	0.033-0.127 (0.001-0.002)	perturbed and re-found	held
1e-3	softplus (f)	8/8	8 (0)	0.050-0.204	perturbed and re-found [construction (f)]	held
1e-3	exp (g)	8/8	8 (0)	0.110-0.165	perturbed and re-found [construction (g)]	held

A non-held softplus or exp label is a property of constructions (f) / (g), not of the class (Amendment 1.2); at 3e-3 most of their continuations were still descending at 100 epochs, so the line between "not held" and "perturbed and re-found" is set partly by the budget. At 1e-3 the registered rule reads "perturbed" only through its 10x-control clause (the SSM control peaks at 0.001-0.002); the |.| peaks lie inside the attention held range (AO_bal 0.024-0.154). Registered joints: |.| at 3e-3 cost, solution perturbed and re-found; the 1e-3 joint was declared cross-rate (from scratch at 3e-3) and read the same, and is superseded by the same-rate joint of block A2 (App. H13: |.| and softplus cost, solution perturbed and re-found; exp none); exp forms none at 3e-3 (RUNAWAY). The registered "search cost" joint (needs held) did not form; on E2' |.| is held at 1e-3 (secondary).

H11. Route or barrier (SIGNSEP; `SIGNSEP_RESULTS.md`, registered, `SIGNSEP_PREREG.md` + Amendments 1-3)

Design. Five observation-step-free rank-4 arms on the ring (32 cells, moves +1/-1; LRSWEEP's recipe: T = 128, 300 epochs, cosine, AdamW 0.05) at lr 1e-3 and 3e-3, fresh seeds 2300-2307, 80 runs; independent check before launch (5 MAJOR, 7 MINOR, 6 NOTE, folded in). Every arm starts from one common start, MECHANISM's balanced-gauge |.| start (App. H3; median |k_c| 0.52 rad), at which signed (S), |.| (A), W and F give bitwise-identical logits. **W** = |.| with one learnable offset per move, a_c(x) = omega_c | w_c . z_x + tau_x pi / |omega_c||, tau = 0 at the start and exempt from weight decay (Amendment 2): it keeps the barrier and has a collective route to the wrap (E8.6). **F** = signed with the move latents frozen: no barrier, no collective route (each channel's clock depends on its own W_out row and omega only). **P** = W's class with tau_x = tau_y = 0.75 at the start, every channel past pi (median |k_c| 5.23): control C1. Gates: G0 k_S >= 6; G1 k_A <= 3 and better(S, A); classes S-LIKE / A-LIKE / PARTIAL / UNMEASURED against the anchors; C1 k_P >= 6 and median final tau_sum of P >= 1.55 (Amendment 3). better(a, b) = Fisher < .05 with k_a > k_b, or permutation p < .05 with d >= +0.05.

Outcome per rate (SOLVED, mean accuracy at T = 128; none diverged).

lr	signed	|.|	W (route, barrier)	F (no route, no barrier)	P (control)	reading
1e-3	5/8, 0.981	0/8, 0.930	2/8, 0.932	3/8, 0.939	5/8, 0.960	UNREADABLE (G0: 5 < 6)
3e-3	8/8, 1.000	3/8, 0.973	**0/8, 0.953**	**4/8, 0.971**	7/8, 0.985	UNMEASURED

At 3e-3: G1 S vs A Fisher p 0.026 (k_A at the bound). W A-LIKE: S vs W Fisher p 0.0002, perm p 0.0002, d +0.048; W vs A Fisher p 0.20, perm p 0.10, d -0.021. F UNMEASURED: S vs F Fisher p 0.077, perm p 0.026 with d +0.029 (below 0.05); F vs A Fisher p 1.0. C1 passes: P 7/8, median tau_sum 1.999. Half-readings: barrier BARRIER BLOCKS (a route is not enough), route UNMEASURED. No definite row of the 2x2, so the rate and the overall reading are UNMEASURED. Scope: ring, this start, 300 epochs, lr 1e-3 and 3e-3. Signed from this start at 1e-3 in earlier batches: 7/8 (MECHANISM), 7/8 (ACCOUNTS T2); from the default start 8/8 (LRSWEEP, SIGNROUTE).

End states (descriptive, medians at 3e-3).

arm	u-ratio	cos(z_x, z_y)	wrapped clock rms	cancelling (through zero / by wrap)	tau_sum
signed	0.002	-1.000	0.001	1.000 (1.000 / 0)	--
|.|	0.878	+0.536	1.228	0.051 (0.032 / 0)	--
W	0.782	+0.210	1.653	0.016 (0 / 0)	0.547 (per seed 0.13-0.70)
F	0.707 (frozen)	0.000 (frozen)	0.962	0.182 (0.170 / 0.013)	--
P	0.036	-0.997	0.011	1.000 (0 / 1.000)	1.999 (2.00 on every seed)

In a |.| class the "through zero" share counts near-blind one-sided channels, not cancellation. P ends at the end point of W's collective route (u -> 0, tau_sum = 2); W's offsets move toward it but stop below 1 on every seed. At 1e-3 W's tau_sum stays at 0.01-0.11 (its two SOLVED runs are off the sheet) and P's moves from 1.5 to 1.63-1.71. r(log10 final loss, accuracy) -0.782 at T = 128 and -0.603 at T = 512 over 80 runs. `experiment_audit` windowed classes (rates pooled) agree with the registered ones in every arm; one DESCENDING run each in |.|, W, F and P at 3e-3.

Per seed (accuracy at T = 128, class), 3e-3: signed 1.000S on every seed; |.| 0.934T 0.960T 1.000S 0.963T 1.000S 0.985D 1.000S 0.945T; W 0.948T 0.972D 0.963T 0.959T 0.957T 0.974T 0.933T 0.913T; F 0.960T 0.916T 1.000S 1.000S 0.955D 1.000S 0.983S 0.956T; P 1.000S x 6, 0.881D (s2306), 1.000S. 1e-3: signed 0.968T 1.000S 0.925T 0.956T 1.000S 1.000S 1.000S 1.000S; the |.| arm all STALLED (0.859-0.988); W 0.876T 0.865T 0.929T 1.000S 0.923T 0.948T 1.000S 0.917T; F 0.901T 0.927T 0.989S 1.000S 0.869T 1.000S 0.913T 0.915T; P 0.899T 1.000S 1.000S 0.874T 1.000S 1.000S 0.911T 1.000S.

H12. Learning rates on the torus (G-4 TORUS_LRSWEEP; registered, `TORUS_LRSWEEP_RESULTS.md`, `TORUS_LRSWEEP_PREREG.md` + Amendment 1)

Design. Observation-step-free shared rank-4 arms, signed (SO) and |.| (AO), on GENERALITY's torus recipe (64 x 64, T = 1024, 900 epochs, cosine, 1 layer, d 128, fast attention) at lr 1e-3, 3e-3 and 1e-2 with AdamW weight decay 0.05 (decay exponent sum(lr_t) x wd = 2.4 / 7.2 / 24.1), fresh seeds 2400-2407, 48 runs; at a seed every arm and rate shares the initial weights and the training stream. Independent check before launch (2 MAJOR, 9 MINOR, folded in). Registered B1 as in H1 (LRSWEEP's rule and code, unchanged): each arm's best rate on training loss only (most SOLVED, then lowest mean final-5% loss, then the lower rate); selection redone inside the null over all 12,870 relabellings of the 16 per-seed rate profiles; decision p = max(selection-aware, naive). Stage 2 (a decay-matched control at |.|'s rate) was registered for a selected |.| rate >= 1e-2. UNSTABLE: a post-warm-up epoch above the seed's map entropy + 0.05, a final move step > 1e3, or a final loss > 1.9. The extension pilot (seeds 2492-2493, outside the batch) found |.| at 3e-2 collapsed to the memoryless loss on both seeds, so the grid was not extended.

learning rate	signed SOLVED, accuracy	|.| SOLVED, accuracy	per-rate label (Holm)
1e-3	**8/8, 1.000**	0/8, 0.855	signed learns better (Fisher and perm p 0.0002)
3e-3	0/8, 0.975 (8 descending)	**0/8, 0.890**	unreadable (signed < 4/8); perm p 0.0002
1e-2	0/8, 0.968 (8 descending)	0/8, 0.768 (2 UNSTABLE)	unreadable (signed < 4/8); perm p 0.0002

Registered B1: |.| at 3e-3 (0/8 at every rate; its lowest loss and highest accuracy are both at 3e-3, an interior rate) against signed at 1e-3 (the grid's lowest rate, at ceiling), d -0.110, selection-aware p 0.0002 (SOLVED) and 0.0095 (accuracy), naive Fisher and permutation 0.0002: SIGNED LEARNS BETTER, reading COST AT BEST RATES, with 5 of the 8 |.| runs at 3e-3 still descending (budget-scoped). Stage 2 not triggered. Declared secondaries agree: SOLVED at loss 0.1, 0/8 vs 8/8 (accuracy p 0.0073); accuracy at T = 2048, 0.817 vs 0.998 (selection-aware p 0.010); cross-fit, both halves 0/4 vs 4/4 (Fisher p 0.029). Revisit strata at T = 1024 (short / long / wrap-only): signed at 1e-3 1.000 / 1.000 / 1.000, |.| at 3e-3 0.957 / 0.618 / 0.495 (floor 0.507). Usage-weighted wrap-cancelling share: signed 1.00 at every rate, |.| 0.36 / 0.66 / 0.61. r(log10 final loss, accuracy) -0.646 at T = 1024 over 48 runs. `experiment_audit` windowed classes agree with the registered ones in every cell.

Per seed (accuracy at T = 1024, class; ! UNSTABLE):

```
seed | 1e-3:SO     | 1e-3:AO     | 3e-3:SO     | 3e-3:AO     | 1e-2:SO     | 1e-2:AO
2400 | 1.0000 SOLV | 0.8605 STAL | 0.9782 DESC | 0.9288 DESC | 0.9466 DESC | 0.8288 DESC
2401 | 1.0000 SOLV | 0.8948 STAL | 0.9622 DESC | 0.9546 DESC | 0.9966 DESC | 0.7768 DESC
2402 | 1.0000 SOLV | 0.8233 STAL | 0.9963 DESC | 0.9028 DESC | 0.9513 DESC | 0.8349 DESC
2403 | 1.0000 SOLV | 0.8421 STAL | 0.9992 DESC | 0.9603 DESC | 0.9518 DESC | 0.8305 DESC
2404 | 1.0000 SOLV | 0.9279 DESC | 0.9625 DESC | 0.8094 STAL | 0.9958 DESC | 0.5137 STAL!
2405 | 1.0000 SOLV | 0.7571 STAL | 0.9642 DESC | 0.8310 STAL | 0.9678 DESC | 0.8226 DESC
```

```
2406 | 1.0000 SOLV | 0.7969 STAL | 0.9624 DESC | 0.8948 DESC | 0.9805 DESC | 0.8237 STAL!
2407 | 1.0000 SOLV | 0.9348 DESC | 0.9771 DESC | 0.8411 STAL | 0.9505 DESC | 0.7110 STAL
```

H13. The SSM at the second rate (G-3 block A2; registered, `G3_SSM_A2_RESULTS.md`, `G3_SSM_PREREG.md` Amendments 2-3)

H10's model, task and recipe from scratch at lr 1e-3 (40 runs, seeds 2100-2107, kalman GPU 0, 2026-10-24/25, code `ab872fb`; model, trainer and library code md5-equal to block A's). Not results-blind: launched because block A's exp cell was RUNAWAY; design and analysis fixed before it ran, and an independent check (Amendment 3) fixed the Holm family ({|.|, softplus, exp}, |.| at raw p) and the exp branches before launch.

arm	SOLVED	classes (S / T / D / R)	accuracy T = 128	T = 512 (descriptive)	RUNAWAY	vs signed (registered)
signed	7/8	7 / 1 / 0 / 0	0.994	0.764	0	-- (G0 passes)
exp	6/8	6 / 2 / 0 / 0	0.985	0.753	0	no sign cost (Fisher 1.0, perm 0.72; d -0.009)
softplus	0/8	0 / 8 / 0 / 0	0.938	0.533	0	signed learns better (Holm: Fisher 0.0042, perm 0.0006; d -0.056)
|.|	0/8	0 / 8 / 0 / 0	0.756	0.431	0	signed learns better (Fisher 0.0014, perm 0.0002; d -0.238)
index	0/8	0 / 6 / 1 / 1	0.709	0.415	0	integration gate passes

Registered reading: SIGN COST IN THE SSM at 1e-3 for |.| and softplus; for exp, per Amendment 3.2, "no sign cost for exp at lr 1e-3", read with the prereg's caveat that exp starts past pi on 13 of 32 channels per head; exp's 3e-3 cell stays instability. The 2-cell Holm sensitivity gives the same labels. Same-rate joints with H2: |.| and softplus cost, solution perturbed and re-found; exp none. r(log10 final loss, accuracy) -0.689 / -0.891 at T = 128 / 512 over blocks A and A2. Signed at 1e-3 solves on loss with partial codes on four seeds (cosine -0.46 to +0.11, T = 512 0.43-0.78); the other four end antipodal.

Mechanism at 1e-3 (registered readouts, medians; the last two columns analysis, `docs/audits/2026-10-25/g3_a2_exp_mech.py`):

arm	u-ratio	cos(z_+, z_-)	cancel share (by wrap)	max move step	cancelling channels, over 8 seeds	of them started past pi
signed	0.261	-0.728	0.74 (0.00)	2.0	329 of 458 (8 by wrap)	-- (through zero)
exp	0.895	+0.601	0.22 (0.21)	5.2	95 of 484 (92 by wrap)	90 (2 omega_0 > pi)
softplus	0.887	+0.569	0.12 (0.11)	3.0	63 of 464 (58 by wrap)	58 (2 ln 2 omega_0 > pi)
|.|	0.640	-0.228	0.05 (0.01)	1.7	20 of 422 (7 by wrap)	--

exp ends with every step positive (minimum 0.004-0.33 per run) and bounded (maximum 4.1-7.8, against up to 3.6e24 at 3e-3), its cancelling channels 1-3 full turns round; omega

ends at 0.44-0.49 of its initial value in every arm (weight decay). Per seed (accuracy at T = 128 and class):

```
arm       2100    2101    2102    2103    2104    2105    2106    2107
signed    0.977T  1.000S  0.999S  0.993S  0.992S  0.997S  0.995S  1.000S
exp       0.998S  0.999S  0.908T  0.999S  0.999S  0.994S  0.988T  0.999S
softplus  0.984T  0.951T  0.886T  0.957T  0.938T  0.899T  0.966T  0.924T
abs       0.736T  0.819T  0.741T  0.744T  0.739T  0.822T  0.720T  0.724T
index     0.693T  0.713T  0.699T  0.709R  0.699D  0.690T  0.720T  0.747T
```

H14. Exp and softplus in attention on the torus (EXPTORUS; registered, `EXPTORUS_RESULTS.md`, `EXPTORUS_PREREG.md` + Amendments 1-2)

Design. H12's recipe and class unchanged (torus 64 x 64, T = 1024, observation-step-free shared rank 4, 1 layer, 900 epochs, cosine, AdamW weight decay 0.05 coupled to the rate, fast attention), four arms that differ only in the function on the step readout s = W_out W_in e(move): signed s, |s|, exp(s), softplus(s); each at lr 1e-3 and 3e-3; fresh seeds 3400-3407, 64 runs (kalman, 2026-10-26, code `782ba4f`). At a seed every arm and rate shares the initial weights and the training stream; the initial *steps* differ (exp(0) = 1, softplus(0) = ln 2), so exp starts with 11 of 32 channels per head past pi (34%; 41% on the SSM ring of H13) and softplus with 8. B1 as in H12 (each arm's best rate on training loss; selection redone over all 12,870 relabellings; decision p = max(selection-aware, naive)), Holm over the two primaries (exp, softplus) per route; anchor gate |.| vs signed must read signed learns better. UNSTABLE as H12, with non-finite loss added. Independent check before launch (2 MAJOR, 5 MINOR, folded in as Amendment 1). *Disclosure (Amendment 2):* a pilot on outside seeds (exp only, 2 seeds per rate: 0/4 solved, both 3e-3 runs unstable) ran before the batch, so the exp readings are partly pilot-informed; design, thresholds and labels were not changed.

arm	best rate	SOLVED	accuracy T = 1024	d vs signed	decision p, SOLVED / accuracy (Holm)	label
signed	1e-3	**8/8**	**0.993**	--	--	--
|.| (gate)	1e-3	0/8	0.858	-0.135	0.0002 / 0.0002 (raw)	signed learns better: gate passes
exp	1e-3	**0/8**	0.916	-0.077	0.0003 / 0.0006	**signed learns better** (6/8 descending)
softplus	3e-3 (grid top: not bracketed)	**0/8**	0.908	-0.085	0.0003 / 0.11	**signed learns better** on SOLVED (5/8 descending)

Registered batch reading: BOTH PAY (the exp and softplus content-dependent increments pay in attention on the torus; the SSM's exp no-cost does not transfer). Per rate (declared; Holm over the two rates):

learning rate	signed	|.|	exp	softplus
1e-3	**8/8, 0.993**	0/8, 0.858	0/8, 0.916 (1 UNSTABLE)	0/8, 0.849
3e-3	0/8, 0.964 (8 descending)	0/8, 0.865	0/8, 0.743 (**6 UNSTABLE**)	0/8, 0.908

At 1e-3 (G-3 A2's rate, no selection) each non-negative arm is signed learns better (Fisher 0.0002, Holm 0.0003); the 3e-3 cells are unreadable (signed 0/8). All 7 UNSTABLE runs are exp, RUNAWAY by loss (none diverged; largest step 35). Declared secondaries agree: SOLVED at loss 0.1, 0/8 vs 8/8 for every arm; accuracy at T = 2048, exp 0.763 and softplus 0.823 vs 0.980 (selection-aware p 0.0002, 0.020); both cross-fit halves 0/4 vs 4/4 (Fisher p 0.029). Revisit strata at T = 1024 (short / long / wrap-only) at the selected rates: signed 0.992 / 0.993 / 0.996, exp 0.977 / 0.695 / 0.524, softplus 0.971 / 0.659 / 0.529 (blank ~0.51). r(log10 final loss, accuracy) -0.618 over 64 runs; `experiment_audit` windowed classes agree with the registered ones in every cell.

Cancelling census at the selected rates (declared secondary; (pair, channel) cells over 8 seeds; non-blind |wrap(first move angle)| > 0.05, cancelling |wrap(pair sum)| < 0.1; past-pi set from each run's initial weights):

arm, rate	non-blind	cancelling	by wrap / through zero	from the past-pi start set	P(cancel | past-pi start) vs not	usage-weighted cancel	opposite-sign
signed, 1e-3	693	693	0 / 693	36	1.00 vs 1.00	1.00	1.00
|.|, 1e-3	439	46	25 / 21	17	0.31 vs 0.08	0.19	0.09
exp, 1e-3	443	107	97 / 10	95	0.31 vs 0.09	0.81	0.26
softplus, 3e-3	159	64	57 / 7	46	0.68 vs 0.20	0.89	0.41

exp cancels as in H13 (by a wrap, on channels that started past pi) but on a minority of non-blind cells and without a map; omega ends at 0.11 (exp) and 0.02 (softplus) of its initial value under the coupled decay. Not separated: whether the SSM escape of H13 is the mixer, the task (ring vs torus) or the start; the two batches are never pooled.

Per seed (accuracy at T = 1024, class; ! UNSTABLE):

```
seed | 1e-3:SO     | 1e-3:AO     | 1e-3:EXP    | 1e-3:SPL    | 3e-3:SO     | 3e-3:AO     |
3e-3:EXP     | 3e-3:SPL
3400 | 0.9988 SOLV | 0.8863 STAL | 0.9274 DESC | 0.7067 STAL | 0.9778 DESC | 0.9363 DESC |
0.8626 DESC! | 0.9207 DESC
3401 | 1.0000 SOLV | 0.9064 STAL | 0.9530 DESC | 0.8102 STAL | 0.9588 DESC | 0.8553 STAL |
0.8046 STAL  | 0.8879 STAL
3402 | 0.9420 SOLV | 0.8790 STAL | 0.9092 STAL | 0.9255 DESC | 0.9595 DESC | 0.7783 STAL |
0.5429 STAL! | 0.9837 DESC
3403 | 1.0000 SOLV | 0.7671 STAL | 0.9231 DESC | 0.9221 DESC | 0.9493 DESC | 0.8987 DESC |
0.8510 STAL! | 0.9501 DESC
3404 | 1.0000 SOLV | 0.8243 STAL | 0.8757 DESC | 0.6904 STAL | 0.9567 DESC | 0.8957 DESC |
0.8651 DESC  | 0.8680 RISI
3405 | 1.0000 SOLV | 0.8395 STAL | 0.8792 STAL!| 0.9104 DESC | 0.9769 DESC | 0.8328 STAL |
0.7891 STAL! | 0.8775 STAL
3406 | 1.0000 SOLV | 0.9158 DESC | 0.9213 DESC | 0.8743 STAL | 0.9672 DESC | 0.8633 STAL |
0.6247 STAL! | 0.9410 DESC
3407 | 1.0000 SOLV | 0.8465 STAL | 0.9380 DESC | 0.9517 STAL | 0.9645 DESC | 0.8583 STAL |
0.6036 STAL! | 0.8336 DESC
```

I. CARoPE, CoPE and Selective RoPE

CARoPE (`model_mono_baselines.py`, `CARoPEF`). Per head h and token t a scalar rate f_h(x_t) = 1 / (softplus(x_t W)_h + 1) in (0, 1) replaces RoPE's base frequency, so channel i advances by f_h(x_t)^i per token; initialised as index RoPE (W = 0, a per-head bias with 1 / (softplus(b) + 1) = 10000^(-2 / d_head)). Every token, observations included, advances the phase; the layer is MapWM's, so only the phase differs. Proposition 8 (E5): no channel can read every return. Registered (GENERALITY): torus 0.667 (0/8), below index RoPE (0.722, -0.055, p 0.040); ring 0.741 against index 0.730 (+0.011, not resolved).

CoPE (`CoPEF`). Golovneva et al.'s Listing 1 in torch: per layer one module shared by the heads; gates g_ij = sigmoid(q_i . k_j / sqrt(d_h)); count p_ij = sum_{k=j}^{i} g_ik, clamped to 2047; position logit by linear interpolation of q_i . e[floor p] and q_i . e[ceil p], table e at zeros initially; added to the attention logits; no rotary position. *Representability (constructed):* gates 17/16 and 15/16 per (move, observation) pair make p_ij mod 2 a function of net displacement, read by a period-2 table (1.000 revisit accuracy). The exact solutions form a lattice (P7 of v3: period 2 or 3).

batch	task	CoPE	SOLVED	comparison
GENERALITY (700-707)	ring	0.887	0/8	signed 1.000, 8/8 (-0.113, p 0.0002); index 0.730 (+0.157, p 0.0002)
WRAP_COPE (900-907)	ring, table at zeros / sinusoid past pi / below pi / slope-matched	0.840 / 0.913 / 0.867 / 0.943	1/8, 2/8, 1/8, 0/8	contrasts not resolved (bimodal seeds)
ACCOUNTS T4 (1100-1115)	256-cell line (never wraps) / ring	0.914 / 0.873	1/16 / 0/16	line: signed 1.000, 8/8 (Fisher p < 1e-4); index 0.772, 0/8

On both line and ring several CoPE seeds reach 0.99-1.00 held-out accuracy without meeting the loss criterion (at a held-out accuracy cut of 0.999 the line reads 5/16 and the ring 2/16; analysis). MapFormer v4 Table 2 reports CoPE failing 1D and 2D navigation (one layer 0.63 +/- 0.21 in 1D and 0.65 in 2D; four layers 0.88 and 0.76).

Selective RoPE (`SROPE_TORUS_RESULTS.md`, registered). Faithful Selective RoPE (Movahedi et al. 2026), the module reproduced from the authors' code to 6e-6, in the authors' best language-model configuration (SRoPE), the same without the SiLU (SRoPE_NoSiLU), the authors' state-tracking configuration (SRoPE_ST: low rank 32, no SiLU, no gate, a learnable bias of +1 rad per token per channel at initialisation) and our earlier generator swap (SRoPEGen), against MapWM (signed, rank 4) and index RoPE in one batch; torus, T = 1024, 900 epochs, seeds 1700-1707; own-stream floors blank 0.509, n-gram 0.578, retrace 0.845.

arm	accuracy T = 1024	SOLVED	classes	short / long / wrap-only	T = 2048
MapWM	1.000	8/8	all SOLVED	1.000 / 1.000 / 0.999	0.992
SRoPEGen	0.985	5/8	5 SOLVED, 2 DESC, 1 STAL	0.997 / 0.975 / 0.865	0.951
SRoPE_NoSiLU	0.969	2/8	2 SOLVED, 6 DESC		0.861

arm	accuracy T = 1024	SOLVED	classes	short / long / wrap-only	T = 2048
				0.987 / 0.952 / 0.805	
SRoPE (best LM config)	0.961	0/8	all DESCENDING	0.992 / 0.937 / 0.659	0.807
SRoPE_ST	0.769	0/8	7 STAL, 1 DESC	0.788 / 0.705 / 0.628	0.724
index RoPE	0.725	0/8	all STALLED	0.775 / 0.470 / 0.465	0.561

Registered: SRoPE vs MapWM fails to learn the map (d -0.039 [-0.050, -0.030], perm p 0.0002; 0/8 vs 8/8, Fisher p 0.0002); SRoPE beats index RoPE (+0.236, p 0.0002); it uses its phase (angle set to zero: 0.961 -> 0.507, p 0.0078). Declared secondaries: the SiLU makes no detected difference (p 0.57); SRoPEGen vs MapWM not resolved (5/8 vs 8/8, p 0.20); SRoPE_ST fails against MapWM and is not distinguished from index RoPE (+0.044, p 0.23); at a loss threshold of 0.1 SRoPE vs MapWM is still 0/8 vs 8/8. Descriptive readouts: an observation's step is 0.86-0.96 of a move's in SRoPE / NoSiLU (0.45 SRoPEGen, 0.027 MapWM); opposite moves cancel (wrap-aware) in 0.75-0.80 of channel pairs (MapWM 1.00); the angles vary with context through the causal convolution (context CV 0.11-0.13, MapWM 0.00); SRoPE_ST's learned per-token angles have negative-mass share 0.00 for every token class and cancel in 0.00 of channels, so its bias keeps the step effectively non-negative.

J. Text world

Task (`TEXTWORLD_RESULTS.md`, `train_textworld.py`). The 64 x 64 torus walk rendered as English: 58 words, three synonyms per direction, movement-free verbs, fillers and asides; T = 1024 words (about 132 steps); held-out map; predict the observation word at revisits. A direction word appears only inside a movement clause (context-free steps). Floors: best constant 0.505, reversal-copy 0.597, retrace 0.934 (analysis; J6). Recipe: the torus recipe (A2), path MapWM r4 at one layer against index RoPE at one and two layers; 200 evaluation walks.

Step tables per seed (declared secondaries, `fig_textworld_out.txt`; "minus common" = opposition after removing the component shared by the four direction steps; ablation columns: held-out accuracy, eval mode without the dropout scale):

seed	type	raw opposition	minus common	common / north	accuracy	non-direction steps zeroed	direction steps zeroed
0	clock	1.730	0.0071	0.711	0.881	0.191	0.388
1	map	0.083	0.0029	0.041	0.998	0.893	0.505
2	clock	1.434	0.0064	1.144	0.999	0.665	0.482
3	clock	1.707	0.0022	0.480	0.995	0.673	0.463
4	clock	1.768	0.0052	0.501	0.999	0.539	0.445
5	map	0.120	0.0050	0.060	1.000	0.898	0.505
6	map	0.080	0.0117	0.040	1.000	0.950	0.505

seed	type	raw opposition	minus common	common / north	accuracy	non-direction steps zeroed	direction steps zeroed
7	map	0.119	0.0057	0.059	0.882	0.586	0.498

On the clock seeds the common component is parallel to the verbs' step (cosine +0.999 to +1.000), and 31-38 of 64 phase channels drift between visits to the same cell (2-4 on the map seeds). Synonym cosine 0.9996-1.0000 on every seed.

The paragraph and table above use the registered batch. The analyses below (`docs/paper/expansion/scripts/language/`) read the 8 registered text-world path runs (seeds 0-7) and the 8 MapWM, 8 NormStep and 8 DirOnly runs of TW_NORMSTEP (seeds 10-17) on a held-out map (np seed 424242, 40 walks of 1024 words unless stated). The manual forward used for the manipulations reproduces every model's logits exactly, and the role-annotated generator reproduces the task generator token for token. "Re-scored" = with the x1.111 attention-dropout scale (C9).

J1. The step table by word class (analysis; `tw_step_account.py`)

Steps omega-scaled, relative to the clause-level direction differential (one move):

word class	map-type models (14 learned-step)	clock-type models (10)
direction words (differential)	1.00 by normalisation; opposites cancel 0.001-0.013	same
adverbs, fillers	0.004-0.03	0.004-0.025
`and` / `a` / `she` (shared by fillers, seeing phrases, asides)	0.02-0.07	0.02-0.08
verbs, seeing words, object words, `.` (once per move)	0.02-0.14	0.24-1.49
aside-only words (`thought`, `about`, `remembered`, `story`)	0.07-0.16 per word	0.41-1.33 per word
aside sentence, net (usage-weighted; equal by construction to the aside noun's offset from its place)	0.05-0.12 of a move	0.04-0.16 of a move
DirOnly oracle: every non-direction word	0 by construction	--

The clause tick (what every move adds whatever its direction) is 0.03-0.23 of a move on map-type models and 3.3-6.8 moves (raw norm, all channels) on clock-type models, parallel to the verbs' step (cosine +0.999 to +1.000 on all 10).

J2. The tick predicts the drift; the clock counts moves (analysis; `tw_step_account.py`, `tw_clock_unit.py`)

Predicted drift per channel = mean |wrap(lag x tau_c)| over consecutive-visit pairs (median lag 8 moves, 1,221 pairs); observed = mean |wrap(dtheta_c)| at those pairs. Classified at 1 rad they agree on 0.98-1.00 of channels on each of the 32 models. On the 22 map and DirOnly models both are near zero, so the agreement there is close to an identity; the informative evidence is the 10 clock models, where the predicted and observed drifting-channel counts match exactly. Clock models drift in 26-38 channels

between consecutive visits (registered text world s0 / s2 / s3 / s4: 26 / 29 / 35 / 36; MapWM s11 / s13 / s14 / s15: 38 / 27 / 28 / 30; NormStep s11 / s16: 27 / 28); map models in 0-3. (The registered 31-38 of the CORRECTED block in `TEXTWORLD_RESULTS.md` counts drift from the first visit.) Regressing each move's net phase in the drifting channels on direction, aside present, filler count and adverb present: an aside sentence adds 0.009-0.062 of a tick, a filler word 0.001-0.007, an adverb word 0.001-0.007 (10 clock models). A word counter would give fillers about 1/8 of a tick each and a sentence counter would give asides about one tick: the clock counts moves. On all 10 clock models 2-6 channels tick by exactly pi per move (e.g. -3.141, -3.140 rad); on the even 64-torus a pi tick encodes the parity of the cell and is invisible to revisits (E7.2), and these channels carry up to 0.24 of retrieval usage. The registered finding that zeroing every non-direction step costs the clock seeds 0.33-0.69 and the map seeds 0.05-0.30 tracks the direction words' common component in the used channels (0.06-0.34 of a move on clock seeds, 0.01-0.03 on map seeds), which the other words cancel within the move (Spearman +0.905 across the 8 registered seeds, exact permutation p 0.0046; `tw_common_used.py`, p from `scripts/whatwhere/whatwhere_stats.py`; the used channels are those whose tick is 0 mod pi, a post hoc rule).

J3. The clock is idle (analysis; `usage_tick.py`, `tw_attention_asides.py`)

79-99% of retrieval usage sits in channels whose tick is 0 mod pi. Removing the drifting channels' q and k components from the score (re-scored accuracy):

model	intact	clock channels removed	as many map channels removed (5 draws)	only clock channels kept
text world s0 (eval mode 0.875, below ceiling)	0.973	0.717	0.43 [0.24, 0.58]	0.07
text world s2	0.999	0.990	0.25 [0.03, 0.63]	0.03
text world s3	0.998	0.993	0.06 (all 29 map channels)	0.06
text world s4	0.999	0.999	0.03 (all 28)	0.03
MapWM s11	0.989	0.985	0.48 (all 26)	0.48
MapWM s13	0.997	0.996	0.65 [0.50, 0.84]	0.48
MapWM s14 (eval mode 0.960)	0.995	0.902	0.25 [0.03, 0.72]	0.03
MapWM s15	0.995	0.994	0.07 [0.02, 0.23]	0.02
NormStep s11	0.998	0.996	0.46 [0.20, 0.60]	0.44
NormStep s16 (eval mode 0.835)	0.982	0.809	0.22 [0.04, 0.48]	0.04

On the 7 clock models at ceiling in eval mode the clock costs at most 0.009 to remove; the 3 below ceiling still lean on it (0.09-0.26).

J4. The used-channel ratio across data (analysis; `crossover_usage.py`, `usage_tick.py`, `tw_alpha_drift.py`)

Usage-weighted median of rho_c (E13) at the training length, all MapWM with a signed step:

data	used-channel rho	usage in clock-dominated channels	seeds
2D torus, 1 layer, matched length (SIGN_MATCHED signed)	0.064-0.079	0.000	8
text world, 1 layer (23 of 24 learned-step models; the unconverged tw s0: 2.5)	0.06-0.23	0.01-0.30	24
code, bytes, 9 layers (trained at 2048)	4.5-6.6	0.50-0.55	3
TinyStories, words, 6 layers (trained at 512)	9.9-12.3	0.42-0.43	3
k-back recency with fillers, 1 layer (signed rank 4)	21.8-22.1	0.93-1.00	8

Clock-dominated: rho_c > 1 and more than half a turn per training window. The pooled growth exponent alpha does not separate map from clock data: it reads 0.61-0.74 on text-world map seeds (the true walk: 0.668), 0.98-1.02 on text-world clock seeds, which carry a perfect map, 1.00-1.01 on TinyStories and 0.99-1.01 on code. The used-channel rho separates the clock seeds (0.07-0.23) from TinyStories (9.9-12.3).

J5. Asides: the offset is what beats the oracle (analysis; `tw_attention_asides.py`, `tw_attention_asides_dironly.py`)

`TW_NORMSTEP_RESULTS.md` (registered) found that DirOnly, whose steps are restricted to the twelve direction words, stops at 0.970-0.974 on 8/8 seeds because an aside's noun sits at the place's phase, and that learned-step models make none of these errors; re-scored, every learned-step arm beats DirOnly (-0.024, p 0.0002). Two manipulations at evaluation, each keeping the phase after the aside unchanged:

- *Collapse* (learned models): every aside word is put at its clause's object-slot phase. The 19 learned-step models at ceiling in eval mode fall from 0.989-1.000 to 0.952-0.975, and 71-100% of their errors then name an object told in an aside at that cell (before: 0-13 errors, one of that kind, in MapWM s13). The 5 models below ceiling respond partly (0.966-0.994). The 19 / 5 split at eval-mode accuracy 0.98 is a post hoc cut (it falls in a gap, 0.960 against 0.989).
- *Lift* (DirOnly): intact, DirOnly scores 0.962-0.981 on this stream and 100% of its errors are aside errors on 8/8 seeds. Offsetting its aside words by half a move in-plane gives 0.999-1.000 on 8/8; by one move off the map plane 1.000 on 7/8 (one seed 0.972); by a quarter move off-plane 0.963-0.996 (a variant added after the main run). The lift sizes are free choices, not fitted.

At each revisit query the 19 learned-step models at ceiling put 0.55-0.76 of their attention on the object slot of the most recent visit to the cell and 0.004-0.035 on aside nouns told there; DirOnly puts 0.041-0.077 on those aside nouns.

J6. Retrace on the text world (analysis; `tw_retrace_floor.py`, `tw_strata_200.py` -> `tw_strata_200walks_out.txt`)

The retrace rule of B, ported to the text world, scores 0.9337 on the registered evaluation stream (0.9365 on TW_NORMSTEP's), where the constant and reversal-copy rules reproduce the registered 0.5054 / 0.5965. On all 200 walks of that stream (6,381 revisits) the rule applies to 5,525 (86.6%) and is never wrong when it applies; on the other 856 (13.4%), places reached by another route than undoing the last run, it predicts 'nothing',

which is right on the 50.6% of them whose cell is blank. Accuracy by stratum (mean over 8 seeds; fresh seeds 2-7 in brackets):

model (registered batch, n = 8)	all	retrace applies (5,525)	retrace cannot reach (856)	... cell holds an object (423)	... blank cell (433)
path, 1 layer, re-scored	0.993	0.999	0.956 (0.978; min 0.786 at s0)	0.925 (0.967; s0 0.617, s7 0.872)	0.985
path, 1 layer, eval mode	0.969	0.975	0.934 (0.958)	0.884 (0.927)	0.984
index RoPE, 2 layers	0.772	0.828	0.411 (0.34-0.55)	0.129 (0.09-0.15)	0.687
index RoPE, 1 layer	0.505	0.513	0.454	0.004	0.894
best constant	0.505 ('nothing')	--	0.506 ('nothing')	0.088 ('cat')	1.000 ('nothing')

The object-cell column is the stratum "retrace wrong" of the first version of this analysis (`tw_strata.py`, first 100 walks: 5.9% of 3,055 revisits, path 0.920 / 0.961 fresh, RoPE 2L 0.138, constant 0.112). Its best constant excludes 'nothing' by construction, and one index layer scores near zero there because it predicts 'nothing'. On the whole non-retrace stratum, which needs no outcome-defined split, two index layers score below the constant. Two-layer index RoPE learns part of retrace (its best seeds reach 0.927 and 0.926 overall, against retrace's 0.934) and nothing that needs the map. Seeds at ceiling score 0.936-1.000 on the object-cell stratum.

K. What and where; the leak

K1. What and where at T = 1024 (analysis; `docs/audits/2026-10-20/WHAT_WHERE_T1024.md`, `causal_whatwhere_T1024.py`)

The causal what/where test re-run in the regime where a map is needed: the SIGN_MATCHED checkpoints (torus, trained and tested at T = 1024, one layer, rank 4, n = 8 per arm; floors on these walks blank 0.506, n-gram 0.575, retrace 0.843). The rebuilt forward equals the model logits exactly on all 32 checkpoints; intact accuracy matches the registered evaluation to <= 0.0007. Held-out accuracy when the score is rebuilt:

condition	signed (learns the map)	|.|	softplus	index RoPE
intact	0.998	0.821	0.780	0.731
token identity within type removed from the score (token type kept)	0.966 (-0.032; 8/8 lower, sign-flip p 0.008, uncorrected)	0.544 (-0.277)	0.476 (-0.305)	0.479 (-0.252)
one shared position kernel x per-pair content gain plus offset	0.998 (-0.000)	0.584 (-0.237)	0.675 (-0.106)	0.439 (-0.293)
additive position + content	0.895 (-0.104)	0.350	0.512	0.492
token type removed too (one global mean)	0.113	0.295	0.302	0.399

condition	signed (learns the map)	|.|	softplus	index RoPE
content only (no rotation)	0.505	0.399	0.385	0.490
SS(P) / SS(S): within-row score sum of squares carried by the content-averaged score	0.925	0.644	0.577	0.618
content x position interaction share	0.061	0.187	0.174	0.353

The model that learns the map factorises what and where: one position kernel per head, scaled by a per-pair content gain plus offset (2 x 441 parameters per head, fitted on the calibration walks, so this condition nests the additive one), reproduces it exactly; removing token identity within type (which also removes action identity within the action class and blank against object within the observation class) costs 0.032 (8/8 seeds lower; sign-flip p 0.008, the design's minimum; paired t p 0.012; Holm over the 24 contrasts 0.08; the effect sits at the MDE of 0.031); token type is essential. The failing models do not factorise: removing identity costs 0.25-0.31 on average (|.|: 7 of 8 runs lose at least 0.15; Abs_r4 s0 loses 0.020), and a shared kernel times a gain does not restore them. Within the non-negative arms, across 16 runs, accuracy correlates with the share of it that survives removing token identity (Spearman 0.84, p 5e-5; a post hoc statistic, `scripts/whatwhere/whatwhere_stats.py`); the two non-negative runs that reach > 0.95 keep 0.97 and 0.99 of their accuracy, the four lowest softplus runs fall to 0.03-0.07 (5-12% of their accuracy), and the |.| runs keep 0.22-0.77 of theirs except Abs_r4 s0 (0.98 of 0.931). (Intact softplus reads 0.780 on this rebuilt forward against the registered 0.781. The `.txt` of the source was written by `--report` after the run; the JSON is the run's.)

K2. What and where at T = 128 (analysis; `docs/WHAT_WHERE_CHECKS.md`; the 48 PAPER2X2 checkpoints)

score used	RoPE	PoPE	MapWM r2	MapWM r4	MapPoPE r2	MapPoPE r4
intact	0.805	0.679	0.971	1.000	0.999	1.000
position only (object identity removed, token type kept)	0.463	0.476	0.837	0.989	0.974	0.986
shared position kernel x content gain	0.430	0.435	0.791	1.000	0.988	1.000

On short walks, where retrace already scores 0.934, converged path models keep 0.974-0.989 with object identity removed and 0.988-1.000 with a shared kernel times a content gain; index models fall below the n-gram floor (0.598). Separation as such is MapFormer's (their Fig. 9).

K3. The what-to-where leak (`LEAK_RESULTS.md`, `GAIN_PHASE_RESULTS.md`, registered)

On a task with a fresh map and 16 fresh objects per sequence (32 x 32 torus, T = 1024, rank 4), object tokens leak a small step into the phase. Removing the step from object tokens (ActOnly) or normalising the step input (NormStep: the step reads LayerNorm(embedding)) lets the model converge: 16/16 SOLVED (unseen-object accuracy 0.9997 / 0.9998) against 0/8 for MapWM (0.9890, still descending). In distribution the remedies gain +0.0107 in unseen-object accuracy (p 0.0002, 8/8), equal

to MapWM's leak; the accuracy and loss are strongly coupled there (r(loss, accuracy) -0.94). On fresh seeds the convergence result replicated (NormStep 8/8 against MapWM 0/8) and the accuracy gain did not (+0.0045, p 0.46; 0.9940 against 0.9895). A score change alone leaves the leak in place (GAIN_PHASE).

K4. Maps from pixels (IMAGE_TORUS; registered, `IMAGE_TORUS_RESULTS.md`, `IMAGE_TORUS_PREREG.md` + Amendments 1-3)

Task and recipe. FACTORIAL's torus (64 x 64, 16 classes, p_empty 0.5, trained and tested at T = 1024, one training map per run, held-out evaluation map) with each observation token replaced by a 28 x 28 image of its class: MNIST digits 0-9 and Fashion-MNIST trouser, dress, sandal, sneaker, bag, ankle boot; a fresh instance at every visit; blank cells are the all-zero image. Training draws only training-split images (96,000; zero test draws recorded per run), every registered readout only test-split images (16,000; zero byte-identical images across splits). Encoder: three strided convolutions and a linear map to d_model (79,968 parameters), trained end to end, its output multiplied by a fixed gain: 30.3729 matches an image embedding's norm to a token embedding's at initialisation (set on seeds 0-15, not the batch seeds), 1.0 is the raw output. Recipe: 1 layer, 2 heads, d 128, 900 epochs, lr 1e-3 warmup + cosine, AdamW wd 0.05, batch 16, --fast-attn; 8 seeds per cell (3100-3107), 48 runs in one batch. Arms: ActOnly (step from action tokens only), MapWM (shared rank 4; every token sets a step), index RoPE. The perception ceiling ceil_vote is a perfect map with a standalone classifier (same encoder and a linear head, 10 epochs; test accuracy 0.980) voting over every earlier visit to the cell.

Disclosure. The design was changed after the pilot's pre-declared stop rule fired (image MapWM at gain 30.37 stayed at the memoryless level over 150 epochs on three pilot seeds; pilots on seeds 3180-3183, outside the batch, never pooled). P1 is partly pilot-informed (image ActOnly seen on one pilot seed at 0.9944; image RoPE's pilot loss known, its accuracy not); P2 and G are pilot-informed, not blind (confirmations on fresh seeds). The original P1 (image MapWM vs image RoPE) reads UNMEASURED under its own rule (-0.033, raw p 0.73) and its prediction (image MapWM 0.97-0.99) was contradicted.

contrast (Holm over P1, P2; G outside)	label	d	95% CI	p
P1 image ActOnly - image RoPE	PATH WINS IN PIXELS	+0.294	[+0.282, +0.306]	perm 0.00016, Holm 0.0003
P2 image ActOnly - image MapWM (gain 30.37)	THE OBSERVATION STEP COSTS IN PIXELS	+0.327	[+0.167, +0.487]	perm 0.00016, Holm 0.0003
G image MapWM gain 1 - gain 30.37	GAIN 1 RESCUES MAPWM	+0.326	[+0.167, +0.487]	perm 0.00016
residual: image ActOnly - MapWM gain 1	(no label)	+0.0007	[+0.0002, +0.0012]	perm 0.091
S2b (ActOnly - MapWM), images minus tokens, paired by seed	(secondary)	+0.287	--	sign-flip 0.039

Per seed (accuracy at T = 1024; floors on the same walks: retrace 0.838-0.848, blank 0.504-0.515, ceil_vote 0.992-0.993):

cell	3100	3101	3102	3103	3104	3105	3106	3107	mean	classes
image ActOnly	0.9954	0.9937	0.9937	0.9933	0.9952	0.9957	0.9945	0.9951	0.9946	8 S
image MapWM, gain 1	0.9930	0.9937	0.9927	0.9944	0.9943	0.9937	0.9945	0.9945	0.9938	8 S
image MapWM, gain 30.37	0.9197	0.5052	0.5076	0.5037	0.5141	0.9616	0.9200	0.5094	0.6677	1 S, 5 T, 2 D
image RoPE	0.6778	0.6957	0.6931	0.7220	0.7025	0.6875	0.7244	0.7047	0.7010	8 T
token ActOnly	1.0000	1.0000	1.0000	0.9986	1.0000	1.0000	1.0000	1.0000	0.9998	8 S
token MapWM	1.0000	0.9988	0.9999	0.9949	1.0000	0.6827	1.0000	1.0000	0.9595	7 S, 1 T

End states of image MapWM at gain 30.37 (thresholds fixed before the batch): LEARNED (above the seed's retrace floor) 3100, 3105, 3106; COLLAPSED (otherwise, encoder probe < 0.5) 3101-3104, 3107; NOISY (per-instance step >= an action step) none. Collapsed runs end at the memoryless loss (2.06-2.10; ln 2 + 0.5 ln 16 = 2.079); four never leave it, and 3103 dipped to 1.91 at epoch 25, during warm-up, and was back at 2.03 by epoch 35. They score exactly their seed's blank floor under every readout, and their encoder outputs one constant (probe 0.0709, the most frequent test class; object step = blank step; per-instance step 0), with that constant observation step 17-48x the action step. Learned runs escape by epoch 50 (3100, 3105) or between epochs 100 and 400 (3106); two null their observation step (0.03x and 0.003x an action step), 3106 keeps a constant one of 9.4x and depends on it (zeroing it: 0.920 -> 0.520). At gain 1 every run is LEARNED with object steps 0.001-0.006 and per-instance steps 0.0006-0.0008 of an action step, and uses them (zeroing costs 0.0005-0.090, median 0.047).

Strata, perception and other readouts (means; retrace on the same revisits in brackets):

cell	short gap (0.911)	long gap (0.509)	wrap-only (0.508)	T = 2048 (OOD)	probe (median)	context destroyed (blank 0.508)
image ActOnly	0.995	0.994	0.994	0.989	0.985	0.269
image MapWM, gain 1	0.994	0.993	0.991	0.970	0.984	0.278
image MapWM, gain 30.37	0.688	0.589	0.544	0.638	0.071	0.425
image RoPE	0.745	0.482	0.483	0.493	0.965	0.359
token ActOnly	1.000	1.000	0.997	0.992	--	0.290
token MapWM	0.961	0.958	0.947	0.943	--	0.292

Cost of pixels for the path model: image ActOnly - token ActOnly -0.0052 (perm p 0.00016); image ActOnly - ceil_vote +0.0019 per seed (min +0.0006). Transfer gap (train-

split minus test-split images at evaluation): ActOnly +0.005, RoPE +0.010 (sign-flip p 0.0078 each). r(final-5% training loss, accuracy) over the 48 runs -0.988.

Scope: class images on a discrete grid, one top-down image per cell visit, allocentric moves; 1 layer, T = 1024, one recipe; block A only (NormStep on images, deeper index models and token RoPE were registered as optional and not run).

L. PoPE's score

L1. Registered

batch	result	file
PAPER2X2 (T = 128, r2, n = 8)	RoPE 0.805, PoPE 0.679, MapWM 0.971, MapPoPE 0.999; interaction +0.154 (8/8); at r4 +0.125 with both path arms at 1.000	`PAPER2X2_RESULTS.md`
MAPPOPE_PAIR (T = 128, r2, n = 16)	the score rule, not the doubled angle count: +0.0243 (p 0.012; SOLVED 16/16 vs 10/16), plausibly convergence speed, +0.009 on fresh seeds; the angle count +0.0002 at a ceiling (0.9995-0.9997), uninformative	`MAPPOPE_PAIR_RESULTS.md`, `GAIN_GRAIN_RESULTS.md`
SCORE_RANK (T = 1024)	no rank-2 rescue: 3/12 vs 3/12 (App. G2)	`SCORE_RANK_RESULTS.md`
DELTA_FREE (T = 128, r2, n = 8, fresh seeds 1600-1607)	PEAK-PINNING SUPPORTED: freeing delta raises the index model by +0.122 [90% CI +0.076, +0.167] (8/8, sign-flip p 0.0078), concentrated in multi-step reversals (R3+ +0.180, Holm p 0.031), and leaves the path model at ceiling (-0.018, p 0.31; SOLVED 6/8 vs 8/8, Fisher p 0.47)	`DELTA_FREE_RESULTS.md`

DELTA_FREE arms (accuracy at T = 128 / 512 / 1024, SOLVED): PoPE-Flat (index, delta pinned: zero initialisation held at 0 by the clamp's dead zone) 0.649 / 0.615 / 0.599, 0/8; PoPE-FreeDelta (delta initialised U(-2 pi, 0), clamp removed) 0.771 / 0.536 / 0.442, 0/8; MapPoPE-Flat (path, pinned) 0.999 / 0.980 / 0.933, 8/8; MapPoPE-FreeDelta 0.981 / 0.901 / 0.831, 6/8; MapPoPE-FrozenDelta (path, delta frozen at the free arms' draw) 0.999 / 0.929 / 0.832, 8/8. The delta x phase interaction is -0.140 [-0.186, -0.092] (0/8 seeds positive). Declared secondaries: with delta pinned 0% of index content pairs prefer offset 3 over offset 1 (kernel argmax at offset 1 in 1074/1088 pairs), freed 50-100% per seed with argmax spread over offsets 3, 4, 7, 8; attention on the newest observation (never the target) 0.287 -> 0.003; index gain by stratum R1 +0.039 (n.s.), R2 +0.117, R3+ +0.180, unvisited-run +0.014; path target alignment C 0.987 pinned vs 0.616 free and 0.614 frozen, with the code term C0 0.993 / 0.798 / 0.884; pinned arms 96-100% in the dead zone; free arms moved mostly by weight decay. Beyond the training length the path robustness contrasts (free or frozen minus pinned at T = 1024: -0.101, -0.102) are below the registered threshold, and the free index arm's retrace kernel does not extrapolate (-0.158 at T = 1024).

L2. The mechanism, probed (analysis; `docs/paper/expansion/scripts/mappope/kernel_probe.py, probe_stats.py`)

Layer 0 rebuilt by hand; every run's logits equal the model's own (difference 0.0); per-seed accuracies reproduce the stored evaluation. T = 128, PAPER2X2 checkpoints, 100 walks per seed:

readout	RoPE	PoPE	MapWM r2	MapPoPE r2
target coherence C (path arms)	--	--	0.703	0.995 (p 0.0002, 8/8)
code term C0 (path arms)	--	--	0.905	0.996 (p 0.0002)
same-cell attention mass (path arms; retrieval head)	--	--	0.747	0.925 (p 0.0008)
attention on action keys (path arms; retrieval head; over both heads 0.223 vs 0.147, p 0.15)	--	--	0.179	0.041 (p 0.0002)
content pairs with K(3) > K(1) (index arms)	53-100% per seed	0% on every seed	--	--
attention on the newest observation (index arms; the more recency-weighted head)	0.002	0.307 (p 0.0002)	--	--
R3+ accuracy (index arms)	0.779	0.566 (p 0.0002)	--	--

The argmax of the content-pair kernel over the odd offsets 1-15, where an observation key can sit, is at offset 1 for 1068/1088 PoPE pairs; for RoPE it is at offsets 3 (569), 7 (342), 11 (156) and 15 (21): every RoPE pair peaks on a retrace offset 4j - 1, a comb (`kernel_argmax_odd.py`; the argmax over all offsets 1-16 also lands on even offsets, where no observation key sits). The PoPE - RoPE accuracy gap decomposes over retrace strata as R1 -0.006, R2 -0.005, R3+ -0.110 and non-retrace -0.004. MapWM's content phase costs C0 - C = +0.20 (7/8 seeds; a statement about target coherence on the retrieval head, not an accuracy prediction). About 80% of the +0.154 interaction is the index side, (MapPoPE - MapWM) - (PoPE - RoPE) = (+0.028) - (-0.126).

T = 1024, SCORE_RANK checkpoints (r2 n = 12, r4 n = 8; 40 of the 100 registered walks per seed; each stratum read on the head with the most same-cell attention in that stratum, `HEAD_RULE=stratum` -> `kernel_probe_score_rank_hstratum_out.txt`, `probe_stats_hstratum_out.txt`; the first version read every stratum on the head chosen over all revisits):

stratum	MapWM r2: C / C0	PoPE score r2: C / C0	MapWM r4: C / C0	PoPE score r4: C / C0
plain gap < 128	0.617 / 0.900	0.960 / 0.960	0.926 / 1.000	0.997 / 1.000
plain gap >= 128	0.538 / 0.795	0.908 / 0.905	0.923 / 0.993	0.991 / 0.993
wrap-only	0.144 / 0.399	0.485 / 0.482	0.904 / 0.967	0.969 / 0.970

On short gaps the score raises C by +0.34 (p 0.0002) and C0 by +0.06 (0.900 vs 0.960, p 0.16): the repair is mostly in the readout term. On gap >= 128 revisits the score raises C

(0.538 to 0.908, p 0.005) and leaves C0 low (0.80 / 0.91, p 0.32, against 0.99 at rank 4); on wrap-only revisits C0 is 0.40 / 0.48 under the two scores (p 0.64), and rank 4 raises it to 0.97 under either (MapWM +0.568, p 0.0009; PoPE score +0.488, p 0.012). In both rank-4 models the wrap-only revisits are carried by the head that is not the short-gap head (C0 0.97 there). At T = 128 MapPoPE's C0 is also higher than MapWM's (0.996 vs 0.905), so on short walks a better readout and a better code co-occur. About 60 contrasts are computed in this appendix without a multiplicity correction; the p 0.0002 contrasts survive Holm over the family, those at p 0.03-0.04 (e.g. MapPoPE - MapWM short-gap accuracy +0.114, p 0.036, consistent with the registered +0.115 on the same checkpoints) do not.

M. Loops and depth on a rank-2 code

M1. Registered (`LOOP_RANK_RESULTS.md`; torus, T = 1024, 900 epochs, n = 8, one batch)

arm	parameters	passes	SOLVED	accuracy	vs A: Fisher / permutation
A: r2, 1 layer	204,373	1	0/8	0.894	--
L: r2, looped x4	204,373	4	2/8	0.973	0.47 / 0.0034 (+0.079, CI [+0.026, +0.131])
L4: r2, 4 real layers	799,189	4	5/8	0.990	0.0256 / 0.0012 (+0.096)
C: r4, 1 layer	204,757	1	8/8	0.998	reference

L and A are identical in parameters and initialisation at every seed (maximum weight difference 0.0). Holm-adjusted p 0.0034 / 0.0028 / 0.0028; L4 - L +0.017 (p 0.0009). The registered verdict is UNMEASURED (it required L to solve >= 4/8). Final losses form three regimes: A 0.051-0.675, L 0.045-0.127, L4 0.011-0.070, C 0.002-0.010. Strata (short / long / wrap-only, registered secondaries): A 0.950 / 0.671 / 0.557; L 1.000 / 0.927 / 0.746; L4 1.000 / 0.990 / 0.887; C 1.000 / 0.995 / 0.977.

M2. Per pass (analysis; `scripts/recurrence/pass_strata.py`, `attn_offsets.py`)

The re-implemented forward matches each model's own forward and its stored evaluation (max |diff| <= 3.1e-4). L at `n_loops = j` is exact; L4's intermediate readouts are a logit lens.

arm	stratum	pass 1	2	3	4	5	6	8
L	all	0.381	0.887	0.964	0.973	0.971	0.954	0.847
L	long	0.363	0.612	0.875	0.927	0.921	0.871	0.731
L	wrap-only	0.393	0.519	0.700	0.747	0.730	0.699	0.623
L4	all	0.507	0.575	0.935	0.990			
L4	wrap-only	0.504	0.507	0.601	0.887			

One pass of L's trained weights scores below the always-blank floor (0.381 vs 0.51; mean, per seed 0.09-0.67, 7 of 8 below); the loop is a fixed-depth computation, and past four passes accuracy falls. Same-cell attention mass (best head per seed,

`pass_strata_besthead.py` -> `out/PASS_STRATA_BESTHEAD.txt`; head labels are a symmetry, and at pass 2 the retrieval head is head 1 on three seeds) rises over L's passes 0.03 -> 0.50 -> 0.64 -> 0.63, and the share of queries whose top-attended key is a same-cell observation 0.35 -> 0.86 -> 0.88 -> 0.92: retrieval forms at passes 2-3 and pass 4 sharpens it. Attention entropy falls 6.1 -> 2.4 nats (head mean, 10 walks per seed; `attn_offsets.py`). On wrap-only revisits L4's last layer puts its top attention on a same-cell observation 81% of the time (best head), from a code whose own one-pass reader finds the cell 12% of the time (the reader's top-1 cell, a different quantity). Mass on same-cell action tokens stays at 0.01-0.15 (best-head means over passes 1-4). With an index code every pass works on short gaps only: the index loop's long-gap and wrap-only accuracy sits at 0.50-0.52 after every pass while short gaps climb 0.41 -> 0.94 (`index_loop_passes.py`, FACTORIAL checkpoints).

M3. Code quality, read without the block (analysis; `phase_reader_soft.py`, `phase_code.py`)

The one-pass reader of E11.1 (token-type phase offsets on query and key, a key bias, a soft vote over observations), fitted to each model's own theta on 24 separate walks and scored on the 50 evaluation walks:

arm	model (all / short / long / wrap)	one-pass reader	model - reader
A	0.894 / 0.950 / 0.671 / 0.558	0.866 / 0.940 / 0.558 / 0.448	+0.028 / +0.010 / +0.113 / +0.110
L	0.973 / 1.000 / 0.927 / 0.747	0.893 / 0.971 / 0.608 / 0.406	+0.080 / +0.029 / +0.319 / +0.341
L4	0.990 / 1.000 / 0.990 / 0.887	0.897 / 0.967 / 0.653 / 0.453	+0.094 / +0.033 / +0.337 / +0.435
C	0.998 / 1.000 / 0.995 / 0.977	1.000 / 1.000 / 1.000 / 0.997	-0.002 / 0.000 / -0.005 / -0.020

Code quality L - A +0.027 (permutation p 0.28), L4 - A +0.031 (p 0.27), both below the exact-t MDE (0.074 / 0.080; long-gap +0.050 / +0.095 against 0.27-0.28; `reader_mde.py` -> `out/READER_MDE.txt`): whether the loop and depth learn a code one pass reads better is unmeasured; C - L +0.107, C - L4 +0.103 (p 0.0002 each). The model is scored on 100 walks and the reader on the first 50 of them; on matched walks the margins change by <= 0.004. The reader under-fits on some runs: with five times the fitting steps on seeds 0, 1 and 3 the independent audit found L's long-gap reader accuracy up by 0.03-0.06, so L's long-gap margin (+0.32) is probably 0.03-0.05 too large; no verdict changes. Model-minus-reader margin against A's: L overall p 0.0036, long 0.0025, wrap-only 0.0002; L4 0.0031, 0.0045, 0.0002. A simpler top-1 kernel reader gives the same verdicts.

M4. Step-map geometry (analysis; `geometry.py`)

Per run, the omega-weighted x- and y-step phase vectors; *2D map*: common mode below 0.1 of the map step and the second singular value above 0.3 of the first; *1D map*: common mode below 0.1, x and y steps collinear (sigma_2 / sigma_1 <= 0.002 in every such run); *drifting*: common mode >= 0.1.

arm	2D map	1D map	drifting
A	1/8	0/8	7/8
L	0/8	4/8 (code x + kappa y, |kappa| 1.38-1.42)	4/8
L4	1/8	4/8 (|kappa| 0.62-1.72)	3/8
C	8/8	0/8	0/8

2D map, aids vs rank 4: 1/16 vs 8/8 (Fisher p 1.2e-5); the aids do shift the class, 1D cancelling 8/16 against 0/8 for plain rank 2 (Fisher p 0.022, post hoc), so they change which code is learned within the family but rarely reach rank 4's. The class thresholds are post hoc and stable (any common-mode cut in (0.08, 0.34) and any second-singular-value cut in (0.002, 0.57) give the same classes; closest call C s0 at 0.08). In the reader's terms the 1D code is the collapsed end state of a rank-D head with the common mode removed. At initialisation the loop multiplies the step map's gradient norm by 2.9 and leaves its batch-to-batch coherence unchanged (0.46 -> 0.41; `grad_at_init.py`): the loop does not deliver a stronger or more consistent signal to the code, and Adam removes the scale.

N. EM's recency deficit: a contrast case

Single-origin MapEM (one p_0, an ablation of the paper-faithful model) on k-back recency (T = 1024, 300 epochs):

leg	result	file
exists	the rewind installed and frozen scores 1.000 on 8/8 at T = 1024 and 2048	`WARM_RESULTS.md` (W1)
kept	installed at the 1/64 scale and trained: 0.642; at 8x the scale 0.941 (4/8 >= 0.95); fully kept (1.000, 8/8) with `w_in`'s content columns held at zero	`WARM_RESULTS.md`, `NOLEAK_RESULTS.md`
from scratch	0.600, against 0.975 for MapWM and 0.837 for paper-faithful MapEM	`RECENCY_EM_RESULTS.md`
found per token	with one offset k every query shares, EM finds a 63-symbol rewind on 7/8 seeds (0.985), faster than MapWM	`SEARCH_RESULTS.md`
budget	four times the budget moves the 64-offset task from 0.578 to 0.928 (6/8 near 1.000)	`SPREAD_RESULTS.md` (an arm of a registered batch; the budget contrast itself was not registered)

EM's deficit is a search problem of a different kind: exposure per query token, which budget closes. Rank 2, by contrast, solves 0/8 with twice the budget on the torus (5/8 runs still descending at 1800 epochs), and |.| on the ring 3/8 at three times the budget (`RING_BUDGET_RESULTS.md`).

O. Analysis scripts

Every number marked (analysis) comes from one of these scripts (CPU, eval-only, on stored checkpoints of registered batches unless stated). "Verification" = the check the

script runs before any new readout. Independent audit: every script below was audited results-blind and re-run on CPU (`docs/paper/review_draft5/ANALYSIS_SCRIPT_AUDIT.md`); every committed output re-derived, except `sign/step_basin.py`, which is not deterministic (its numbers are labelled exploratory). The audit's corrections are applied in this version; the scripts marked "new" were added to apply them, so that every quoted number has a committed generator.

script (under `docs/paper/expansion/scripts/`)	**what it computes**	**used in**	**verification**
`depth/torus_depth.py`, `summarize_torus.py` (`--segments`: new)	torus revisit classes, runs spanned, attention masses; fixed-gap segments and runs spanned by long-gap revisits	Sec. 6, Fig. 4; F1-F2 (the revisit-class table, formerly main-text Table 3)	stored accuracy to 4 decimals (16 runs); forward = model logits
`depth/torus_probe.py`	ridge probes of the residual stream	F3	same stream
`depth/torus_runlen.py`	run-length-shifted evaluation	F4	-- (distribution-shifted)
`depth/ring_depth.py`, `summarize_ring.py`	ring classes, probes, one-layer mixture test	F5	stored accuracy for all 128 runs (< 2e-3)
`depth/ring_probe_window.py`, `ring_class_floors.py`, `ring_plain_bins.py` (new)	late-position probes and old-displacement targets; per-class floors; plain-only run bins	Sec. 6; F5	same stream as `ring_depth.py`
`depth/dyck_distance.py`, `dyck_crosstab.py`	Dyck closing accuracy by blocks and distance	F6	arm means equal the registered ones
`sign/clock_table.py` (new)	the clock / map ratio of E7.1' for every sign and rank arm, per run and per head	Sec. 7.1; Table G3a, Fig. G2; G3; H7	reproduces `sign_clock.py`'s one-sided ratios as a check
`sign/sign_clock.py`, `rank_clock.py`	cancelling usage share; observation and latent clocks	G3; H7	cone inequality as an identity (0 violations)
`sign/latent_geometry.py`, `lattice_p.py`	latent end states; +1/+2 relations	H4; H7	initialisation rebuilt bit-exact
`sign/latent_reach.py`, `aobal_geometry.py`	latent-reachable share; readout targets	H7	--
`sign/prop_checks.py`	numerical checks of E8.3-E8.4	E8	--
`sign/path_profiles.py`	loss along straight paths	H7	--
`sign/step_basin.py`, `basin_summary.py`	step-map-only training from fixed content (CPU; not deterministic: exploratory)	Sec. 7.3; H7	exact solutions held (pilot)
		G4	

script (under docs/paper/expansion/scripts/)	what it computes	used in	verification
rank/rank_geom.py, sweep_final.py, summarise_final.py, c1_runlevel.py	off-plane share, readgain, r_beta		drift and frame conditioning equal the registered head statistics to 4 decimals
rank/escape_start.py, init_frame.py, grad_hiding.py	escape re-read; initial conditions	G4; G6	--
rank/clock_decomp.py, c1_runlevel_once.py (new)	move / observation parts of the leftover clock; escape-arm clock sizes; C1 counting each network once	Sec. 7.2; G4	--
rank/train_snap.py, analyze_snap.py, train_nd_cpu.py	CPU pilots on seeds 3000-3002, 4000-4002	G6	outside every registered batch
recurrence/pass_strata.py, attn_offsets.py, index_loop_passes.py; pass_strata_besthead.py (new)	per-pass accuracy and attention; best head per seed	M2	forward and stored evaluation to <= 3.1e-4 (1e-6 on FACTORIAL)
recurrence/reader_mde.py (new)	MDE of the reader's code-quality contrasts	Sec. 7.4; M3	--
recurrence/phase_reader_soft.py, phase_code.py	one-pass reader fits	Sec. 7.4; M3	fitted on separate walks
recurrence/geometry.py, training_curves.py, grad_at_init.py	step-map classes; curves; gradients at initialisation	M4	--
mappope/kernel_probe.py, probe_stats.py (HEAD_RULE=stratum / SUFFIX=_hstratum: new); kernel_argmax_odd.py (new)	kernel, coherence C and C0, attention by offset; per-stratum head; kernel argmax over odd offsets	Sec. 7.4; L2	rebuilt logits equal the model's (0.0); per-seed accuracies reproduce the stored evaluation
language/tw_lib.py and the tw_*.py scripts; tw_strata_200.py (new: all 200 walks, the non-retrace stratum and its floors)	step table, drift, clock unit, asides, attention, retrace, strata	Table 6; Sec. 8; J1-J6	manual forward logit-identical; generator token-identical; N_WALKS=100 reproduces tw_strata.py
language/usage_tick.py, crossover_usage.py, tw_alpha_drift.py	used-channel rho; growth exponent	Sec. 9; J4	--
docs/audits/2026-10-20/causal_whatwhere_T1024.py; whatwhere/whatwhere_stats.py (new)	what/where at T = 1024; its sign-flip p, SS(P)/SS(S) and Spearman, and the text-world Spearman p	Sec. 8; Sec. 9; J2; K1	rebuilt forward exact on 32 checkpoints; intact accuracy within 0.0007

Post hoc choices (rule 7; from the audit's Sec. 5): the best-head rule of the kernel probes and the per-stratum head rule that replaces it at T = 1024; 40 of the 100 registered walks at T = 1024 (L2); the r_beta < 0.5 cut, the 10x criteria for t_kappa / t_lam and the off-plane share as the escape statistic (G4, G6); geometry.py's class thresholds (M4, stable

over wide ranges); `sign_clock.py`'s 0.1 / 0.05 rad cuts and the near-pi cut 2.6 rad of `clock_table.py` (H7); the `lattice_p.py` tolerance (H4); the path-profile rise statistic (H7); the depth run-length bins (F1, F4); the text-world 0.98 ceiling cut, the blank exclusion of the object-cell stratum and the lift sizes (J5, J6); the what/where Spearman (K1); the 0 mod pi used-channel rule (J2-J4). The audit found that none of its corrections depends on a threshold choice.